

A Compact Paravector Form of Space-Time Algebra

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Abstract—We develop a notation-light paravector presentation of the real Clifford algebra $\text{Cl}_{3,0}$, equivalently the Pauli algebra and the even algebra of Minkowski space-time. A central pseudoscalar supplies the complex structure, so every algebra element is written as one complex scalar plus one complex three-vector. Within this single product calculus we derive the determinant quadratic form, spatial rotations, Lorentz boosts, relativistic kinematics, Maxwell electrodynamics, Klein–Gordon and Dirac factorizations, and the Pauli limit. The construction is a self-contained reformulation of established flat-space physics, not a proposal for new dynamics. Particular care is given to the distinction between event, field, and spinor transformation laws and to the minimal-left-ideal restriction required for one Dirac field.

Index Terms—space-time algebra, geometric algebra, Lorentz transformations, Maxwell electrodynamics, Dirac equation, Pauli equation.

Introduction

This manuscript develops a compact paravector language for several standard parts of flat-space physics. The algebra generated by three elements σ_n with $\sigma_n^2 = 1$ and $\sigma_n \sigma_m = -\sigma_m \sigma_n$ is the associative real Clifford algebra $\text{Cl}_{3,0}$. It is isomorphic to the Pauli matrix algebra $M_2(\mathbb{C})$ and to the even subalgebra of the space-time algebra $\text{Cl}_{1,3}$. It is therefore an eight-real-dimensional realization of even space-time algebra, not the full sixteen-real-dimensional space-time algebra.

The central pseudoscalar $i = \sigma_1 \sigma_2 \sigma_3$, with $i^2 = -1$, lets every algebra element be written uniquely as

$$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}, \quad S \in \mathbb{C}, \quad \mathbf{V} \in \mathbb{C}^3.$$

The notation $\mathbb{C} \oplus \mathbb{C}^3$ used below records this vector-space decomposition; multiplication is always the Pauli product, not componentwise multiplication in a direct-sum algebra. Keeping scalar/vector and real/imaginary parts visible makes the calculations short without suppressing their physical content.

Three involutions organize the presentation: the grade involution $(\cdot)^*$, reversion $(\cdot)^R$, and their composition $(\cdot)^{*R}$. They distinguish the positive coefficient norm from the complex determinant form $ZZ^{*R} = S^2 - \mathbf{V}^2$. On real event paravectors the latter restricts to the Minkowski interval. The same product calculus then gives rotations and boosts, relativistic kinematics, Maxwell theory, and Klein–Gordon, Dirac, and Pauli equations.

The physical results derived here are established results in a deliberately economical notation. The aim is not to introduce a new interaction or change the underlying physics, but to show precisely how familiar constructions share one algebraic carrier. This perspective follows the STA and paravector literature of Hestenes, Baylis, and Doran–Lasenby [12–14], with historical roots in Hamilton’s quaternions and Clifford’s geometric algebra [1, 3].

Bold symbols denote ordinary three-vectors. Natural units $\hbar = c = 1$, rationalized Heaviside–Lorentz electromagnetic units, and metric signature $(+, -, -, -)$ are used throughout. The product formula for scalar coefficients extends directly to complex-valued functions; when differential operators occur, their displayed order and the Leibniz rule are part of the definition.

Reading Guide

The reference sheet on the next page collects the main numbered equations. Section 1 develops the algebra, its functions, transformations, and matrix realizations. Sections 2–4 treat relativistic kinematics, Lorentz actions, and differential operators. Sections 5 and 6 then develop Maxwell theory and relativistic wave equations, including the minimal spinor carrier and Pauli limit. Section 7 states the scope, and Appendix A records the symbolic checks and literature correspondence.

Equation Reference Sheet

Natural units: $\hbar = c = 1$. Signature: $(+, -, -, -)$.

Name	Formula	Ref.
Custom functions and maps		
Real and imaginary parts	$\text{real}(Z) := X, \quad \text{img}(Z) := Y, \quad Z = X + iY$	(18)–(19)
Scalar and vector parts	$\text{scalar}(Z) := S, \quad \text{vector}(Z) := \mathbf{V}, \quad Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$	(20)–(21)
Order functions	$\text{com}(A, B) := AB - BA, \quad \text{acom}(A, B) := AB + BA$	(34)
Axis functions	$\text{par}(\mathbf{A}) := (\mathbf{A} \cdot \mathbf{u})\mathbf{u},$ $\text{prp}(\mathbf{A}) := \mathbf{A} - \text{par}(\mathbf{A}), \quad \mathbf{u}^2 = 1$	(31)
Auxiliary block map	$\mathbf{W}(Z) := \{0, Z; Z^{*\text{R}}, 0\}$	(74)
Algebra		
Sigma product	$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i \sigma_k$	(2)
Involutions	$\sigma_n^* = -\sigma_n, \quad \sigma_n^{\text{R}} = \sigma_n, \quad i^* = i^{\text{R}} = -i$	(6), (7)
Paravector	$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}, \quad S \in \mathbb{C}, \quad \mathbf{V} \in \mathbb{C}^3$	(15)
Dot–cross product	$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$	(32)
Product	$ZZ' = SS' + \mathbf{V} \cdot \mathbf{V}' + (S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}$	(33)
Positive norm	$\ Z\ ^2 = \text{scalar}(ZZ^{\text{R}}) = S ^2 + \ \mathbf{V}\ ^2$	(41)
Determinant form	$\langle\langle Z \rangle\rangle^2 = ZZ^{*\text{R}} = S^2 - \mathbf{V}^2$	(42)
Inverse	$Z^{-1} = Z^{*\text{R}}/(S^2 - \mathbf{V}^2)$	(47)
Projectors	$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$	(50)
Vector exponential	$\exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = \cosh(\theta) + \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}$	(53)
Geometry and fields		
Similarity transport	$TZ' = ZT \Rightarrow Z' = T^{-1}ZT$	(58), (59)
Event congruence	$X' = T^{*\text{R}}XT^*, \quad TT^{*\text{R}} = 1$	(62)
Active spatial rotor	$R = \exp(\theta \mathbf{u} \cdot i\boldsymbol{\sigma}/2), \quad X' = R^{\text{R}}XR$	(57)
Event boost	$L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2), \quad X' = L^*XL^*$	(54)
Interval	$\text{d}s^2 = \text{d}X\text{d}X^* = \text{d}t^2 - \text{d}\mathbf{r}^2$	(77)
Four-velocity and momentum	$U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}$	(78), (79)
Derivative and wave operator	$\partial = \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, \quad \square = \partial\partial^* = \partial_t^2 - \boldsymbol{\nabla}^2$	(86), (89)
Electromagnetic field	$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad \partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$	(90), (93)
Potential identity	$\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}, \quad \partial\Phi = S + F^*$	(90), (91)
Field boost	$F' = L^*FL$	(96)
Lorentz force	$m \text{d}U/\text{d}s = q \text{real}(FU)$	(100)
Wave equations		
Frequency split	$\Omega = \sqrt{m^2 - \boldsymbol{\nabla}^2}, \quad (\partial_t \pm i\Omega)\psi(\pm) = 0$	(104)
Klein–Gordon	$(-E^2 + \mathbf{P}^2 + m^2)\psi = 0$	(106)
Dirac system	$(\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi = 0$	(113)
Leading Pauli limit	$E\phi = (\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})\phi/(2m) + O(m^{-2})$	(120)

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1 Algebraic Foundations

This section fixes the algebra before assigning physical meanings to particular elements. The product is associative and unital. All coefficients in the master product formula are central complex scalars unless an ordered differential-operator calculation is stated explicitly.

1.1 Basis and pseudoscalar

For $n \in \{1, 2, 3\}$, let the algebra be generated over $\mathbb{R}$ by three basis elements σ_n . Their products generate scalars, vectors, bivectors, and pseudoscalars, giving the eight-real-dimensional algebra $\text{Cl}_{3,0} \cong M_2(\mathbb{C}) \cong \text{Cl}_{1,3}^+$. The basis relations are

$$\begin{aligned} \sigma_n^2 &= 1, & \sigma_n \sigma_m &= -\sigma_m \sigma_n \quad (n \neq m), \\ i &:= \sigma_1 \sigma_2 \sigma_3, & i^2 &= -1. \end{aligned} \quad (1)$$

The element i is a pseudoscalar because $i\sigma_n = \sigma_n i$, and squaring i returns the scalar -1 :

$$\begin{aligned} (\sigma_1 \sigma_2 \sigma_3)^2 &= \sigma_1 \sigma_2 (\sigma_3 \sigma_1) \sigma_2 \sigma_3, \\ &= -\sigma_1 (\sigma_2 \sigma_1) (\sigma_3 \sigma_2) \sigma_3, \\ &= -\sigma_1^2 \sigma_2^2 \sigma_3^2, \\ &= -1. \end{aligned}$$

With the Kronecker delta and Levi-Civita symbol,

$$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i \sigma_k, \quad (2)$$

where

$$\begin{aligned} \delta_{nm} &:= \begin{cases} 1, & n = m, \\ 0, & n \neq m, \end{cases} \\ \epsilon_{nmk} &:= \begin{cases} +1, & (n, m, k) \text{ even permutation of } (1, 2, 3), \\ -1, & (n, m, k) \text{ odd permutation of } (1, 2, 3), \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

1.2 The Conjugate and Reverse Operators

The two basic involutions are

$$(\cdot)^* = \text{grade involution: flip the signs of } \sigma_n, \quad (3)$$

$$(\cdot)^R = \text{reverse order of multiplied } \sigma_n. \quad (4)$$

The star is a real-linear automorphism and reversion is a real-linear anti-automorphism. Because each sends the central pseudoscalar i to $-i$, both are antilinear with respect to the internal complex structure. Their composition is a complex-linear anti-automorphism. In operator form,

$$\begin{aligned} (XY)^* &= X^* Y^*, & (XY)^R &= Y^R X^R, \\ X^{*R} &= (X^R)^* = (X^*)^R, & (X^*)^* &= (X^R)^R = X. \end{aligned} \quad (5)$$

Applied to the generators and the pseudoscalar, these rules reduce to

$$\sigma_n^* = -\sigma_n, \quad \sigma_n^R = \sigma_n, \quad \sigma_n^{*R} = -\sigma_n, \quad (6)$$

$$i^* = -i, \quad i^R = -i, \quad i^{*R} = i. \quad (7)$$

For a two-generator bivector pair with $n \neq m$,

$$\begin{aligned} (\sigma_n \sigma_m)^* &= \sigma_n^* \sigma_m^* = (-\sigma_n)(-\sigma_m) = \sigma_n \sigma_m, \\ (\sigma_n \sigma_m)^R &= \sigma_m^R \sigma_n^R = \sigma_m \sigma_n = -\sigma_n \sigma_m, \\ (\sigma_n \sigma_m)^{*R} &= (\sigma_m \sigma_n)^* = \sigma_m^* \sigma_n^* \\ &= (-\sigma_m)(-\sigma_n) = \sigma_m \sigma_n = -\sigma_n \sigma_m. \end{aligned} \quad (8)$$

Equation (8) records the same sign pattern on the sigma-number-two elements. These are the standard intrinsic involutions of $\text{Cl}_{3,0}$. Under the usual identification $\sigma_k = \gamma_k \gamma_0$ with $\text{Cl}_{1,3}^+$, the combined operation $(\cdot)^{*R}$, rather than $(\cdot)^R$ alone, corresponds to space-time reversion restricted to the even subalgebra [12, 14].

The sigma number of a reduced basis product is the number of sigma generators that remain after simplification. Thus 1, σ_n , $\sigma_n \sigma_m$ with $n \neq m$, and i have sigma numbers 0, 1, 2, and 3. Any unreduced product with sigma number larger than 3 simplifies to one of these four cases. Tables 1 and 2 summarize the basis elements and operator signs by sigma number.

Table 1. Sigma basis elements by sigma number.

sigma number	representative basis elements
0 (scalar)	1
1 (vector)	$\sigma_1, \sigma_2, \sigma_3$
2 (bivector)	$\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2$
3 (pseudoscalar)	$\sigma_1 \sigma_2 \sigma_3$

Table 2. Operator signs by sigma number.

sigma number	$(\cdot)^*$	$(\cdot)^R$	$(\cdot)^{*R}$
0 (scalar)	+	+	+
1 (vector)	−	+	−
2 (bivector)	+	−	−
3 (pseudoscalar)	−	−	+

1.3 Generic Complex Paravector

Collect the three generators into the sigma-vector

$$\sigma := \{\sigma_1, \sigma_2, \sigma_3\}, \quad (9)$$

which is a vector of algebraic objects rather than a vector of scalar components. Multiplication by the pseudoscalar gives the associated bivector triple

$$i\sigma := \{i\sigma_1, i\sigma_2, i\sigma_3\} = \{\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2\}, \quad (10)$$

because $i = \sigma_1 \sigma_2 \sigma_3$ and each product $i\sigma_n$ reduces to the complementary ordered two-generator product. Equation (10) therefore records the canonical sigma-number-two basis built from the

generators. With this basis fixed, for real scalars $a, b \in \mathbb{R}$ and real vectors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, with $\mathbf{A} \cdot \boldsymbol{\sigma} = A_1\sigma_1 + A_2\sigma_2 + A_3\sigma_3$, define the real paravectors $X, Y \in \mathbb{R} \oplus \mathbb{R}^3$ by

$$X = a + \mathbf{A} \cdot \boldsymbol{\sigma}, \quad (11)$$

$$Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}. \quad (12)$$

Every algebra element can be written as a complex paravector $Z \in \mathbb{C} \oplus \mathbb{C}^3$, either as a real paravector plus an imaginary paravector,

$$Z = X + iY, \quad (13)$$

or as a complex scalar plus a complex vector component,

$$Z = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} \quad (14)$$

$$= S + \mathbf{V} \cdot \boldsymbol{\sigma}, \quad (15)$$

where the complex scalar and vector components are

$$S = a + ib \in \mathbb{C}, \quad (16)$$

$$\mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3. \quad (17)$$

On coefficient objects, the induced actions of either basic involution are $S^* = S^R = a - ib$ and $\mathbf{V}^* = \mathbf{V}^R = \mathbf{A} - i\mathbf{B}$. When an involution acts on the full algebra term $\mathbf{V} \cdot \boldsymbol{\sigma}$, its action on the sigma basis is included as well, according to Eqs. (5) and (6). Define the real, imaginary, scalar, and vector functions by

$$\text{real}(Z) := X = \frac{Z + Z^R}{2}, \quad (18)$$

$$\text{img}(Z) := Y = \frac{Z - Z^R}{2i}, \quad (19)$$

$$\text{scalar}(Z) := S = \frac{Z + Z^{*R}}{2}, \quad (20)$$

$$\text{vector}(Z) := \mathbf{V}, \quad \mathbf{V} \cdot \boldsymbol{\sigma} = \frac{Z - Z^{*R}}{2}. \quad (21)$$

For the scalar part and vector part of the paravector Z , we also write $Z_S := \text{scalar}(Z)$ and $Z_V := \text{vector}(Z)$.

The conjugate and reverse operators act on the generic complex paravector by

$$Z^* = (a - ib) + (-\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (22)$$

$$Z^R = X - iY = (a - ib) + (\mathbf{A} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (23)$$

$$Z^{*R} = (a + ib) - (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (24)$$

For the real paravector X from Eq. (11), these reduce to

$$X^* = a - \mathbf{A} \cdot \boldsymbol{\sigma}, \quad (25)$$

$$X^R = X = a + \mathbf{A} \cdot \boldsymbol{\sigma}, \quad (26)$$

$$X^{*R} = a - \mathbf{A} \cdot \boldsymbol{\sigma} = X^*. \quad (27)$$

Equations (22)–(24) are the generic formulas used below. Notice in particular that coefficient conjugation is already encoded by reversion; no overbar notation is needed. Equations (25)–(27) give the real specialization. Equations (11)–(21) fix the real and complex paravector pieces and extractors used throughout the later

algebra. Table 3 records the type assignments, Table 4 records the recovery formulas for the extractor definitions, and Table 5 records the same split in components.

Table 3. Type assignments for the generic complex paravector notation.

Type name	Underlying set	Example
real scalar	$\mathbb{R}$	a, b
real vector	$\mathbb{R}^3$	$\mathbf{A}, \mathbf{B}$
complex scalar	$\mathbb{C}$	S
complex vector	$\mathbb{C}^3$	$\mathbf{V}$
real paravector	$\mathbb{R} \oplus \mathbb{R}^3$	X, Y
complex paravector	$\mathbb{C} \oplus \mathbb{C}^3$	Z

Table 4. Components of Z .

scalar:	$S = (Z + Z^{*R})/2$
vector:	$\mathbf{V} \cdot \boldsymbol{\sigma} = (Z - Z^{*R})/2$
real:	$X = (Z + Z^R)/2$
img:	$Y = (Z - Z^R)/(2i)$

Table 5. Real and imaginary scalar/vector parts of Z .

real scalar:	$a = (S + S^R)/2$ $= (X + X^*)/2$
img scalar:	$b = (S - S^R)/(2i)$ $= (Y + Y^*)/2$
real vector:	$\mathbf{A} \cdot \boldsymbol{\sigma} = (\mathbf{V} + \mathbf{V}^R) \cdot \boldsymbol{\sigma}/2$ $= (X - X^*)/2$
img vector:	$\mathbf{B} \cdot \boldsymbol{\sigma} = (\mathbf{V} - \mathbf{V}^R) \cdot \boldsymbol{\sigma}/(2i)$ $= (Y - Y^*)/2$

1.4 Dot products and cross products

For real vectors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, the dot product is defined by

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &:= \sum_{n=1}^3 \sum_{m=1}^3 \delta_{nm} A_n B_m \\ &= A_1 B_1 + A_2 B_2 + A_3 B_3. \end{aligned} \quad (28)$$

Define the square of a vector by

$$\mathbf{A}^2 := \mathbf{A} \cdot \mathbf{A}. \quad (29)$$

For the cross product, the Levi-Civita form gives

$$(\mathbf{A} \times \mathbf{B})_k := \sum_{n=1}^3 \sum_{m=1}^3 \epsilon_{nmk} A_n B_m,$$

which gives the vector

$$\mathbf{A} \times \mathbf{B} = \begin{Bmatrix} A_2 B_3 - A_3 B_2, \\ A_3 B_1 - A_1 B_3, \\ A_1 B_2 - A_2 B_1 \end{Bmatrix}. \quad (30)$$

Dot products are commutative, whereas cross products are anti-symmetric:

$$\begin{aligned}\mathbf{A} \cdot \mathbf{B} &= +\mathbf{B} \cdot \mathbf{A}, \\ \mathbf{A} \times \mathbf{B} &= -\mathbf{B} \times \mathbf{A}.\end{aligned}$$

For a fixed unit vector $\mathbf{u}$, define the parallel and perpendicular component functions once for later use:

$$\begin{aligned}\text{par}(\mathbf{A}) &:= (\mathbf{A} \cdot \mathbf{u})\mathbf{u}, \\ \text{prp}(\mathbf{A}) &:= \mathbf{A} - \text{par}(\mathbf{A}), \quad \mathbf{u}^2 = 1.\end{aligned}\quad (31)$$

With Eqs. (1) and (2),

$$\begin{aligned}(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) &= \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \sigma_n \sigma_m \\ &= \sum_{n=1}^3 A_n B_n \\ &\quad + \sum_{n < m} (A_n B_m - A_m B_n) \sigma_n \sigma_m \\ &= A_1 B_1 + A_2 B_2 + A_3 B_3 \\ &\quad + (A_2 B_3 - A_3 B_2) i \sigma_1 \\ &\quad - (A_1 B_3 - A_3 B_1) i \sigma_2 \\ &\quad + (A_1 B_2 - A_2 B_1) i \sigma_3.\end{aligned}$$

This can be efficiently expressed in terms of the dot product and cross product as

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (32)$$

Table 6 records the same product in component form. The dot and cross products extend complex-bilinearly to $\mathbb{C}^3$. Formula (32) may also be used for commuting scalar-valued functions. Differential-operator coefficients need a separate ordered calculation because derivatives act on everything to their right and obey the Leibniz rule.

The reversed order flips only the antisymmetric term:

$$\begin{aligned}(\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \mathbf{A} \cdot \mathbf{B} - i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) + (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) &= 2\mathbf{A} \cdot \mathbf{B}, \\ (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) - (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) &= 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

The same product immediately gives

$$\begin{aligned}(\mathbf{A} \cdot \boldsymbol{\sigma})^2 &= \mathbf{A}^2, \\ (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{A}^2)^n, \\ (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n+1} &= (\mathbf{A}^2)^n (\mathbf{A} \cdot \boldsymbol{\sigma}), \\ (\mathbf{A} \cdot \boldsymbol{\sigma})\boldsymbol{\sigma} &= \mathbf{A} + i\mathbf{A} \times \boldsymbol{\sigma}.\end{aligned}$$

These are the basic sigma-product identities from which the later paravector formulas are built.

1.5 Products

With Eq. (32) fixing the scalar/vector split for real vectors, the fully complex product follows as the master formula from which the simpler cases are recovered by restriction.

For Z from Eq. (15) and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3$ defined analogously to Eqs. (15) and (17),

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + i(\mathbf{V} \times \mathbf{V}')) \cdot \boldsymbol{\sigma}. \quad (33)$$

Equations (15) and (33) fix the generic complex paravector and its master multiplication rule: pure scalar, real vector, real paravector, and mixed products all follow by setting the unused parts to zero.

1.5.1 Products of real scalars, vectors, and paravectors

Let X and Y be the real paravectors from Eqs. (11) and (12). Using Eq. (32),

$$\begin{aligned}XY &= (a + \mathbf{A} \cdot \boldsymbol{\sigma})(b + \mathbf{B} \cdot \boldsymbol{\sigma}) \\ &= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + i(\mathbf{A} \times \mathbf{B})) \cdot \boldsymbol{\sigma}, \\ YX &= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} - i(\mathbf{A} \times \mathbf{B})) \cdot \boldsymbol{\sigma} \\ &= (XY)^R.\end{aligned}$$

Table 7 gives the same product in component form. The scalar part is symmetric, while the imaginary-vector part changes sign with the order.

Two immediate consequences are

$$\begin{aligned}\text{com}(X, Y) &= 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ XY \in \mathbb{R} \oplus \mathbb{R}^3 &\iff \mathbf{A} \times \mathbf{B} = 0.\end{aligned}$$

So two real paravectors commute exactly when their vector parts are parallel.

Setting $Y = X$ removes the antisymmetric term automatically:

$$X^2 = a^2 + \mathbf{A}^2 + 2a\mathbf{A} \cdot \boldsymbol{\sigma}.$$

If $X^2 = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, then matching scalar and vector parts gives

$$\begin{aligned}b &= a^2 + \mathbf{A}^2, \\ \mathbf{B} &= 2a\mathbf{A},\end{aligned}$$

hence

$$\begin{aligned}a^2 &= \frac{1}{2}(b \pm \sqrt{b^2 - \mathbf{B}^2}), \\ \mathbf{A}^2 &= b - a^2.\end{aligned}$$

These equations are necessary, but the direction equation $\mathbf{B} = 2a\mathbf{A}$ must also be retained. When $\mathbf{B} \neq \mathbf{0}$, real paravector roots exist precisely when $b \geq \|\mathbf{B}\|$. For either choice of the inner sign, choose one of the two signs of the resulting nonzero a and set

$$\mathbf{A} = \frac{\mathbf{B}}{2a}.$$

When $\mathbf{B} = \mathbf{0}$, a real root requires $b \geq 0$; the scalar roots are $X = \pm\sqrt{b}$, and the degenerate pure-vector roots satisfy $a = 0$ and $\mathbf{A}^2 = b$. Thus the displayed quadratic relation is a branch condition, rather than by itself a complete square-root formula.

1.5.2 Products of complex paravectors

The master product law (33) is the complex extension of the real formulas above. Writing Z as in Eqs. (15)–(14) and $Z' = X' + iY' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, one obtains every later specialization by setting the unused scalar or vector parts to zero. Here S, S' and the components of $\mathbf{V}, \mathbf{V}'$ are central complex scalars. This hypothesis permits scalar coefficients to pass through one another and through the sigma generators. Table 8 gives the full component decomposition of ZZ' , Table 9 gives Z^2 , and Tables 10–15 record the order-sensitive products built from the conjugate and reverse operators.

Within this finite-dimensional scalar-coefficient algebra, the scalar part is cyclic:

$$\text{scalar}(ZZ') = \text{scalar}(Z'Z).$$

In particular, if $Z' = MZ^{*\text{R}}$, then

$$\begin{aligned} \text{scalar}(ZMZ^{*\text{R}}) &= \text{scalar}(MZ^{*\text{R}}Z) \\ &= \text{scalar}(M)(Z^{*\text{R}}Z), \end{aligned}$$

because $Z^{*\text{R}}Z$ is scalar.

The order-sensitive variants fall into three algebraic types:

ZZ^* is a real scalar plus a complex vector,

ZZ^{R} is a real scalar plus a real vector,

$ZZ^{*\text{R}}$ is a complex scalar.

Those three cases are precisely the ones needed later for positive norms, determinant forms, inverses, and transformation formulas.

If a component is instead a differential or another noncommuting operator, the ordered expansion must be used before any simplification: in general $\mathbf{V}S' \neq S'\mathbf{V}$, Leibniz terms may occur, and cyclicity of the algebraic scalar projection alone does not justify cycling the operator coefficients. The operator applications later in the manuscript therefore require their displayed ordering and are not automatic substitutions into the central-coefficient formulas of this subsection.

The following tables collect the basic product decompositions and the order-dependent component expansions used throughout this subsection.

Table 6. Components of $(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma})$.

real scalar:	$\mathbf{A} \cdot \mathbf{B}$
img scalar:	0
real vector:	$\mathbf{0}$
img vector:	$\mathbf{A} \times \mathbf{B}$

Table 7. Components of XY .

real scalar:	$ab + \mathbf{A} \cdot \mathbf{B}$
img scalar:	0
real vector:	$a\mathbf{B} + b\mathbf{A}$
img vector:	$\mathbf{A} \times \mathbf{B}$

Tables 8 and 9 collect the full component expansions for the generic product ZZ' and the special case Z^2 .

Table 8. Components of ZZ' .

real scalar:	$aa' - bb' + \mathbf{A} \cdot \mathbf{A}' - \mathbf{B} \cdot \mathbf{B}'$
img scalar:	$ab' + ba' + \mathbf{A} \cdot \mathbf{B}' + \mathbf{B} \cdot \mathbf{A}'$
real vector:	$a\mathbf{A}' - b\mathbf{B}' + a'\mathbf{A} - b'\mathbf{B} - \mathbf{A} \times \mathbf{B}' - \mathbf{B} \times \mathbf{A}'$
img vector:	$a\mathbf{B}' + b\mathbf{A}' + a'\mathbf{B} + b'\mathbf{A} + \mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}'$

Table 9. Components of Z^2 .

real scalar:	$a^2 - b^2 + \mathbf{A}^2 - \mathbf{B}^2$
img scalar:	$2ab + 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$2(a\mathbf{A} - b\mathbf{B})$
img vector:	$2(a\mathbf{B} + b\mathbf{A})$

Tables 10 and 11 make the order dependence of ZZ^* and Z^*Z explicit.

Table 10. Components of ZZ^* .

real scalar:	$a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2$	
img scalar:	0	
real vector:	$-2\mathbf{A} \times \mathbf{B}$	$\leftarrow$ differs
img vector:	$2(a\mathbf{B} - b\mathbf{A})$	

Table 11. Components of Z^*Z .

real scalar:	$a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2$	
img scalar:	0	
real vector:	$2\mathbf{A} \times \mathbf{B}$	$\leftarrow$ differs
img vector:	$2(a\mathbf{B} - b\mathbf{A})$	

Tables 12 and 13 play the same role for products involving the reverse operator.

Table 12. Components of ZZ^{R} .

real scalar:	$a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2$	
img scalar:	0	
real vector:	$2(a\mathbf{A} + b\mathbf{B} + \mathbf{A} \times \mathbf{B})$	$\leftarrow$ differs
img vector:	$\mathbf{0}$	

Table 13. Components of $Z^{\text{R}}Z$.

real scalar:	$a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2$	
img scalar:	0	
real vector:	$2(a\mathbf{A} + b\mathbf{B} - \mathbf{A} \times \mathbf{B})$	$\leftarrow$ differs
img vector:	$\mathbf{0}$	

Tables 14 and 15 show that the products formed with the reverse-conjugate operator collapse to purely scalar and imaginary-scalar parts.

Table 14. Components of ZZ^{R*} .

real scalar:	$a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	$2ab - 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
img vector:	$\mathbf{0}$

Table 15. Components of $Z^{R*}Z$.

real scalar:	$a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	$2ab - 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
img vector:	$\mathbf{0}$

1.5.3 Order dependence and commutators

For arbitrary algebra elements A and B , define the order functions

$$\text{com}(A, B) := AB - BA, \quad \text{acom}(A, B) := AB + BA. \quad (34)$$

The part of a product that depends on order is its commutator function. For generic complex paravectors $Z, Z' \in \mathbb{C} \oplus \mathbb{C}^3$,

$$\text{com}(Z, Z') = 2i(\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma} \quad (35)$$

Equation (35) is the compact commutator law. In real and imaginary vector components it expands to

$$\begin{aligned} \text{com}(Z, Z') = & -2(\mathbf{A} \times \mathbf{B}' + \mathbf{B} \times \mathbf{A}') \cdot \boldsymbol{\sigma} \\ & + 2i(\mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}') \cdot \boldsymbol{\sigma}. \end{aligned}$$

The real-paravector commutator

$$\text{com}(X, Y) = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$$

is the corresponding specialization. In every case the scalar terms cancel, so the commutator isolates the cross-product part of the multiplication law.

The symmetric companion is the anticommutator. For pure vectors,

$$\begin{aligned} \text{acom}(\mathbf{A} \cdot \boldsymbol{\sigma}, \mathbf{B} \cdot \boldsymbol{\sigma}) &= (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) \\ &+ (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= 2\mathbf{A} \cdot \mathbf{B}. \end{aligned} \quad (36)$$

So $\text{acom}(\cdot, \cdot)$ isolates the symmetric dot-product part, just as $\text{com}(\cdot, \cdot)$ isolates the antisymmetric cross-product part.

More generally, for generic complex paravectors $Z, Z' \in \mathbb{C} \oplus \mathbb{C}^3$, define

$$\text{acom}(Z, Z') = ZZ' + Z'Z. \quad (37)$$

Using Eq. (33), one finds

$$\text{acom}(Z, Z') = 2(SS' + \mathbf{V} \cdot \mathbf{V}') + 2(S\mathbf{V}' + S'\mathbf{V}) \cdot \boldsymbol{\sigma}. \quad (38)$$

The cross-product contribution to the anticommutator cancels identically. The result keeps only the symmetric scalar and vector couplings.

The same order dependence appears already in the sigma basis itself. The pseudoscalar commutes with the generators,

$$\begin{aligned} \text{com}(i, \sigma_n) &= 0, \\ i\boldsymbol{\sigma} &= \boldsymbol{\sigma}i = \{\sigma_2\sigma_3, \sigma_3\sigma_1, \sigma_1\sigma_2\}, \end{aligned}$$

while distinct sigma generators satisfy

$$\text{com}(\sigma_n, \sigma_m) = 2\sigma_n\sigma_m = 2 \sum_{k=1}^3 \epsilon_{nmk} i\sigma_k \quad (n \neq m).$$

Thus the oriented sigma-number-two basis is exactly the commutator closure of the sigma-number-one generators.

Tables 12–15 display the same product types in full component form. Their positive-norm and determinant-form consequences are already fixed by Eqs. (41), (43), and (42), so no new quadratic definitions are introduced here.

1.6 Norm and Determinant Form

Two quadratic constructions will be used, and they should not be conflated: a positive coefficient norm and a generally complex determinant form. For a complex scalar $S = a + ib \in \mathbb{C}$, define the scalar norm by

$$\begin{aligned} |S| &:= \sqrt{SS^*}. \\ |S|^2 &= SS^* \\ &= a^2 + b^2. \end{aligned} \quad (39)$$

If $S \in \mathbb{R}$, then Eq. (39) reduces to the ordinary absolute value $|S| = \sqrt{S^2}$. For a complex vector $\mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3$, define the vector norm by

$$\begin{aligned} \|\mathbf{V}\| &:= \sqrt{\mathbf{V} \cdot \mathbf{V}^*}. \\ \|\mathbf{V}\|^2 &= \mathbf{V} \cdot \mathbf{V}^* \\ &= \mathbf{A}^2 + \mathbf{B}^2. \end{aligned} \quad (40)$$

The quadratic product $\mathbf{V}^2 = \mathbf{V} \cdot \mathbf{V}$ is not the same object as the positive norm square $\mathbf{V} \cdot \mathbf{V}^*$. For the complex paravector Z from Eqs. (15)–(14), define the paravector norm by

$$\begin{aligned} \|Z\| &:= \sqrt{|S|^2 + \|\mathbf{V}\|^2}, \\ \|Z\|^2 &= |S|^2 + \|\mathbf{V}\|^2 \\ &= a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 \\ &= \text{scalar}(ZZ^R) = \text{scalar}(Z^R Z). \end{aligned} \quad (41)$$

This is a nonnegative real norm. In the Pauli matrix realization it is $\|Z\|^2 = \frac{1}{2} \text{tr}(ZZ^\dagger)$. The second construction is the complex determinant quadratic form

$$\begin{aligned} \langle\langle Z \rangle\rangle &:= \sqrt{S^2 - \mathbf{V}^2}, \\ \langle\langle Z \rangle\rangle^2 &= S^2 - \mathbf{V}^2 \\ &= (a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2) + 2i(ab - \mathbf{A} \cdot \mathbf{B}) \\ &= ZZ^{*R} = Z^{*R}Z. \end{aligned} \quad (42)$$

The product order does not matter because $Z^{*R}Z = ZZ^{*R}$ is central. Under the Pauli realization this quantity is $\det(Z)$. Its

principal square root in Eq. (42) is branch-dependent and is not itself a globally multiplicative norm; the squared determinant form is the primary algebraic object.

For the real paravector X from Eq. (11), $X^{*R} = X^*$, and the determinant form restricts to the Minkowski quadratic form

$$\begin{aligned}\langle\langle X \rangle\rangle^2 &= a^2 - \mathbf{A}^2 \\ &= XX^* = X^*X.\end{aligned}\quad (43)$$

For $Z, Z' \in \mathbb{C} \oplus \mathbb{C}^3$, the squared determinant form is multiplicative:

$$\begin{aligned}\langle\langle ZZ' \rangle\rangle^2 &= ZZ'(ZZ')^{*R} = ZZ'Z'^{*R}Z^{*R} = Z(Z'Z'^{*R})Z^{*R} \\ &= Z\langle\langle Z' \rangle\rangle^2 Z^{*R} = \langle\langle Z' \rangle\rangle^2 ZZ^{*R} = \langle\langle Z \rangle\rangle^2 \langle\langle Z' \rangle\rangle^2 \\ &= \langle\langle Z'Z \rangle\rangle^2.\end{aligned}\quad (44)$$

Here $(ZZ')^{*R} = Z'^{*R}Z^{*R}$, and both determinant factors are central scalars. Under pure phase multiplication, the norm and determinant form behave as

$$\begin{aligned}\|\exp(i\theta)Z\|^2 &= \|Z\|^2, \\ \langle\langle \exp(i\theta)Z \rangle\rangle^2 &= \exp(2i\theta)\langle\langle Z \rangle\rangle^2.\end{aligned}$$

More generally, under complex scalar multiplication by $r \exp(i\theta)$ with $r \in \mathbb{R}$, the two quadratic constructions scale differently:

$$\|r \exp(i\theta)Z\|^2 = r^2 \|Z\|^2, \quad (45)$$

$$\langle\langle r \exp(i\theta)Z \rangle\rangle^2 = r^2 \exp(2i\theta)\langle\langle Z \rangle\rangle^2. \quad (46)$$

The positive norm is phase-insensitive, while the determinant form retains the phase square. The term ‘‘Minkowski metric’’ below is reserved for the real-paravector restriction (43) and its polarization.

1.7 Inverse formulas

The inverse follows from the determinant identities (42) and (44). For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, when $S^2 - \mathbf{V}^2 \neq 0$,

$$\begin{aligned}Z^{-1} &:= \frac{Z^{*R}}{\langle\langle Z \rangle\rangle^2} = \frac{Z^{*R}}{ZZ^{*R}} = \frac{Z^{*R}}{Z^{*R}Z} \\ &= \frac{S - \mathbf{V} \cdot \boldsymbol{\sigma}}{S^2 - \mathbf{V}^2}.\end{aligned}\quad (47)$$

If the determinant form vanishes, the inverse does not exist.

Differentiating the identity $Z^{-1}Z = 1$ gives

$$d(Z^{-1}) = -Z^{-1}(dZ)Z^{-1}. \quad (48)$$

For $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3$, if $\mathbf{V} \times \mathbf{V}' = \mathbf{0}$, then

$$Z^{-1}Z' = Z'Z^{-1}. \quad (49)$$

1.8 Supporting Algebraic Derivations

The supporting derivations record the explicit sigma-basis and operator calculations, together with one generic map decomposition, that support the earlier basis and operator formulas without repeating the product development already collected above.

1.8.1 Basis-Element Classification

Table 1 lists the four sigma numbers and the corresponding basis-element types, while Table 2 lists the signs acquired under the conjugate operator, the reverse operator, and the reverse-conjugate operator.

1.8.2 Complex Paravector-Valued Maps

This notation records how real scalar and vector functions combine into a generic complex paravector-valued map without imposing any physical interpretation:

$$\begin{aligned}Z(t, \mathbf{r}) &= (a(t, \mathbf{r}) + i b(t, \mathbf{r})) \\ &\quad + (\mathbf{A}(t, \mathbf{r}) + i \mathbf{B}(t, \mathbf{r})) \cdot \boldsymbol{\sigma}, \\ a, b &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}, \\ \mathbf{A}, \mathbf{B} &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}^3, \\ Z &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{C} \oplus \mathbb{C}^3.\end{aligned}$$

1.9 Projector Algebra

Projectors are introduced next because they organize the later function formulas and ladder constructions. Once Eqs. (2) and (32) are fixed, the projector calculus gives the cleanest route from simple unit-direction elements to more general paravector decompositions.

If $\mathbf{A} \in \mathbb{R}^3$ is written as $\mathbf{A} = \theta \mathbf{u}$ with $\theta = \|\mathbf{A}\|$, $\mathbf{u} \in \mathbb{R}^3$, and $\mathbf{u}^2 = 1$, define the idempotents

$$\Pi(\pm) := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad (50)$$

satisfying $\Pi(\pm)^2 = \Pi(\pm)$, $\Pi(+)\Pi(-) = 0$, $\Pi(+)+\Pi(-) = 1$, and $(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm\Pi(\pm)$. Algebraically, $\Pi(+)$ and $\Pi(-)$ isolate the parts aligned and anti-aligned with the direction $\mathbf{u}$. Their role here is purely structural: they split an object into two parts adapted to a chosen axis.

For a scalar-valued function $f : \{\pm 1\} \rightarrow \mathbb{C}$,

$$f(\mathbf{u} \cdot \boldsymbol{\sigma}) = f(+)\Pi(+)+f(-)\Pi(-). \quad (51)$$

This gives a compact projector-based formula for roots, inverses, exponentials, logarithms, and related functions. Once the minimal polynomial of an element of this Pauli algebra is known, polynomial and suitably analytic functions can be reduced within the algebraic framework, with derivative data required at a defective root. The detailed formulas begin with projector and eigenvalue identities and continue to ladder operations associated with a chosen axis.

About the z axis, define the ladder elements

$$\begin{aligned}\ell(\pm) &:= \frac{1}{2}(\sigma_1 \pm i\sigma_2), \quad \text{com}(\sigma_3, \ell(\pm)) = \pm 2\ell(\pm), \\ \ell(\pm)^2 &= 0, \quad \ell(+)\ell(-) = \Pi(+), \quad \ell(-)\ell(+)=\Pi(-).\end{aligned}\quad (52)$$

Equation (52) records the nilpotent combinations of σ_1 and σ_2 that exchange the two projector parts selected by Eq. (50). The projector eigenvalue formulas, mixed projector products, and ladder identities built from Eqs. (50), (51), and (52) are collected next.

1.9.1 Functions of $\mathbf{u} \cdot \boldsymbol{\sigma}$

The projector pair from Eq. (50), together with the functional-calculus rule in Eq. (51), gives the two-root decomposition of the unit-direction element $\mathbf{u} \cdot \boldsymbol{\sigma}$. The corresponding eigenvalue data are

$$\begin{aligned}\Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \Pi(\pm)^2 &= \Pi(\pm), \\ \Pi(+)\Pi(-) &= 0, \\ \Pi(+)+\Pi(-) &= 1, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) &= \pm\Pi(\pm),\end{aligned}$$

so $\mathbf{u} \cdot \boldsymbol{\sigma}$ has the two eigenvalues $+1$ and -1 . Specified scalar values at those two roots determine the corresponding projector function:

$$\begin{aligned}f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(+1)\Pi(+) + f(-1)\Pi(-) \\ &= \frac{1}{2}(f(+1) + f(-1)) + \frac{1}{2}(f(+1) - f(-1))\mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned}$$

These are the basic projector formulas for a unit-direction element. The first examples below collect the square root, logarithm branch data, rational inverse, Cayley transform, and the two exponential families for later reuse:

$$\begin{aligned}\sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} &= \epsilon(+)\Pi(+) + \epsilon(-)i\Pi(-), \\ \log(\mathbf{u} \cdot \boldsymbol{\sigma}) &= 2i\pi(n\Pi(+) + (\tfrac{1}{2} + m)\Pi(-)), \\ (z + \mathbf{u} \cdot \boldsymbol{\sigma})^{-1} &= \frac{1}{z+1}\Pi(+) + \frac{1}{z-1}\Pi(-) \\ &= \frac{z - \mathbf{u} \cdot \boldsymbol{\sigma}}{z^2 - 1}, \\ \frac{i - \mathbf{u} \cdot \boldsymbol{\sigma}}{i + \mathbf{u} \cdot \boldsymbol{\sigma}} &= i\mathbf{u} \cdot \boldsymbol{\sigma}, \\ \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) &= \cosh(\theta) + \sinh(\theta)\mathbf{u} \cdot \boldsymbol{\sigma}, \\ \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}) &= \cos(\theta) + i\sin(\theta)\mathbf{u} \cdot \boldsymbol{\sigma},\end{aligned}$$

where $m, n \in \mathbb{Z}$ index independent logarithm branches, and the inverse formula requires $z \neq \pm 1$. The signs $\epsilon(+), \epsilon(-) \in \{+1, -1\}$ are likewise independent; the conventional scalar branches $\sqrt{+1} = 1$ and $\sqrt{-1} = i$ correspond to $\epsilon(+)=\epsilon(-)=+1$. These identities follow by applying the chosen scalar value separately on the two projector ranges. The later exponential and logarithm subsections revisit the same formulas with their domains stated for real and complex paravectors.

1.9.2 Functions of X

Let

$$\begin{aligned}X &= t + \mathbf{r} \cdot \boldsymbol{\sigma}, \\ t &= \text{scalar}(X) \in \mathbb{R}, \\ \mathbf{r} &= \text{vector}(X) \in \mathbb{R}^3, \\ \mathbf{u} &:= \mathbf{r}/\|\mathbf{r}\| \quad \text{when } \mathbf{r} \neq 0.\end{aligned}$$

If $\mathbf{r} \neq 0$, write $\mathbf{r} = \|\mathbf{r}\|\mathbf{u}$ with $\mathbf{u}^2 = 1$. Then the minimal polynomial of X is

$$\begin{aligned}X^2 - 2tX + (t^2 - \mathbf{r}^2) &= 0, \\ \lambda(\pm) &= t \pm \|\mathbf{r}\|, \\ 0 &= (X - \lambda(-))(X - \lambda(+)) \\ &= X^2 - \lambda(-)X \\ &\quad - X\lambda(+) + \lambda(-)\lambda(+).\end{aligned}$$

With

$$a := \lambda(+), \quad b := \lambda(-),$$

the corresponding projector decomposition is

$$\begin{aligned}\Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ X &= a\Pi(+) + b\Pi(-), \\ \Pi(\pm)X &= \lambda(\pm)\Pi(\pm).\end{aligned}$$

If f is defined at the two eigenvalues, then

$$\begin{aligned}f(X) &= f(a)\Pi(+) + f(b)\Pi(-) \\ &= \frac{1}{2}(f(a) + f(b)) + \frac{1}{2}(f(a) - f(b))\mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned}$$

Writing $X' := f(X)$, introduce its scalar and signed axial coefficients by

$$\begin{aligned}t' &= \frac{1}{2}(f(a) + f(b)), \\ \rho' &:= \frac{1}{2}(f(a) - f(b)), \\ X' &= t' + \rho'\mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned}$$

For complex-valued f , both t' and ρ' may be complex. Even when they are real, ρ' is signed; the magnitude of the vector part is $|\rho'|$, not ρ' itself. The projector summary may therefore be recorded as

$$\begin{aligned}\lambda(\pm) &= t \pm \|\mathbf{r}\|, \\ \mathbf{u} &= \mathbf{r}/\|\mathbf{r}\|, \\ \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \Pi(\pm)X &= \lambda(\pm)\Pi(\pm), \\ X &= \lambda(+)\Pi(+) + \lambda(-)\Pi(-), \\ f(X) &= f(\lambda(+))\Pi(+) + f(\lambda(-))\Pi(-).\end{aligned}$$

The standard examples then follow immediately:

$$\begin{aligned}\exp(X) &= \exp(t) \left(\cosh(\|\mathbf{r}\|) + \sinh(\|\mathbf{r}\|)\mathbf{u} \cdot \boldsymbol{\sigma} \right), \\ \sqrt{X} &= \epsilon(+)\sqrt{a}\Pi(+) + \epsilon(-)\sqrt{b}\Pi(-), \\ \epsilon(+), \epsilon(-) &\in \{+1, -1\}, \\ \log(X) &= \log(a)\Pi(+) + \log(b)\Pi(-), \\ X^{-1} &= \frac{\Pi(+)}{a} + \frac{\Pi(-)}{b} \\ &= \frac{X^*}{t^2 - \mathbf{r}^2}, \\ X^n &= a^n\Pi(+) + b^n\Pi(-) \\ &= \frac{1}{2}(a^n + b^n) + \frac{1}{2}(a^n - b^n)\mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned}$$

Here the scalar square roots and logarithms must be taken on specified branches. A square root that is itself a real paravector exists when $a, b \geq 0$, equivalently $t \geq \|\mathbf{r}\|$; otherwise the displayed roots are complex. The displayed logarithm requires $a, b \neq 0$; it is the real principal logarithm when $a, b > 0$. The inverse requires $ab \neq 0$. The power formula holds for $n \geq 0$, and also for negative integers when $ab \neq 0$. For two paravectors diagonalized along possibly different axes,

$$f(X)g(X') = (f(a)\Pi(+) + f(b)\Pi(-)) \\ \times (g(a')\Pi'(+) + g(b')\Pi'(-)),$$

so

$$f(X)g(X') = f(a)g(a')\Pi(+)\Pi'(+) \\ + f(a)g(b')\Pi(+)\Pi'(-) \\ + f(b)g(a')\Pi(-)\Pi'(+) \\ + f(b)g(b')\Pi(-)\Pi'(-).$$

The mixed projector products are

$$\begin{aligned} \Pi(+)\Pi'(+) &= \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') \\ &\quad + \frac{1}{4}(\mathbf{u}' + \mathbf{u} + i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}, \\ \Pi(+)\Pi'(-) &= \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') \\ &\quad + \frac{1}{4}(-\mathbf{u}' + \mathbf{u} - i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}, \\ \Pi(-)\Pi'(+) &= \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') \\ &\quad + \frac{1}{4}(\mathbf{u}' - \mathbf{u} - i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}, \\ \Pi(-)\Pi'(-) &= \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') \\ &\quad + \frac{1}{4}(-\mathbf{u}' - \mathbf{u} + i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}. \end{aligned}$$

As a useful example, the Cayley-type transform is itself a projector exponential:

$$\begin{aligned} \frac{1 - \mathbf{r} \cdot \boldsymbol{\sigma}}{1 + \mathbf{r} \cdot \boldsymbol{\sigma}} &= \exp(-2 \operatorname{arctanh}(\mathbf{r} \cdot \boldsymbol{\sigma})) \\ &= \exp(-2 \operatorname{arctanh}(\|\mathbf{r}\|))\Pi(+) \\ &\quad + \exp(+2 \operatorname{arctanh}(\|\mathbf{r}\|))\Pi(-). \end{aligned}$$

The rational expression requires $\|\mathbf{r}\| \neq 1$. For a real-valued $\operatorname{arctanh}$ and the displayed real exponentials, one further assumes $\|\mathbf{r}\| < 1$; otherwise a complex branch must be selected.

1.9.3 Minimal Polynomial and Projector Decomposition

The two-root data of a complex paravector are collected here because the later function formulas, projector identities, and exponential identities all depend on this algebraic factorization; see Eqs. (33) and (51):

$$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \\ Z^2 - 2SZ + (S^2 - \mathbf{V}^2) = 0.$$

When $\mathbf{V}^2 \neq 0$, choose z with $z^2 = \mathbf{V}^2$ and set

$$\begin{aligned} \lambda(\pm) &:= S \pm z, \\ \Pi_Z(\pm) &:= \frac{1}{2} \left(1 \pm \frac{\mathbf{V} \cdot \boldsymbol{\sigma}}{z} \right). \end{aligned}$$

Then

$$\begin{aligned} (Z - \lambda(-))(Z - \lambda(+)) &= 0, \\ Z &= \lambda(+)\Pi_Z(+) + \lambda(-)\Pi_Z(-), \\ f(Z) &= f(\lambda(+))\Pi_Z(+) \\ &\quad + f(\lambda(-))\Pi_Z(-) \end{aligned}$$

for any scalar function defined on the two eigenvalues. Changing the sign of z merely exchanges the two labels.

The case $\mathbf{V}^2 = 0$ must be treated separately. If $\mathbf{V} = \mathbf{0}$, then $Z = S$ is scalar. If $\mathbf{V} \neq \mathbf{0}$, let $N := \mathbf{V} \cdot \boldsymbol{\sigma}$; then $N^2 = 0$, $Z = S + N$ is not diagonalizable, and for a function analytic at S ,

$$f(Z) = f(S) + f'(S)N.$$

This nilpotent formula is the continuous replacement for the two-projector formula when the two roots coalesce.

1.9.4 Spin-Axis Projectors

The projector family from Eq. (50) resolves the left action of a chosen unit axis $\mathbf{u} \in \mathbb{R}^3$. The basic identities are

$$\begin{aligned} \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \Pi(\pm)^2 &= \Pi(\pm), \\ \Pi(+)\Pi(-) &= 0, \\ \Pi(+) + \Pi(-) &= 1, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) &= \pm\Pi(\pm), \\ \Pi(\pm)^R &= \Pi(\pm), \\ \Pi(\pm)^* &= \Pi(\mp). \end{aligned}$$

Accordingly,

$$\mathbf{u} \cdot \boldsymbol{\sigma} = (+1)\Pi(+) + (-1)\Pi(-),$$

so the two projector parts carry the eigenvalues $+1$ and -1 of the axis operator.

In the Pauli matrix realization, each $\Pi(\pm)$ has rank one. A general element of $\mathbb{C} \oplus \mathbb{C}^3$, however, is a full 2×2 complex matrix, not one irreducible two-component spinor. The following projection formulas therefore hold column by column. They describe a single Pauli spinor after one fixes a minimal left ideal by placing a fixed projector on the right; without that restriction, the full paravector describes two spinor columns or may be interpreted as a density amplitude.

If the axis is rotated by the rotor R from Eq. (57), define

$$\mathbf{u}' \cdot \boldsymbol{\sigma} = R^R(\mathbf{u} \cdot \boldsymbol{\sigma})R, \quad \Pi'(\pm) := \frac{1}{2}(1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma}) = R^R\Pi(\pm)R.$$

Then

$$\mathbf{u}' \cdot \boldsymbol{\sigma} = (+1)\Pi'(+) + (-1)\Pi'(-).$$

For an algebra element $\psi \in \mathbb{C} \oplus \mathbb{C}^3$, define its left-projected parts

$$\psi(\pm) := \Pi(\pm)\psi.$$

These satisfy

$$(\mathbf{u} \cdot \boldsymbol{\sigma})\psi(\pm) = \pm\psi(\pm),$$

$$\psi = \psi(+) + \psi(-).$$

Thus each spinor column splits into the two eigenspaces of the chosen axis. Reversion acts as

$$\psi(\pm)^R = \psi^R \Pi(\pm)^R$$

$$= \psi^R \Pi(\pm),$$

and therefore

$$\psi(\pm)^R \psi(\pm) = \psi^R \Pi(\pm) \psi,$$

$$\psi(\pm)^R \psi(\mp) = \psi^R \Pi(\pm) \Pi(\mp) \psi = 0,$$

$$\psi(\mp)^R \psi(\pm) = \psi^R \Pi(\mp) \Pi(\pm) \psi = 0.$$

For two algebra elements ϕ, ψ , define the mixed pairing

$$\langle \phi | M | \psi \rangle := \text{scalar}(\phi^R M \psi).$$

For a nonzero ψ , an expectation value is the normalized ratio

$$\langle M \rangle_\psi := \frac{\langle \psi | M | \psi \rangle}{\text{scalar}(\psi^R \psi)}.$$

For a full matrix ψ , this is the Hilbert–Schmidt or density-amplitude pairing; for a vector in a fixed minimal left ideal, it is the ordinary Pauli-spinor pairing. The unnormalized numerator for the rotated axis expands as

$$\begin{aligned} \langle \psi | \mathbf{u}' \cdot \boldsymbol{\sigma} | \psi \rangle &= \langle \psi(+) + \psi(-) | \mathbf{u}' \cdot \boldsymbol{\sigma} | \psi(+) + \psi(-) \rangle \\ &= \langle \psi(+) | \mathbf{u}' \cdot \boldsymbol{\sigma} | \psi(+) \rangle \\ &\quad + \langle \psi(+) | \mathbf{u}' \cdot \boldsymbol{\sigma} | \psi(-) \rangle \\ &\quad + \langle \psi(-) | \mathbf{u}' \cdot \boldsymbol{\sigma} | \psi(+) \rangle \\ &\quad + \langle \psi(-) | \mathbf{u}' \cdot \boldsymbol{\sigma} | \psi(-) \rangle. \end{aligned}$$

The algebra governing these same-side and cross-side terms is

$$(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) = \mathbf{u}' \cdot \mathbf{u} + i(\mathbf{u}' \times \mathbf{u}) \cdot \boldsymbol{\sigma},$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) = \mathbf{u}' \cdot \mathbf{u} - i(\mathbf{u}' \times \mathbf{u}) \cdot \boldsymbol{\sigma},$$

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) = (2(\mathbf{u}' \cdot \mathbf{u})\mathbf{u} - \mathbf{u}') \cdot \boldsymbol{\sigma}.$$

Write the decomposition of the second axis into parts parallel and perpendicular to $\mathbf{u}$ as

$$\mathbf{u}'_{\parallel} := (\mathbf{u} \cdot \mathbf{u}')\mathbf{u}, \quad \mathbf{u}'_{\perp} := \mathbf{u}' - \mathbf{u}'_{\parallel}.$$

Then the projector products are

$$\Pi(\pm)(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm(\mathbf{u} \cdot \mathbf{u}')\Pi(\pm),$$

$$\Pi(\pm)(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi(\mp) = \frac{1}{2}(\mathbf{u}'_{\perp} \pm i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma} \Pi(\mp).$$

The diagonal terms are therefore weighted by the axis overlap $\mathbf{u} \cdot \mathbf{u}'$, while the off-diagonal terms are controlled by the perpendicular and commutator parts of the rotated axis.

If θ is the angle between $\mathbf{u}$ and $\mathbf{u}'$, then

$$\mathbf{u}' \cdot \mathbf{u} = \cos(\theta),$$

and the rank-one projector overlap is

$$2 \text{ scalar}(\Pi(s)\Pi(s')) = \frac{1}{2}(1 + ss'\mathbf{u} \cdot \mathbf{u}'), \quad s, s' \in \{+1, -1\}.$$

For a normalized irreducible axis eigenstate this overlap is the transition probability. In particular, the same-side and opposite-side probabilities are

$$\frac{1}{2}(1 + \mathbf{u}' \cdot \mathbf{u}) = \cos^2(\theta/2), \quad \frac{1}{2}(1 - \mathbf{u}' \cdot \mathbf{u}) = \sin^2(\theta/2).$$

If $\mathbf{u}'$ is obtained by rotating $\mathbf{u}$ through angle ϕ about a unit axis $\mathbf{n}$, Rodrigues's formula gives

$$\begin{aligned} \mathbf{u}' \cdot \mathbf{u} &= (\mathbf{u} \cdot \mathbf{u}) \cos(\phi) + (\mathbf{u} \cdot (\mathbf{n} \times \mathbf{u})) \sin(\phi) \\ &\quad + (\mathbf{n} \cdot \mathbf{u})(1 - \cos(\phi))(\mathbf{n} \cdot \mathbf{u}) \\ &= (\mathbf{n} \cdot \mathbf{u})^2 + (1 - (\mathbf{n} \cdot \mathbf{u})^2) \cos(\phi). \end{aligned}$$

In the perpendicular case $\mathbf{n} \cdot \mathbf{u} = 0$, this reduces to

$$\mathbf{u}' \cdot \mathbf{u} = \cos(\phi),$$

hence

$$\frac{1}{2}(1 + \mathbf{u}' \cdot \mathbf{u}) = \cos^2(\phi/2), \quad \frac{1}{2}(1 - \mathbf{u}' \cdot \mathbf{u}) = \sin^2(\phi/2).$$

These are the familiar spin- $\frac{1}{2}$ transition probabilities. Without the normalization and irreducible-state qualification above, they should be read only as projector overlaps.

1.9.5 Ladder Operations About the z Axis

For raising and lowering about the z axis, choose

$$\mathbf{u} = \mathbf{e}_3 = (0, 0, 1), \quad \Pi(\pm) = \frac{1}{2}(1 \pm \sigma_3).$$

Define the ladder operators by

$$\ell(\pm) := \frac{1}{2}(\sigma_1 \pm i\sigma_2).$$

They satisfy the commutator relation

$$\text{com}(\sigma_3, \ell(\pm)) = \pm 2\ell(\pm),$$

so $\ell(+)$ raises the σ_3 eigenvalue and $\ell(-)$ lowers it.

Their projector products are

$$\ell(+)\ell(-) = \Pi(+), \quad \ell(-)\ell(+) = \Pi(-),$$

and each ladder operator is nilpotent:

$$\ell(\pm)^2 = 0.$$

The one-sided projector actions are

$$\begin{aligned} \ell(+)\Pi(+) &= 0, & \ell(+)\Pi(-) &= \ell(+), \\ \Pi(+)\ell(+) &= \ell(+), & \Pi(-)\ell(+) &= 0, \\ \ell(-)\Pi(-) &= 0, & \ell(-)\Pi(+) &= \ell(-), \\ \Pi(-)\ell(-) &= \ell(-), & \Pi(+)\ell(-) &= 0, \end{aligned}$$

which make the raising/lowering interpretation explicit. The original sigma generators are recovered from the same ladder/projector data:

$$\begin{aligned}\sigma_1 &= \ell(+) + \ell(-), & \sigma_2 &= \frac{\ell(+) - \ell(-)}{i}, \\ \sigma_3 &= \Pi(+) - \Pi(-).\end{aligned}$$

1.9.6 Spin-Axis Noncommutativity

For unit directions $\mathbf{u}, \mathbf{u}' \in \mathbb{R}^3$,

$$\begin{aligned}\text{acom}(\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}) &= 2 \mathbf{u} \cdot \mathbf{u}', \\ \text{com}(\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}) &= 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}.\end{aligned}$$

The anticommutator records the axis overlap, while the commutator records the oriented area spanned by the two axes. Thus the axis operators commute exactly when the directions are parallel or antiparallel. These identities are purely algebraic. A Robertson or Robertson–Schrödinger uncertainty relation additionally requires a normalized state, expectation functional, and state-dependent variances; no such inequality is inferred from the commutator alone [11].

1.10 Exponential and Logarithmic Algebra

With the projector calculus fixed, exponential and logarithmic formulas can be developed in one place. The sigma-number-one exponential produces hyperbolic half-angle transform elements, and the sigma-number-two exponential produces the circular half-angle elements used later for rotors and for the quaternion correspondence.

1.10.1 Real Exponential and Half-Angle Elements

With the projector calculus in hand, the sigma-number-one exponential becomes especially transparent. For the real paravector

$$X = a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3,$$

write

$$\begin{aligned}\mathbf{A} &= \theta \mathbf{u}, & \theta &= \|\mathbf{A}\|, \\ \mathbf{u} &= \frac{\mathbf{A}}{\|\mathbf{A}\|} \quad (\mathbf{A} \neq 0), & \mathbf{u} &= (u_1, u_2, u_3), \\ \mathbf{u}^2 &= 1.\end{aligned}$$

Then Eq. (51) gives

$$\begin{aligned}\exp(\pm \mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\pm \theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= \exp(\pm \theta) \Pi(+) + \exp(\mp \theta) \Pi(-) \\ &= \cosh(\theta) \pm \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned}\tag{53}$$

Equivalently, the same result follows directly from the power series by separating the even and odd powers:

$$\begin{aligned}\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= \sum_{n=0}^{\infty} \frac{\theta^n (\mathbf{u} \cdot \boldsymbol{\sigma})^n}{n!} \\ &= \sum_{n=0}^{\infty} \frac{\theta^{2n}}{(2n)!} + \sum_{n=0}^{\infty} \frac{\theta^{2n+1}}{(2n+1)!} \mathbf{u} \cdot \boldsymbol{\sigma} \\ &= \cosh(\theta) + \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned}$$

With

$$\begin{aligned}\gamma &= \cosh(\theta), \\ \mathbf{v} &= \tanh(\theta) \mathbf{u},\end{aligned}$$

Eq. (53) becomes the basic hyperbolic parametrization used repeatedly below:

$$\begin{aligned}\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\ \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) &= \gamma(1 - \mathbf{v} \cdot \boldsymbol{\sigma}).\end{aligned}$$

For the real paravector X from Eq. (11), the scalar part commutes with the vector part, so

$$\begin{aligned}\exp(X) &= \exp(a) \exp(\mathbf{A} \cdot \boldsymbol{\sigma}), \\ \exp(X^*) &= \exp(a) \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) = \exp(X)^*, \\ \exp(-X) &= \exp(-a) \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) = \exp(X)^{-1}.\end{aligned}$$

The corresponding half-angle element is

$$L(\theta, \mathbf{u}) = \exp\left(\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right), \quad X' = L^* X L^*. \tag{54}$$

Equation (54) is the $T = L$ specialization of the general event congruence (62), derived below from the carrier transformation. Because $L^{\mathbb{R}} = L$ and $L^* = L^{-1}$, it reads

$$X' = L^{-1} X (L^{-1})^{\mathbb{R}}.$$

It preserves the real-paravector subspace and mixes the scalar component with the vector component parallel to $\mathbf{u}$. It must be distinguished from the similarity $T^{-1} X T$, which preserves the scalar part and is used below as an algebra and basis automorphism. For rotations the two actions have the same appearance because the rotor is unitary; for hyperbolic elements they do not. Finally, the conjugate operator simply exchanges the two projector weights:

$$\begin{aligned}\exp(\mathbf{A} \cdot \boldsymbol{\sigma})^* &= \exp(-\theta) \Pi(+) + \exp(\theta) \Pi(-) \\ &= \gamma(1 - \mathbf{v} \cdot \boldsymbol{\sigma}) \\ &= \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= \exp((\mathbf{A} \cdot \boldsymbol{\sigma})^*) \\ &= \exp(\mathbf{A} \cdot \boldsymbol{\sigma})^{-1}.\end{aligned}$$

1.10.2 Real Logarithm of a Paravector

The logarithm is the inverse problem for the same projector formulas. Let Y be the real paravector from Eq. (12), and assume $b > \|\mathbf{B}\|$, so that the two real eigenvalues are positive and the principal logarithm is defined. For $\mathbf{B} \neq 0$, write $\mathbf{B} = \|\mathbf{B}\| \mathbf{u}$. Then

$$\begin{aligned}\lambda(\pm) &= b \pm \|\mathbf{B}\|, \\ Y &= \lambda(+) \Pi(+) + \lambda(-) \Pi(-),\end{aligned}$$

with the same projector pair $\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$. The logarithm therefore acts separately on the two eigenvalues:

$$\begin{aligned}\log(Y) &= \log(\lambda(+)) \Pi(+) + \log(\lambda(-)) \Pi(-) \\ &= \frac{1}{2} \log(\lambda(+)\lambda(-)) + \frac{1}{2} \log(\lambda(+)/\lambda(-)) \mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned}\tag{55}$$

When $\mathbf{B} = \mathbf{0}$, the axis is unnecessary and this formula reduces directly to $\log(Y) = \log(b)$. For $\mathbf{B} \neq \mathbf{0}$, if $Y = \exp(X)$ with X from Eq. (11) and $\mathbf{A} = \alpha \mathbf{u}$, then $\lambda(\pm) = \exp(a \pm \alpha)$, so the recovered parameters are

$$\begin{aligned} Y &= \exp(X) \\ &= \exp(a + \mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= \exp(a) \exp(\mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= \exp(a) (\cosh(\alpha) + \sinh(\alpha) \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= b + \|\mathbf{B}\| \mathbf{u} \cdot \boldsymbol{\sigma}, \end{aligned}$$

so one reads off

$$\begin{aligned} \alpha &= \operatorname{arctanh}(\|\mathbf{B}\|/b), \\ \exp(a) &= \sqrt{b^2 - \mathbf{B}^2}, \\ a &= \frac{1}{2} \log(b^2 - \mathbf{B}^2), \\ \mathbf{A} &= \alpha \mathbf{u} = \operatorname{arctanh}(\|\mathbf{B}\|/b) \frac{\mathbf{B}}{\|\mathbf{B}\|}. \end{aligned}$$

Thus the principal logarithm of a future-directed time-like real paravector is again a real paravector,

$$\log(Y) = a + \mathbf{A} \cdot \boldsymbol{\sigma}.$$

1.10.3 Circular Exponentials and Half-Angle Elements

Replacing θ by $i\theta$ turns the same projector calculation into the trigonometric form of the sigma-number-two exponential:

$$\begin{aligned} \exp(\pm i\mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\pm i\theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= \exp(\pm i\theta) \Pi(+) + \exp(\mp i\theta) \Pi(-) \\ &= \cos(\theta) \pm i \sin(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (56)$$

In particular, the negative-sign form may be written as

$$\exp(-i\mathbf{A} \cdot \boldsymbol{\sigma}) = \cos(\theta) + \sin(\theta) \mathbf{u} \cdot (-i\boldsymbol{\sigma}).$$

Later, after introducing the quaternion basis $\mathbf{q} = -i\boldsymbol{\sigma}$, this same formula becomes

$$\exp(-i\mathbf{A} \cdot \boldsymbol{\sigma}) = \cos(\theta) + \sin(\theta) \mathbf{u} \cdot \mathbf{q}.$$

For a real angle $\theta \in \mathbb{R}$ and a unit real vector $\mathbf{u} \in \mathbb{R}^3$, the corresponding circular half-angle exponential is

$$\begin{aligned} R(\theta, \mathbf{u}) &:= \exp\left(\frac{\theta}{2} \mathbf{u} \cdot i\boldsymbol{\sigma}\right), \\ R^R R &= 1, \quad X' = R^R X R. \end{aligned} \quad (57)$$

Equation (57) is the circular transform built from Eq. (56). Because $R^* = R$, it is simultaneously the $T = R$ specialization of the general similarity (59) and of the event congruence (62). Their common derivation and the detailed transform consequences are given in the generalized transform section. The quaternion restatement of the same sigma-number-two algebra is collected later, after the transform discussion has been completed.

1.10.4 Exponentials of Complex Paravectors

Write the real paravectors

$$\begin{aligned} X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Y &= b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \\ &= X + iY, \end{aligned}$$

where

$$S := a + ib \in \mathbb{C}, \quad \mathbf{V} := \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3.$$

Because the scalar part S commutes with $\mathbf{V} \cdot \boldsymbol{\sigma}$, the exponential separates into scalar and directional factors:

$$\begin{aligned} \exp(Z) &= \exp(S + \mathbf{V} \cdot \boldsymbol{\sigma}), \\ &= \exp(S) \exp(\mathbf{V} \cdot \boldsymbol{\sigma}). \end{aligned}$$

$$\begin{aligned} \exp(Z^*) &= \exp(S^*) \exp(-\mathbf{V}^* \cdot \boldsymbol{\sigma}), \\ \exp(Z)^{-1} &= \exp(-S) \exp(-\mathbf{V} \cdot \boldsymbol{\sigma}), \\ \exp(Z^*)^{-1} &= \exp(-S^*) \exp(\mathbf{V}^* \cdot \boldsymbol{\sigma}), \\ (\mathbf{V} \cdot \boldsymbol{\sigma})^2 &= \mathbf{V} \cdot \mathbf{V} = \mathbf{V}^2, \\ &= \mathbf{A}^2 - \mathbf{B}^2 + 2i\mathbf{A} \cdot \mathbf{B}. \end{aligned}$$

The key simplification is that all higher powers collapse onto the two basis elements 1 and $\mathbf{V} \cdot \boldsymbol{\sigma}$:

$$\begin{aligned} (\mathbf{V} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{V}^2)^n, \\ z &= \sqrt{\mathbf{V}^2} \quad \text{complex scalar,} \\ \exp(\mathbf{V} \cdot \boldsymbol{\sigma}) &= \sum_{n=0}^{\infty} \frac{(\mathbf{V} \cdot \boldsymbol{\sigma})^n}{n!}, \\ &= \sum_{n=0}^{\infty} \frac{(\mathbf{V}^2)^n}{(2n)!} + \sum_{n=0}^{\infty} \frac{(\mathbf{V}^2)^n}{(2n+1)!} (\mathbf{V} \cdot \boldsymbol{\sigma}), \\ &= \cosh(z) + \sinh(z) \frac{\mathbf{V} \cdot \boldsymbol{\sigma}}{z}. \end{aligned}$$

The final expression is independent of the sign chosen for z . When $\mathbf{V}^2 = 0$, the quotient $\sinh(z)/z$ is understood by its continuous value 1, and the formula becomes

$$\exp(Z) = \exp(S)(1 + \mathbf{V} \cdot \boldsymbol{\sigma}).$$

1.11 Generalized Transform Algebra

With the projector and exponential formulas fixed, the remaining purely mathematical transform material can be collected in one place. The order is generic first and special-case second: the similarity law comes first, then the basis-transport identities, then the circular and hyperbolic transforms built from the earlier exponential elements:

$$TZ' = ZT, \quad (58)$$

$$Z' = T^{-1}ZT. \quad (59)$$

Here $Z \in \mathbb{C} \oplus \mathbb{C}^3$ is a generic complex paravector and T is a generic invertible algebra element. Equation (58) is the intertwining requirement that defines what it means for the primed and unprimed algebra actions to describe the same calculation. Its unique solution is the inner automorphism (59), useful for basis changes, rotations, and bivector fields. The derivation below also separates this similarity from the Hermitian congruence required for a real event paravector.

1.11.1 Intertwining and the General Similarity Transform

The starting relation (58) expresses compatibility between two descriptions of the same algebraic action. Let T carry a primed carrier into the unprimed description. If an algebra element acts as Z' before that identification and as Z afterward, then applying the action and changing description must commute:

$$T(Z'\psi') = Z(T\psi') \quad \text{for every carrier } \psi'.$$

Removing the arbitrary carrier gives $TZ' = ZT$. Because T is invertible, its unique solution is

$$Z' = T^{-1}ZT.$$

Thus Eq. (59) is not an additional dynamical law: it is the change-of-description law forced by the intertwining requirement. Defining T to carry descriptions in the opposite direction would replace it everywhere by T^{-1} and would give the equivalent convention $Z' = TZT^{-1}$. The inverse-first convention is retained here because the positive exponent in Eq. (57) then gives the displayed positive sign in Rodrigues's formula.

Now take the fully general complex paravectors

$$\begin{aligned} Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma}, & S &\in \mathbb{C}, & \mathbf{V} &\in \mathbb{C}^3, \\ T &= f + \mathbf{g} \cdot \boldsymbol{\sigma}, & f &\in \mathbb{C}, & \mathbf{g} &\in \mathbb{C}^3, \end{aligned}$$

where the dot and cross products are extended complex-bilinearly. Define the central determinant of T by

$$\Delta_T := TT^{*\mathbf{R}} = f^2 - \mathbf{g}^2.$$

When $\Delta_T \neq 0$, the general inverse formula (47) gives

$$T^{-1} = \frac{T^{*\mathbf{R}}}{\Delta_T} = \frac{f - \mathbf{g} \cdot \boldsymbol{\sigma}}{f^2 - \mathbf{g}^2}.$$

Because the scalar part S is central,

$$Z' = S + \frac{(f - \mathbf{g} \cdot \boldsymbol{\sigma})(\mathbf{V} \cdot \boldsymbol{\sigma})(f + \mathbf{g} \cdot \boldsymbol{\sigma})}{\Delta_T}.$$

The first multiplication is

$$(f - \mathbf{g} \cdot \boldsymbol{\sigma})(\mathbf{V} \cdot \boldsymbol{\sigma}) = -\mathbf{g} \cdot \mathbf{V} + (f\mathbf{V} - i\mathbf{g} \times \mathbf{V}) \cdot \boldsymbol{\sigma}.$$

After multiplication by the final factor, the scalar terms cancel and the remaining vector terms give

$$\begin{aligned} Z' &= S + \mathbf{V}' \cdot \boldsymbol{\sigma}, \\ \mathbf{V}' &= \frac{(f^2 + \mathbf{g}^2)\mathbf{V} - 2(\mathbf{g} \cdot \mathbf{V})\mathbf{g} - 2if(\mathbf{g} \times \mathbf{V})}{f^2 - \mathbf{g}^2}. \end{aligned} \quad (60)$$

This is the similarity law for every complex paravector Z and every invertible complex paravector T . It shows directly that the complex scalar part is fixed. Multiplying T by any nonzero central complex scalar changes neither Eq. (59) nor Eq. (60).

The same intertwining controls products, inverses, and functions. For a second algebra element W ,

$$\begin{aligned} (ZW)' &= T^{-1}ZWT = T^{-1}ZT T^{-1}WT = Z'W', \\ (Z')^{-1} &= T^{-1}Z^{-1}T \quad \text{when } Z^{-1} \text{ exists,} \\ (Z')^n &= T^{-1}Z^nT, \\ h(Z') &= T^{-1}h(Z)T. \end{aligned} \quad (61)$$

The last line holds for every convergent power-series function h , and therefore in particular

$$\exp(Z') = T^{-1} \exp(Z) T.$$

Logarithms, roots, and rational functions obey the same relation wherever the same branch and domain are defined.

The reverse conjugation also commutes with this similarity. Since $T^{*\mathbf{R}} = \Delta_T T^{-1}$ and $(T^{-1})^{*\mathbf{R}} = T/\Delta_T$,

$$\begin{aligned} Z'^{*\mathbf{R}} &= T^{*\mathbf{R}} Z^{*\mathbf{R}} (T^{-1})^{*\mathbf{R}} = T^{-1} Z^{*\mathbf{R}} T, \\ Z' Z'^{*\mathbf{R}} &= T^{-1} (ZZ^{*\mathbf{R}}) T = ZZ^{*\mathbf{R}}. \end{aligned}$$

Thus similarity preserves the determinant form $S^2 - \mathbf{V}^2$, the spectrum, and the inverse framework for every invertible T . The positive coefficient norm is a different construction and is not generally invariant under a nonunitary similarity.

The similarity law transports algebra elements that act by multiplication; it is not automatically the Lorentz law for every physical representation. A real event paravector satisfies $X^{\mathbf{R}} = X$. Such paravectors are spanned by Hermitian carrier bilinears $\psi\psi^{\mathbf{R}}$: concretely, in the Pauli realization every Hermitian 2×2 matrix is a real linear combination of bilinears of this form. The same carrier identification used above is $\psi' = T^{-1}\psi$, and therefore

$$\psi'\psi'^{\mathbf{R}} = T^{-1}(\psi\psi^{\mathbf{R}})(T^{-1})^{\mathbf{R}}.$$

By real linearity, the event law is consequently forced to be the inverse-carried Hermitian congruence. For a normalized Lorentz element $TT^{*\mathbf{R}} = 1$, it is

$$\begin{aligned} X' &= T^{-1}X(T^{-1})^{\mathbf{R}} \\ &= T^{*\mathbf{R}}XT^*. \end{aligned} \quad (62)$$

Here the involutions have distinct roles. In the Pauli realization, $^{\mathbf{R}}$ is Hermitian adjunction, whereas $^{*\mathbf{R}}$ is the adjugate. The latter equals T^{-1} only after the determinant-one normalization $TT^{*\mathbf{R}} = 1$ has been imposed. It preserves the real subspace because $(X')^{\mathbf{R}} = X'$, and it preserves the event determinant because T^{-1} has unit determinant. When X is invertible, the transformed inverse follows without a new definition:

$$(X')^{-1} = T^{\mathbf{R}}X^{-1}T, \quad (63)$$

because direct multiplication with $X' = T^{-1}X(T^{-1})^{\mathbf{R}}$ gives the identity. This complementary law is also the algebraic pattern

used later by the dual derivative paravector. Multiplying Eq. (62) on the left by T gives the event intertwiner

$$TX' = XT^*. \quad (64)$$

This becomes $TX' = XT$ for a circular rotor because $R^* = R$, but not for a boost. Deriving Rodrigues's formula first can therefore make the similarity intertwiner look universal even though the boost event belongs to a different representation.

In contrast, a bivector such as the electromagnetic field acts by left multiplication on a carrier. Applying the original intertwining argument to that operator forces the similarity

$$F' = T^{-1}FT. \quad (65)$$

For a circular rotor, $R^* = R$ and $R^{-1} = R^R$, so Eq. (62) reduces to $X' = R^R X R$. For a hyperbolic element, $L^R = L$ and $L^{-1} = L^* = L^{*R}$, so the same equation reduces to $X' = L^* X L^*$, whereas Eq. (65) reduces to $F' = L^* F L$. The unequal right factors therefore follow from the representations being transported, not from an inconsistent choice of inverse. With the present definition of $L(\theta, \mathbf{u})$, the sandwich $L^R X L = L X L$ is the inverse-direction event boost, obtained by replacing θ with $-\theta$. It is physically legitimate, but it is not the passive $+\theta$ convention used in Eq. (54).

The hyperbolic reduction of the general complex formula can also be read off directly. With

$$f = \cosh(\theta/2), \quad \mathbf{g} = \sinh(\theta/2)\mathbf{u}, \quad \mathbf{u}^2 = 1,$$

Eq. (60) becomes

$$\mathbf{V}' = (\mathbf{u} \cdot \mathbf{V})\mathbf{u} + \cosh(\theta)(\mathbf{V} - (\mathbf{u} \cdot \mathbf{V})\mathbf{u}) - i \sinh(\theta) \mathbf{u} \times \mathbf{V}. \quad (66)$$

For $\mathbf{V} = \mathbf{E} + i\mathbf{B}$, its real and imaginary parts are exactly the electric-magnetic boost formulas derived later. For a real $\mathbf{V}$, however, the cross term is generally imaginary; this again shows why hyperbolic similarity is a bivector or basis transformation and not an event boost.

The infinitesimal forms make the distinction especially transparent. If $T = 1 + K + O(K^2)$, then

$$\begin{aligned} Z' &= Z + \text{com}(Z, K) + O(K^2), \\ X' &= X - KX - XK^R + O(K^2). \end{aligned}$$

For a circular generator $K^R = -K$, the event law reduces to the same commutator that generates the similarity rotation. For a hyperbolic generator $K^R = K$, it instead contains an anticommutator: $X' = X - \text{acom}(K, X) + O(K^2)$. The former rotates spatial vectors; the latter mixes the scalar time component with the vector component parallel to the boost. This is why the rotation can first be derived as a similarity while the event boost cannot.

For a one-parameter family $\lambda \mapsto T(\lambda)$, introduce

$$\begin{aligned} \Omega &:= T^{-1}\dot{T}, \quad \dot{T}^{-1} = -T^{-1}\dot{T}T^{-1}, \\ \dot{Z}' &= T^{-1}\dot{Z}T + \text{com}(Z', \Omega), \\ \ddot{Z}' &= T^{-1}\ddot{Z}T, \\ &\quad + 2\text{com}(\dot{Z}', \Omega), \\ &\quad + \text{com}(Z', \dot{\Omega}), \\ &\quad + \text{com}(\Omega, \text{com}(Z', \Omega)). \end{aligned}$$

Dots here mean ordinary derivatives with respect to the real parameter λ . This avoids confusing the transport calculation with exterior differentiation, for which $d^2 = 0$.

1.11.2 Transformed Basis

Let the parameter-dependent element be

$$\begin{aligned} T(\lambda) &= f(\lambda) + \mathbf{g}(\lambda) \cdot \boldsymbol{\sigma}, \quad f \in \mathbb{C}, \quad \mathbf{g} \in \mathbb{C}^3, \\ T^{-1} &= \frac{f - \mathbf{g} \cdot \boldsymbol{\sigma}}{f^2 - \mathbf{g}^2}, \quad f^2 - \mathbf{g}^2 \neq 0, \\ \sigma'_n &:= T^{-1}\sigma_n T, \quad \sigma_n = T\sigma'_n T^{-1}. \end{aligned}$$

Allowing complex $\mathbf{g}$ includes both real hyperbolic elements and imaginary-vector rotors. Similarity immediately gives

$$\begin{aligned} (\sigma'_n)^2 &= 1, \\ \sigma'_n \sigma'_m &= \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i \sigma'_k, \\ \sigma'_1 \sigma'_2 \sigma'_3 &= i. \end{aligned}$$

Thus the transformed generators obey the same Clifford algebra for every invertible T .

For every invertible T , reverse conjugation commutes with the similarity. Using $T^{*R} = (f^2 - \mathbf{g}^2)T^{-1}$ and $(T^{-1})^{*R} = T/(f^2 - \mathbf{g}^2)$,

$$\begin{aligned} (\sigma'_n)^{*R} &= (f^2 - \mathbf{g}^2)T^{-1}(-\sigma_n) \frac{T}{f^2 - \mathbf{g}^2} \\ &= -T^{-1}\sigma_n T = -\sigma'_n. \end{aligned}$$

Consequently a paravector written in the transformed basis,

$$Z = Z_s + Z_v \cdot \boldsymbol{\sigma}',$$

has the same determinant quadratic form,

$$ZZ^{*R} = Z_s^2 - Z_v^2.$$

For differentiation with respect to λ , let

$$\Omega := T^{-1}\dot{T}.$$

Then

$$\dot{\sigma}'_n = \text{com}(\sigma'_n, \Omega), \quad \dot{\boldsymbol{\sigma}}' = \text{com}(\boldsymbol{\sigma}', \Omega).$$

The coefficient form follows directly from

$$\Omega = \frac{f\dot{f} - \mathbf{g} \cdot \dot{\mathbf{g}}}{f^2 - \mathbf{g}^2} + \frac{f\dot{\mathbf{g}} - \dot{f}\mathbf{g} - i\mathbf{g} \times \dot{\mathbf{g}}}{f^2 - \mathbf{g}^2} \cdot \boldsymbol{\sigma}.$$

Relative to the primed basis, write its vector coefficient uniquely as

$$\Omega = \Omega_s + \frac{1}{2}(\mathbf{k} + i\boldsymbol{\omega}) \cdot \boldsymbol{\sigma}', \quad \mathbf{k}, \boldsymbol{\omega} \in \mathbb{R}^3.$$

The scalar part commutes with the basis, while

$$\begin{aligned} \dot{\boldsymbol{\sigma}}' &= (-\boldsymbol{\omega} + i\mathbf{k}) \times \boldsymbol{\sigma}', \\ \dot{Z} &= \dot{Z}_s + (\dot{Z}_v - Z_v \times \boldsymbol{\omega} + iZ_v \times \mathbf{k}) \cdot \boldsymbol{\sigma}'. \end{aligned}$$

These are algebraic moving-basis identities. No space-time, energy, or proper-time interpretation follows from similarity alone; such an interpretation requires a specified Lorentz congruence and frame geometry.

1.11.3 Rotation

The identities in this subsection unpack Eq. (57) as the rotation case of the general transform law, with the transform element specialized to $T = R$. Here the generator has sigma number two, so the exponential is circular rather than hyperbolic:

$$\begin{aligned} R(\theta, \mathbf{u}) &:= \exp(\tfrac{1}{2}\theta(\mathbf{u} \cdot i\boldsymbol{\sigma})), \\ &= \cos(\tfrac{1}{2}\theta) + \sin(\tfrac{1}{2}\theta)(\mathbf{u} \cdot i\boldsymbol{\sigma}) \quad \text{rotor.} \end{aligned}$$

$$R^{-1} = \exp(-\tfrac{1}{2}\theta(\mathbf{u} \cdot i\boldsymbol{\sigma})).$$

Here $\theta \in \mathbb{R}$, $\mathbf{u} \in \mathbb{R}^3$ is a unit real vector, and R is generated by the sigma-number-two element $\mathbf{u} \cdot i\boldsymbol{\sigma}$. Its inverse satisfies $R^{-1} = R^R = R^{*R}$. To derive the sign rather than choose it afterward, substitute

$$f = \cos(\theta/2), \quad \mathbf{g} = i\sin(\theta/2)\mathbf{u}$$

into Eq. (60). Then

$$\begin{aligned} f^2 - \mathbf{g}^2 &= 1, \\ f^2 + \mathbf{g}^2 &= \cos(\theta), \\ -2(\mathbf{g} \cdot \mathbf{r})\mathbf{g} &= (1 - \cos(\theta))(\mathbf{u} \cdot \mathbf{r})\mathbf{u}, \\ -2if(\mathbf{g} \times \mathbf{r}) &= \sin(\theta)(\mathbf{u} \times \mathbf{r}). \end{aligned}$$

Therefore the transformed paravector is

$$\begin{aligned} X' &= R^R X R, \\ &= t' + \mathbf{r}' \cdot \boldsymbol{\sigma}. \end{aligned}$$

$$\begin{aligned} t' &= t, \\ \mathbf{r}' &= \mathbf{r} \cos(\theta) + (\mathbf{u} \times \mathbf{r}) \sin(\theta), \\ &\quad + (\mathbf{u} \cdot \mathbf{r})(1 - \cos(\theta))\mathbf{u}, \end{aligned}$$

Rodrigues's formula.

For a parameter-dependent rotor, define the right- and left-carried generators

$$\Omega := \dot{R}R^R, \quad \Omega' := R^R \dot{R} = R^R \Omega R.$$

Differentiating $R^R R = 1$ gives

$$\Omega^R = -\Omega, \quad (\Omega')^R = -\Omega'.$$

The rotation transport law is the specialization of the general one-parameter similarity formula:

$$\begin{aligned} \dot{X}' &= R^R \dot{X} R + \text{com}(X', \Omega') \\ &= R^R (\dot{X} + \text{com}(X, \Omega)) R. \end{aligned}$$

1.11.4 Spin Rotors

The pseudoscalar identifies each oriented complementary bivector without an index-order ambiguity:

$$i\sigma_1 = \sigma_2\sigma_3, \quad i\sigma_2 = \sigma_3\sigma_1, \quad i\sigma_3 = \sigma_1\sigma_2.$$

For unit $\mathbf{n}, \mathbf{u} \in \mathbb{R}^3$, let

$$\begin{aligned} R(\phi, \mathbf{n}) &= \exp\left(\frac{\phi}{2} \mathbf{n} \cdot i\boldsymbol{\sigma}\right) \\ &= \cos(\phi/2) + \sin(\phi/2) \mathbf{n} \cdot i\boldsymbol{\sigma}. \end{aligned}$$

Then

$$R^{-1} = R^R = R^{*R}, \quad R^R R = 1,$$

and

$$R^R(\mathbf{u} \cdot \boldsymbol{\sigma}) R = \mathbf{u}' \cdot \boldsymbol{\sigma}$$

with $\mathbf{u}'$ given by Rodrigues's formula in the preceding subsection. The rotor itself is single-valued and 4π -periodic:

$$R(\phi + 2\pi, \mathbf{n}) = -R(\phi, \mathbf{n}), \quad R(\phi + 4\pi, \mathbf{n}) = R(\phi, \mathbf{n}).$$

The induced spatial rotation is two-to-one because R and $-R$ give the same conjugation.

In the Pauli matrix realization, reversion is Hermitian adjunction. A general $\psi \in \mathbb{C} \oplus \mathbb{C}^3$ is therefore a full 2×2 matrix, not one irreducible two-component spinor: left multiplication rotates its two columns independently. It may instead be used as a density amplitude, or restricted to a minimal left ideal, implemented by a fixed projector on the right, to represent one Pauli spinor. With the active convention consistent with the vector action above,

$$\psi' = R^R \psi, \quad \psi' \psi'^R = R^R (\psi \psi^R) R.$$

For a nonzero state, a normalized expectation numerator is converted into an expectation value by

$$\langle M \rangle_\psi := \frac{\text{scalar}(\psi^R M \psi)}{\text{scalar}(\psi^R \psi)}.$$

These qualifications separate the algebra action from the choice of an irreducible spin state.

1.12 Quaternion Correspondence

Now that the circular exponential and rotor law have both been developed, the same sigma-number-two algebra can be restated in Hamilton's quaternion language without introducing a new multiplication rule. The point of this subsection is to collect that correspondence in one place after the rotation discussion has been completed.

1.12.1 Quaternion Basis and Involutions

The same sigma-number-two basis used in Eq. (56) also reproduces Hamilton's quaternion units, so the rotation algebra can be written in quaternion form without leaving the STA multiplication table [1, 3]:

$$\begin{aligned} q_1 &= -\sigma_2\sigma_3, & q_2 &= -\sigma_3\sigma_1, \\ q_3 &= -\sigma_1\sigma_2, & \mathbf{q} &:= (q_1, q_2, q_3) = -i\sigma. \end{aligned} \quad (67)$$

These basis elements satisfy the standard quaternion identities

$$q_1^2 = q_2^2 = q_3^2 = q_1q_2q_3 = -1.$$

So the quaternion units are not new generators; they are the sigma-number-two basis already present in the STA algebra, only relabeled.

If $\mathbf{B} = (B_1, B_2, B_3) \in \mathbb{R}^3$, then $\mathbf{B} \cdot \mathbf{q} = -i\mathbf{B} \cdot \sigma$. Since $q_k = -i\sigma_k$, the abstract operator rules immediately give

$$\mathbf{q}^* = \mathbf{q}, \quad \mathbf{q}^R = -\mathbf{q}.$$

Thus the conjugate operator fixes the quaternion basis, while the reverse operator changes its sign.

1.12.2 Quaternion Products

For $a \in \mathbb{R}$ and $\mathbf{B} \in \mathbb{R}^3$, define the quaternion

$$Q := a + \mathbf{B} \cdot \mathbf{q} = a - i\mathbf{B} \cdot \sigma. \quad (68)$$

In the manuscript's complex-paravector notation, the same quaternion may be written as

$$Q = X - iY, \quad X = a + \mathbf{A} \cdot \sigma, \quad Y = b + \mathbf{B} \cdot \sigma,$$

with the quaternion specialization

$$\mathbf{A} = 0, \quad b = 0.$$

So here $X = a$ is purely scalar and $Y = \mathbf{B} \cdot \sigma$ is purely vector. Because $\mathbf{q}^* = \mathbf{q}$, one has

$$Q^* = Q.$$

Therefore

$$\begin{aligned} QQ^* &= Q^2 \\ &= (a - i\mathbf{B} \cdot \sigma)^2 \\ &= a^2 - 2ia\mathbf{B} \cdot \sigma + (-i)^2(\mathbf{B} \cdot \sigma)^2 \\ &= a^2 - \mathbf{B}^2 - 2ia\mathbf{B} \cdot \sigma. \end{aligned} \quad (69)$$

The linear term must carry the factor a . This is just the sigma-number-two multiplication law written in quaternion language.

Since $\mathbf{q}^R = -\mathbf{q}$, the reverse of Q is

$$Q^R = a - \mathbf{B} \cdot \mathbf{q} = a + i\mathbf{B} \cdot \sigma.$$

Multiplying Q by its reverse gives

$$QQ^R = a^2 + \mathbf{B}^2. \quad (70)$$

So the reverse product recovers the standard positive quaternion norm square, while Eq. (69) records the ordinary algebra square.

1.12.3 Quaternion Form of the Rotor Law

Now that Eq. (57) and its transform consequences have been developed, the quaternion rotation law can be read off directly. For a unit quaternion Q satisfying $QQ^R = 1$, and for a real vector $\mathbf{r} \in \mathbb{R}^3$, one has

$$\mathbf{r}' \cdot \mathbf{q} = Q(\mathbf{r} \cdot \mathbf{q})Q^R. \quad (71)$$

This is the quaternion version of the rotor action $\mathbf{r}' \cdot \sigma = R^R(\mathbf{r} \cdot \sigma)R$. In particular, choosing

$$Q = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)\mathbf{u} \cdot \mathbf{q}$$

gives $Q = R^R$, because $\mathbf{q} = -i\sigma$. Consequently

$$Q(\mathbf{r} \cdot \mathbf{q})Q^R = R^R(\mathbf{r} \cdot \mathbf{q})R,$$

which has exactly the same orientation as the sigma-vector rotor law. If one instead chooses $Q = R$, the equivalent quaternion sandwich must be $Q^R(\mathbf{r} \cdot \mathbf{q})Q$. Thus quaternion rotations are the same circular half-angle algebra, with the order fixed by the selected orientation convention.

1.13 Matrix and Auxiliary Realizations

The algebra above is complete on its own, but two realizations are useful. The Pauli representation identifies one paravector with a 2×2 complex matrix and identifies ZZ^{*R} with its determinant [9]. The auxiliary map $\mathbf{W}$ is instead a 2×2 block matrix whose entries are paravectors; after the Pauli realization it is a 4×4 complex matrix. It is a convenient packaging map, not a second 2×2 representation or an algebra homomorphism.

1.13.1 Standard 2×2 Matrix Realization

The real associative algebra generated by the three sigma elements is $Cl_{3,0}(\mathbb{R})$. It is isomorphic both to the even space-time algebra $Cl_{1,3}^+(\mathbb{R})$ and to 2×2 complex matrices. The internal pseudoscalar makes the same algebra a four-dimensional complex vector space, written in this manuscript as $\mathbb{C} \oplus \mathbb{C}^3$; this is not the eight-complex-dimensional algebra $Cl_3(\mathbb{C})$. The following faithful matrix model makes these identifications concrete [9].

The standard realization is

$$\sigma_1 = \{0, 1; 1, 0\}, \quad \sigma_2 = \{0, -i; i, 0\}, \quad \sigma_3 = \{1, 0; 0, -1\}, \quad (72)$$

Here the i appearing in the entries of σ_2 is strictly the ordinary complex unit $i := \sqrt{-1}$. The matrix entries are not defined recursively from the STA pseudoscalar; rather, the abstract

pseudoscalar is represented afterward by the matrix product $\sigma_1\sigma_2\sigma_3 = i\mathbf{I}$. In this realization the basis identities become

$$\sigma_n^2 = \mathbf{I}, \quad \sigma_m\sigma_n = -\sigma_n\sigma_m \quad (m \neq n), \quad \sigma_1\sigma_2\sigma_3 = i\mathbf{I}.$$

So Eq. (72) gives a concrete 2×2 model for the generators and their pseudoscalar.

For a real paravector $X = t + \mathbf{r} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$, with $\mathbf{r} = (r_1, r_2, r_3) \in \mathbb{R}^3$, the matrix form is

$$X = t + \mathbf{r} \cdot \boldsymbol{\sigma} = \{t + r_3, r_1 - ir_2; r_1 + ir_2, t - r_3\},$$

$$\det(X) = t^2 - \mathbf{r}^2. \quad (73)$$

Equation (73) reproduces exactly the real Minkowski quadratic form already encoded by Eq. (43). The same matrix also carries the familiar trace and characteristic data:

$$\begin{aligned} \text{tr}(X) &= 2t, \\ \lambda^2 - \text{tr}(X)\lambda + \det(X) &= 0, \\ \lambda(\pm) &= t \pm \|\mathbf{r}\|. \end{aligned}$$

These eigenvalues are the matrix version of the split into scalar part t and vector magnitude $\|\mathbf{r}\|$.

More generally, $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ represents an arbitrary 2×2 complex matrix, and

$$\begin{aligned} \text{scalar}(Z) &= \frac{1}{2} \text{tr}(Z), & \det(Z) &= S^2 - \mathbf{V}^2, \\ Z^{*\text{R}} &= \text{adj}(Z). \end{aligned}$$

Consequently, the manuscript's determinant quadratic form is the ordinary matrix determinant in this realization:

$$ZZ^{*\text{R}} = \det(Z)\mathbf{I}.$$

Within the same realization, the matrix involutions can also be written explicitly. Let $(\cdot)^c$ denote entrywise complex conjugation and let $(\cdot)^{cT}$ denote complex conjugate transpose. Then

$$\begin{aligned} i^c &= -i, & \sigma_1^c &= \sigma_1, & \sigma_2^c &= -\sigma_2, & \sigma_3^c &= \sigma_3, \\ (\sigma_n)^{cT} &= \sigma_n, \\ (\sigma_n\sigma_m)^{cT} &= \sigma_m\sigma_n = -\sigma_n\sigma_m \quad (n \neq m), \\ (i\mathbf{I})^{cT} &= -i\mathbf{I}, \\ (i\sigma_n)^{cT} &= ((\sigma_1\sigma_2\sigma_3)\sigma_n)^{cT} \\ &= \sigma_n^{cT}(\sigma_1\sigma_2\sigma_3)^{cT} \\ &= -i\sigma_n, \end{aligned}$$

and therefore

$$\begin{aligned} \mathbf{I}^{\text{R}} &= \mathbf{I}, & \sigma_n^{\text{R}} &= \sigma_n, \\ (i\mathbf{I})^{\text{R}} &= -i\mathbf{I}, & (i\sigma_n)^{\text{R}} &= -i\sigma_n, \\ (\cdot)^{\text{R}} &= (\cdot)^{cT}. \end{aligned}$$

So in the matrix model the reverse operator coincides with the Hermitian adjoint. The sigma conjugation formulas are generated by the fixed matrix

$$\mathbf{M} := \{0, 1; -1, 0\}, \quad \mathbf{M}^{-1} = -\mathbf{M}.$$

With this matrix one finds

$$\begin{aligned} \mathbf{I}^* &= \mathbf{I}, & \sigma_n^* &= -\sigma_n, \\ (i\mathbf{I})^* &= -i\mathbf{I}, & (i\sigma_n)^* &= i\sigma_n, \end{aligned}$$

$$\begin{aligned} (\cdot)^* &= \mathbf{M}(\cdot)^c\mathbf{M}^{-1}, \\ (\cdot)^{*R} &= (\cdot)^{\text{R}*} = \mathbf{M}(\cdot)^T\mathbf{M}^{-1}. \end{aligned}$$

Thus the matrix section now contains, in one place, the basis matrices, the spacetime paravector matrix, the determinant/trace/eigenvalue formulas, complex conjugation, complex-conjugate transpose, reverse, sigma conjugation, and reverse sigma conjugation. These concrete formulas match the abstract operator rules defined earlier in the algebraic development.

The operator types are worth making explicit. Both $(\cdot)^*$ and $(\cdot)^{\text{R}}$ are real-linear but complex-antilinear because each sends the central pseudoscalar i to $-i$. The first is an automorphism, whereas the second is an anti-automorphism. Their composition $(\cdot)^{*R}$ is a complex-linear anti-automorphism. In the matrix model, reversion is Hermitian adjunction and reverse conjugation is the adjugate map displayed above.

1.13.2 Auxiliary Block Map

It is also useful to package a scalar-coefficient complex paravector $Z \in \mathbb{C} \oplus \mathbb{C}^3$ into the 2×2 block map

$$\mathbf{W}(Z) := \{0, Z; Z^{*\text{R}}, 0\}. \quad (74)$$

$$\mathbf{W}(Z)^2 = \text{diag}(ZZ^{*\text{R}}, Z^{*\text{R}}Z) = \langle\langle Z \rangle\rangle^2 \mathbf{I}. \quad (75)$$

Here $\mathbf{W}(Z)$ is a 2×2 matrix over the paravector algebra. After each block is replaced by its Pauli matrix, it is a 4×4 complex matrix. It is a useful linear packaging map, not an algebra homomorphism. Equation (75) follows because its diagonal block products reduce to the determinant quadratic form in Eq. (42); the off-diagonal blocks vanish [12, 14].

If $\langle\langle Z \rangle\rangle^2 \neq 0$, the annihilating polynomial $x^2 - \langle\langle Z \rangle\rangle^2$ has distinct roots, so the block matrix is diagonalizable after a square-root branch is chosen. If $\langle\langle Z \rangle\rangle^2 = 0$ and $Z \neq 0$, then $\mathbf{W}(Z)^2 = 0$ while $\mathbf{W}(Z) \neq 0$: it is nilpotent and not diagonalizable. For differential or other noncommuting operator-valued blocks, only the first ordered diagonal identity in Eq. (75) is automatic; reduction to one scalar multiple of the identity requires a separate operator calculation.

2 Space-Time Kinematics

On real event paravectors, the determinant form becomes the Minkowski quadratic form and normalized timelike paravectors become four-velocities. This is standard special-relativistic kinematics in paravector form [4–6, 12, 13].

2.1 Differential space-time and four-velocity

Specializing the real paravector X from Eq. (11) to differentials gives

$$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma} \quad (76)$$

and the interval is the differential version of Eq. (42):

$$ds^2 = \langle\langle dX \rangle\rangle^2 = \langle\langle dt + \boldsymbol{\sigma} \cdot d\mathbf{r} \rangle\rangle^2 = dX dX^* = dt^2 - d\mathbf{r}^2. \quad (77)$$

Here $t, s, \gamma \in \mathbb{R}$, $\mathbf{r}, \mathbf{v}, \mathbf{P} \in \mathbb{R}^3$, and $X, U \in \mathbb{R} \oplus \mathbb{R}^3$. Equation (77) is the real-event restriction of the determinant form in Eq. (42). For a future-directed timelike worldline, $dt > 0$ and $ds = \sqrt{dt^2 - d\mathbf{r}^2} > 0$.

If $\mathbf{v} = d\mathbf{r}/dt \in \mathbb{R}^3$ and $\gamma = (1 - \mathbf{v}^2)^{-1/2} \in \mathbb{R}$, then

$$U := \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad \langle\langle U \rangle\rangle^2 = 1. \quad (78)$$

Multiplying Eq. (78) by the real rest mass $m > 0$ gives the four-momentum

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}, \quad E = m\gamma, \quad \mathbf{P} = m\gamma\mathbf{v}, \quad (79)$$

which immediately yields $E^2 - \mathbf{P}^2 = m^2$. The normalized tangent U is conventionally called the four-velocity; “proper velocity” often denotes only its spatial part $\gamma\mathbf{v}$. Multiplication by $m > 0$ gives the four-momentum of a massive particle. For a massless particle $ds = 0$, so U is undefined and a null momentum paravector must instead be introduced directly.

2.2 Four-acceleration and free motion

The four-acceleration is the real paravector

$$A := \frac{dU}{ds} = \frac{d^2 X}{ds^2}, \quad \text{scalar}(U^* A) = 0. \quad (80)$$

The orthogonality condition in Eq. (80) is the standard four-acceleration orthogonality relation. The invariant proper acceleration is the magnitude $\sqrt{-AA^*}$, not the paravector A itself.

A free-particle Lagrangian is

$$\mathcal{L}_{\text{free}} = -\frac{m}{\gamma}, \quad (81)$$

whose Euler–Lagrange equations reproduce inertial motion and the usual nonrelativistic expansion

$$\mathcal{L}_{\text{free}} = -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4). \quad (82)$$

Equation (82) is worth displaying explicitly because it shows how the relativistic action reduces to the classical kinetic expression plus an additive rest-energy constant when $\|\mathbf{v}\| \ll 1$.

2.3 Supporting Kinematic Derivations

The supporting derivations expand Eqs. (77), (78), and (79), then record the generalized-coordinate and metric remarks associated with the geometric part of the development.

2.3.1 Space-Time Numbers

A spacetime increment is the real paravector

$$dX = dt + dr_1\sigma_1 + dr_2\sigma_2 + dr_3\sigma_3 = dt + d\mathbf{r} \cdot \boldsymbol{\sigma},$$

with conjugate

$$dX^* = dt - dr_1\sigma_1 - dr_2\sigma_2 - dr_3\sigma_3 = dt - d\mathbf{r} \cdot \boldsymbol{\sigma}.$$

Here $dt \in \mathbb{R}$ is the scalar part and $d\mathbf{r} \in \mathbb{R}^3$ is the spatial vector part. Their recovery formulas are

$$\frac{dX + dX^*}{2} = dt, \quad \frac{dX - dX^*}{2} = d\mathbf{r} \cdot \boldsymbol{\sigma},$$

so

$$\text{scalar}(dX) = dt, \quad \text{vector}(dX) = d\mathbf{r}.$$

The paravector metric is

$$\langle\langle dX \rangle\rangle^2 = dX dX^* = dX^* dX = dt^2 - d\mathbf{r}^2 = ds^2.$$

It obeys the familiar Lorentzian identities

$$\begin{aligned} dt^2 > d\mathbf{r}^2, \quad dX \neq 0 &\Rightarrow \langle\langle dX \rangle\rangle^2 > 0, \\ \langle\langle dX + dX' \rangle\rangle^2 - \langle\langle dX \rangle\rangle^2 &- \langle\langle dX' \rangle\rangle^2 = 2(dt dt' - d\mathbf{r} \cdot d\mathbf{r}'), \\ \langle\langle a dX \rangle\rangle^2 &= a^2 \langle\langle dX \rangle\rangle^2, \quad a \in \mathbb{R}. \end{aligned}$$

The commutator of two spacetime increments is purely spatial:

$$\begin{aligned} \text{com}(dX, dX') &= \text{com}(d\mathbf{r} \cdot \boldsymbol{\sigma}, d\mathbf{r}' \cdot \boldsymbol{\sigma}) \\ &= 2i(d\mathbf{r} \times d\mathbf{r}') \cdot \boldsymbol{\sigma}. \end{aligned}$$

Along a future-directed time-like path, so that $dt > 0$, the positive arc length satisfies

$$\begin{aligned} ds &= \sqrt{dt^2 - d\mathbf{r}^2} \\ &= \sqrt{1 - d\mathbf{r}^2/dt^2} dt \\ &= \sqrt{1 - \mathbf{v}^2} dt, \end{aligned}$$

where

$$\mathbf{v} := \frac{d\mathbf{r}}{dt} = (v_1, v_2, v_3) \in \mathbb{R}^3, \quad \gamma := \frac{dt}{ds} = \frac{1}{\sqrt{1 - \mathbf{v}^2}}.$$

For a real non-null increment, $dX^R = dX$, so the inverse exists and

$$\begin{aligned} (dX)^{-1} &= \frac{dX^*}{\langle\langle dX \rangle\rangle^2} \\ &= \frac{dX^*}{dX dX^*} \\ &= \frac{dX^*}{ds^2}. \end{aligned}$$

Null displacements are singled out algebraically because this inverse breaks down exactly when $ds^2 = 0$.

2.3.2 Interval Classes and Proper Time

For a spacetime increment $dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma}$, the interval is

$$\langle\langle dX \rangle\rangle^2 = dX dX^* = dt^2 - d\mathbf{r}^2 = ds^2.$$

This identity gives the three standard interval classes without any additional geometric machinery:

$$\begin{aligned} ds^2 > 0 &\Rightarrow \text{time-like}, \\ ds^2 = 0 &\Rightarrow \text{null}, \\ ds^2 < 0 &\Rightarrow \text{space-like}. \end{aligned}$$

If $\langle\langle dX \rangle\rangle^2 > 0$, the interval is time-like. Then the arc length s equals the proper time τ measured along the path:

$$\begin{aligned}\tau = s &= \int ds = \int d\tau = \int \sqrt{1 - \mathbf{v}^2} dt \\ &= \int \frac{1}{\gamma} dt.\end{aligned}$$

This is the time-dilation relation in natural units. In SI units the same statement is

$$ds = c d\tau.$$

If $\langle\langle dX \rangle\rangle^2 = 0$, the interval is null or light-like. Then

$$dt^2 = d\mathbf{r}^2 \quad \Rightarrow \quad dt = \pm \|d\mathbf{r}\|,$$

and $(dX)^{-1}$ does not exist.

If $\langle\langle dX \rangle\rangle^2 < 0$, the interval is space-like.

The advantage of the paravector notation is that the interval classification, the proper-time formula, and the null singularity of $(dX)^{-1}$ all come from the single algebraic identity $dX dX^* = dt^2 - d\mathbf{r}^2$.

2.3.3 Four-Velocity

The four-velocity paravector is the normalized tangent to a time-like worldline:

$$\begin{aligned}\frac{dX}{ds} &= \frac{dt}{ds} + \frac{d\mathbf{r}}{ds} \cdot \boldsymbol{\sigma} \\ &= \frac{dt + \boldsymbol{\sigma} \cdot d\mathbf{r}}{\sqrt{dt^2 - d\mathbf{r}^2}} \\ &= \frac{dt/dt + \boldsymbol{\sigma} \cdot (d\mathbf{r}/dt)}{\sqrt{(dt/dt)^2 - (d\mathbf{r}/dt)^2}} \\ &= \frac{1 + \mathbf{v} \cdot \boldsymbol{\sigma}}{\sqrt{1 - \mathbf{v}^2}} \\ &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}).\end{aligned}$$

Its scalar and vector parts are therefore

$$\text{scalar}\left(\frac{dX}{ds}\right) = \gamma, \quad \text{vector}\left(\frac{dX}{ds}\right) = \gamma \mathbf{v}.$$

Its norm is unity:

$$\begin{aligned}\left\langle\left\langle \frac{dX}{ds} \right\rangle\right\rangle^2 &= \gamma^2(1 + \mathbf{v} \cdot \boldsymbol{\sigma})(1 - \mathbf{v} \cdot \boldsymbol{\sigma}) \\ &= \gamma^2(1 - \mathbf{v}^2) \\ &= 1.\end{aligned}$$

Also,

$$\frac{dX}{dt} = \frac{dX}{ds} \frac{ds}{dt} = \frac{1}{\gamma} \frac{dX}{ds}.$$

Define

$$\begin{aligned}U &:= \frac{dX}{ds} \\ &= \gamma + \gamma \mathbf{v} \cdot \boldsymbol{\sigma} \\ &= U_s + U_{\mathbf{v}} \cdot \boldsymbol{\sigma}, \\ U_s &= \gamma, \quad U_{\mathbf{v}} = \gamma \mathbf{v}.\end{aligned}$$

Since $ds^2 = dX dX^*$ with $ds > 0$,

$$\begin{aligned}\frac{dX^*}{ds} &= (dX)^{-1} ds \\ &= \left(\frac{dX}{ds}\right)^{-1},\end{aligned}$$

so $U^{-1} = U^*$ and

$$\langle\langle U \rangle\rangle^2 = UU^* = 1.$$

Differentiating this normalization gives

$$\begin{aligned}\frac{d}{ds} \langle\langle U \rangle\rangle^2 &= \frac{dU}{ds} U^* + U \left(\frac{dU}{ds}\right)^* \\ &= \left(\frac{dU}{ds} U^*\right) + \left(\frac{dU}{ds} U^*\right)^{*R},\end{aligned}$$

hence

$$\text{scalar}\left(U^* \frac{dU}{ds}\right) = 0.$$

Finally,

$$\begin{aligned}\langle\langle dX \rangle\rangle^2 &= dX dX^* \\ &= \left(\frac{dX}{ds} ds\right) \left(\left(\frac{dX}{ds}\right)^* ds\right) \\ &= UU^* ds^2 \\ &= \langle\langle U \rangle\rangle^2 ds^2 \\ &= ds^2.\end{aligned}$$

This normalization and orthogonality structure is the starting point for the proper-acceleration formulas below. The following table resolves the four-velocity paravector into its scalar and spatial components.

Table 16. Components of dX/ds .

scalar:	γ
vector:	$\gamma \mathbf{v}$

2.3.4 Four-Acceleration

Let

$$\mathbf{a} := \frac{d\mathbf{v}}{dt} = \{a_1, a_2, a_3\} \in \mathbb{R}^3.$$

With $\gamma = (1 - \mathbf{v}^2)^{-1/2}$, one has

$$\begin{aligned}\frac{d\gamma}{dt} &= \gamma^3(\mathbf{v} \cdot \mathbf{a}), \\ \frac{d\gamma}{ds} &= \gamma^4(\mathbf{v} \cdot \mathbf{a}).\end{aligned}$$

Let $\partial_{\mathbf{v}} := \partial/\partial \mathbf{v}$ denote the gradient with respect to velocity, and let $\mathbf{I}_3$ denote the 3×3 identity. The velocity derivatives needed below are

$$\begin{aligned}\partial_{\mathbf{v}}\left(-\frac{1}{\gamma}\right) &= \gamma \mathbf{v}, \\ \partial_{\mathbf{v}}(\gamma \mathbf{v}) &= \gamma \mathbf{I}_3 + \gamma^3 \mathbf{v} \mathbf{v}^T,\end{aligned}$$

so acting on $d\mathbf{v}/dt = \mathbf{a}$ gives

$$\begin{aligned}\frac{d(\gamma\mathbf{v})}{dt} &= (\partial_{\mathbf{v}}(\gamma\mathbf{v})) \frac{d\mathbf{v}}{dt} \\ &= \gamma\mathbf{a} + \gamma^3(\mathbf{v} \cdot \mathbf{a})\mathbf{v}, \\ \frac{d(\gamma\mathbf{v})}{ds} &= \frac{d(\gamma\mathbf{v})}{dt} \frac{dt}{ds} \\ &= \gamma^2\mathbf{a} + \gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v}.\end{aligned}$$

Recall that the four-velocity paravector decomposes as

$$\begin{aligned}U &= U_s + U_{\mathbf{v}} \cdot \boldsymbol{\sigma}, \\ U_s &= \gamma, \quad U_{\mathbf{v}} = \gamma\mathbf{v}.\end{aligned}$$

Therefore the four-acceleration is

$$\begin{aligned}A &:= \frac{dU}{ds} \\ &= \frac{d\gamma}{ds} + \frac{d(\gamma\mathbf{v})}{ds} \cdot \boldsymbol{\sigma} \\ &= \gamma^4(\mathbf{v} \cdot \mathbf{a}) + \left(\gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma^2\mathbf{a} \right) \cdot \boldsymbol{\sigma}.\end{aligned}$$

Equivalently,

$$A = A_s + A_{\mathbf{v}} \cdot \boldsymbol{\sigma}, \quad A_s = \gamma^4(\mathbf{v} \cdot \mathbf{a}), \quad A_{\mathbf{v}} = \gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma^2\mathbf{a}.$$

For $\mathbf{v} \neq 0$, let $\mathbf{u} = \mathbf{v}/\|\mathbf{v}\|$ with $\mathbf{u}^2 = 1$. The axis functions in Eq. (31) then give

$$\begin{aligned}\text{par}(\mathbf{a}) &= (\mathbf{a} \cdot \mathbf{u})\mathbf{u}, \\ \text{prp}(\mathbf{a}) &= \mathbf{a} - \text{par}(\mathbf{a}).\end{aligned}$$

Then

$$\begin{aligned}\mathbf{a}^2 &= \text{par}(\mathbf{a})^2 + \text{prp}(\mathbf{a})^2, \\ (\mathbf{v} \cdot \mathbf{a})^2 &= \mathbf{v}^2 \text{par}(\mathbf{a})^2,\end{aligned}$$

so the proper-acceleration metric is

$$\begin{aligned}\langle\langle A \rangle\rangle^2 &= -\gamma^4 \left(\mathbf{a}^2 + \gamma^2(\mathbf{v} \cdot \mathbf{a})^2 \right) \\ &= -\gamma^4 \left(\text{prp}(\mathbf{a})^2 + \gamma^2 \text{par}(\mathbf{a})^2 \right).\end{aligned}$$

This is strictly space-like unless the acceleration vanishes. Its invariant magnitude $\sqrt{-\langle\langle A \rangle\rangle^2}$ is the proper acceleration measured in the instantaneous rest frame.

The products with U are

$$\begin{aligned}UA &= (U_s A_s + U_{\mathbf{v}} \cdot A_{\mathbf{v}}) \\ &\quad + (U_s A_{\mathbf{v}} + A_s U_{\mathbf{v}} + i(U_{\mathbf{v}} \times A_{\mathbf{v}})) \cdot \boldsymbol{\sigma}, \\ U^* A &= (U_s A_s - U_{\mathbf{v}} \cdot A_{\mathbf{v}}) \\ &\quad + (U_s A_{\mathbf{v}} - A_s U_{\mathbf{v}} - i(U_{\mathbf{v}} \times A_{\mathbf{v}})) \cdot \boldsymbol{\sigma}.\end{aligned}$$

Their component forms are

$$\begin{aligned}UA &= 2\gamma^5(\mathbf{v} \cdot \mathbf{a}) \\ &\quad + (\gamma^3\mathbf{a} + 2\gamma^5(\mathbf{v} \cdot \mathbf{a})\mathbf{v}) \cdot \boldsymbol{\sigma} \\ &\quad + i\gamma^3(\mathbf{v} \times \mathbf{a}) \cdot \boldsymbol{\sigma}, \\ U^* A &= \gamma^3\mathbf{a} \cdot \boldsymbol{\sigma} - i\gamma^3(\mathbf{v} \times \mathbf{a}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

In particular,

$$\text{scalar}(U^* A) = 0,$$

which is the orthogonality relation between four-velocity and four-acceleration. Table 16 records the scalar and vector parts of U , while Tables 17 and 18 compare the corresponding scalar and vector parts of UA and $U^* A$. The following two tables compare the products UA and $U^* A$, making the orthogonality structure of proper acceleration explicit.

Table 17. Components of UA .

real scalar:	$2\gamma^5(\mathbf{v} \cdot \mathbf{a})$
img scalar:	0
real vector:	$\gamma^3\mathbf{a} + 2\gamma^5(\mathbf{v} \cdot \mathbf{a})\mathbf{v}$
img vector:	$\gamma^3(\mathbf{v} \times \mathbf{a})$

Table 18. Components of $U^* A$.

real scalar:	0
img scalar:	0
real vector:	$\gamma^3\mathbf{a}$
img vector:	$-\gamma^3(\mathbf{v} \times \mathbf{a})$

2.3.5 Four-Momentum

For a massive particle with rest mass $m > 0$, multiplying four-velocity by m produces the energy–momentum paravector,

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}.$$

Its scalar part is the relativistic energy,

$$\begin{aligned}E &:= \text{scalar}(mU) \\ &= m\gamma.\end{aligned}$$

If $\mathbf{v} = 0$, then

$$E = m \quad \text{rest energy} \quad (= mc^2 \text{ in SI units}).$$

The kinetic energy is therefore $E - m$.

Its vector part is the relativistic momentum,

$$\begin{aligned}\mathbf{P} &:= \text{vector}(mU) \\ &= m\gamma\mathbf{v}.\end{aligned}$$

Because $\langle\langle U \rangle\rangle^2 = 1$, the mass shell follows immediately:

$$\begin{aligned}\langle\langle mU \rangle\rangle^2 &= m^2 \\ &= (E + \mathbf{P} \cdot \boldsymbol{\sigma})(E - \mathbf{P} \cdot \boldsymbol{\sigma}) \\ &= E^2 - \mathbf{P}^2.\end{aligned}$$

Hence

$$E^2 = \mathbf{P}^2 + m^2.$$

The mass-shell equation also has the null branch $E^2 = \mathbf{P}^2$, appropriate to massless motion. That branch must be introduced directly as a null energy–momentum paravector: Eq. $mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}$ applies only to massive time-like motion because $ds = 0$ and $U = dX/ds$ is undefined on a null worldline.

2.3.6 Flat Paravector Coordinate Map

Let

$$\begin{aligned} f &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}, \\ \mathbf{g} &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}^3, \end{aligned}$$

and define a real paravector-valued map into the fixed flat space-time algebra,

$$X = f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}.$$

This is a map of the original coordinates into flat Minkowski paravector components; it is not an assertion that an arbitrary coordinate change leaves the Cartesian metric coefficients unchanged. Its differential and conjugate are

$$dX = df + d\mathbf{g} \cdot \boldsymbol{\sigma}, \quad dX^* = df - d\mathbf{g} \cdot \boldsymbol{\sigma}.$$

Their product is

$$\begin{aligned} \langle\langle dX \rangle\rangle^2 &= dX dX^* \\ &= (df + d\mathbf{g} \cdot \boldsymbol{\sigma})(df - d\mathbf{g} \cdot \boldsymbol{\sigma}) \\ &= df^2 - d\mathbf{g}^2. \end{aligned}$$

Along a future-directed time-like path, assume $df > 0$ and $df^2 > d\mathbf{g}^2$. Then

$$\begin{aligned} ds &= \sqrt{df^2 - d\mathbf{g}^2} \\ &= \sqrt{1 - \left(\frac{d\mathbf{g}}{df}\right)^2} df. \end{aligned}$$

The corresponding four-velocity is

$$\begin{aligned} U &:= \frac{dX}{ds} \\ &= \frac{df + d\mathbf{g} \cdot \boldsymbol{\sigma}}{\sqrt{df^2 - d\mathbf{g}^2}} \\ &= \frac{1 + \left(\frac{d\mathbf{g}}{df}\right) \cdot \boldsymbol{\sigma}}{\sqrt{1 - \left(\frac{d\mathbf{g}}{df}\right)^2}} \\ &= \frac{1 + \mathbf{v} \cdot \boldsymbol{\sigma}}{\sqrt{1 - \mathbf{v}^2}} \\ &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \end{aligned}$$

where the velocity relative to the scalar component f and its Lorentz factor are

$$\begin{aligned} \mathbf{v} &:= \frac{d\mathbf{g}}{df} \\ &= \frac{d\mathbf{g}/dt}{df/dt} \\ &= \frac{d\mathbf{g}/ds}{df/ds}, \\ \gamma &:= \frac{df}{ds} \end{aligned}$$

$$\begin{aligned} &= \frac{df}{\sqrt{df^2 - d\mathbf{g}^2}} \\ &= \frac{df/dt}{ds/dt} \\ &= \frac{1}{\sqrt{1 - \mathbf{v}^2}}. \end{aligned}$$

Thus $\mathbf{v}^2 < 1$. For a massive particle, the energy–momentum relation is unchanged:

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}.$$

Use the coordinate acceleration relative to f , in direct analogy with $\mathbf{a} = d\mathbf{v}/dt$ above:

$$\begin{aligned} \mathbf{a} &:= \frac{d\mathbf{v}}{df} \\ &= \frac{d^2\mathbf{g}}{df^2}. \end{aligned}$$

Because $d/ds = \gamma d/df$, the four-acceleration has the same form as before:

$$\frac{dU}{ds} = \gamma^4(\mathbf{v} \cdot \mathbf{a}) + (\gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma^2\mathbf{a}) \cdot \boldsymbol{\sigma}.$$

The commutator of four-velocity and four-acceleration is

$$\begin{aligned} \text{com}\left(U, \frac{dU}{ds}\right) &= \text{com}\left(U, \frac{dU}{df}\right) \frac{df}{ds} \\ &= \gamma \text{com}\left(U, \frac{dU}{df}\right) \\ &= 2\gamma^3(\mathbf{v} \times \mathbf{a}) \cdot (i\boldsymbol{\sigma}). \end{aligned}$$

For a massive free particle in these flat target components,

$$m \frac{dU}{ds} = 0,$$

which is equivalently the straight-worldline equation

$$\frac{d^2X}{ds^2} = 0.$$

After division by the nonzero factor γ , its scalar and vector parts are

$$\begin{aligned} \gamma^3(\mathbf{v} \cdot \mathbf{a}) &= 0, \\ \gamma^3(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma\mathbf{a} &= 0. \end{aligned}$$

The second relation may be written as

$$\gamma M \mathbf{a} = 0, \quad M := \mathbf{I}_3 + \gamma^2 \mathbf{v} \mathbf{v}^T = \mathbf{I}_3 + \frac{\mathbf{v} \mathbf{v}^T}{1 - \mathbf{v}^2}.$$

For $\mathbf{v}^2 < 1$, the matrix M is positive definite, so $\gamma M \mathbf{a} = 0$ implies $\mathbf{a} = 0$. Hence

$$\frac{d^2\mathbf{g}}{df^2} = 0,$$

and along an inertial worldline

$$\mathbf{g} = \mathbf{c}_1 f + \mathbf{c}_2, \quad \mathbf{v} = \frac{d\mathbf{g}}{df} = \mathbf{c}_1,$$

so inertial motion is linear in the flat target components $(f, \mathbf{g})$ with constant velocity. If the same straight worldline is described in arbitrary curvilinear coordinates, its coordinate equation generally contains the corresponding connection terms; no curved-space geodesic equation is being asserted here.

2.3.7 Lagrangian Mechanics and Conserved Quantities

For a particle trajectory written in terms of time, position, and velocity, let the action be

$$S = \int \mathcal{L}(t, \mathbf{r}, \mathbf{v}) dt.$$

The Euler–Lagrange equation is then

$$\left(\nabla - \frac{d}{dt} \partial_{\mathbf{v}} \right) \mathcal{L} = 0, \\ \frac{d}{dt} \partial_{\mathbf{v}} \mathcal{L} = \nabla \mathcal{L}.$$

This makes the standard conserved quantities immediate. The canonical momentum is

$$\partial_{\mathbf{v}} \mathcal{L},$$

so if $\nabla \mathcal{L} = 0$, then

$$\frac{d}{dt} \partial_{\mathbf{v}} \mathcal{L} = 0,$$

and the canonical momentum is constant.

The Hamiltonian is defined by

$$\mathcal{H} = \mathbf{v} \cdot (\partial_{\mathbf{v}} \mathcal{L}) - \mathcal{L} \\ = (\mathbf{v} \cdot \partial_{\mathbf{v}} - 1) \mathcal{L}.$$

Its time derivative is

$$\begin{aligned} \frac{d\mathcal{H}}{dt} &= \frac{d}{dt} (\partial_{\mathbf{v}} \mathcal{L}) \cdot \mathbf{v} + (\partial_{\mathbf{v}} \mathcal{L}) \cdot \frac{d\mathbf{v}}{dt} - \frac{d\mathcal{L}}{dt} \\ &= (\nabla \mathcal{L}) \cdot \mathbf{v} + (\partial_{\mathbf{v}} \mathcal{L}) \cdot \frac{d\mathbf{v}}{dt} - \frac{d\mathcal{L}}{dt} \\ &= (\nabla \mathcal{L}) \cdot \mathbf{v} + (\partial_{\mathbf{v}} \mathcal{L}) \cdot \frac{d\mathbf{v}}{dt} \\ &\quad - \left(\partial_t \mathcal{L} + (\nabla \mathcal{L}) \cdot \mathbf{v} + (\partial_{\mathbf{v}} \mathcal{L}) \cdot \frac{d\mathbf{v}}{dt} \right) \\ &= -\partial_t \mathcal{L}. \end{aligned}$$

Therefore, if $\partial_t \mathcal{L} = 0$, then

$$\frac{d\mathcal{H}}{dt} = 0,$$

so the Hamiltonian is constant. The free-particle and minimally coupled electromagnetic examples below are direct applications of these identities.

2.3.8 Action of a Free Particle

The free-particle action is the mass-weighted proper time along the world line. With $\tau = \int ds$, $U = dX/ds$, and $\gamma = dt/ds$,

$$\begin{aligned} S_{\text{free}} &= \int \mathcal{L}_{\text{free}} dt = -m\tau \\ &= -m \int \langle\langle dX \rangle\rangle \\ &= -m \int ds \\ &= -m \int \left\langle \left\langle \frac{dX}{dt} \right\rangle \right\rangle dt \\ &= -m \int \frac{ds}{dt} dt \\ &= -m \int \left\langle \left\langle \frac{U}{\gamma} \right\rangle \right\rangle dt \\ &= -m \int \frac{1}{\gamma} dt \\ &= -m \int \sqrt{1 - \mathbf{v}^2} dt. \end{aligned}$$

Therefore

$$\mathcal{L}_{\text{free}} = -\frac{m}{\gamma}.$$

Differentiating with respect to velocity gives the canonical momentum,

$$\begin{aligned} \partial_{\mathbf{v}} \mathcal{L}_{\text{free}} &= m\gamma \mathbf{v} \\ &= \mathbf{P}, \\ (\partial_{\mathbf{v}} \cdot \boldsymbol{\sigma}) \mathcal{L}_{\text{free}} &= \mathbf{P} \cdot \boldsymbol{\sigma}. \end{aligned}$$

The Hamiltonian is

$$\begin{aligned} \mathcal{H} &= (\partial_{\mathbf{v}} \mathcal{L}_{\text{free}}) \cdot \mathbf{v} - \mathcal{L}_{\text{free}} \\ &= m\gamma \mathbf{v}^2 + \frac{m}{\gamma} \\ &= m\gamma \left(\mathbf{v}^2 + \frac{1}{\gamma^2} \right) \\ &= m\gamma \\ &= E. \end{aligned}$$

Because the free Lagrangian has no explicit time or position dependence,

$$\partial_t \mathcal{L}_{\text{free}} = 0, \quad \nabla \mathcal{L}_{\text{free}} = 0,$$

so the conserved quantities are the relativistic energy $E = m\gamma$ and momentum $\mathbf{P} = m\gamma \mathbf{v}$.

2.3.9 Non-Relativistic Approximation

For the non-relativistic approximation, expand the free Lagrangian directly:

$$\begin{aligned} \mathcal{L}_{\text{free}} &= -\frac{m}{\gamma} \\ &= -m + \frac{1}{2} m \mathbf{v}^2 + O(\mathbf{v}^4). \end{aligned}$$

Then the canonical momentum becomes

$$\begin{aligned}\partial_{\mathbf{v}}\mathcal{L}_{\text{free}} &= \frac{\partial}{\partial \mathbf{v}} \left(\frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4) \right) \\ &= m\mathbf{v} + O(\mathbf{v}^3) \\ &\approx m\mathbf{v}.\end{aligned}$$

The total energy is the kinetic term plus the rest-mass term:

$$\begin{aligned}\mathcal{H} &= (m\mathbf{v} + O(\mathbf{v}^3)) \cdot \mathbf{v} + m - \frac{1}{2}m\mathbf{v}^2 - O(\mathbf{v}^4) \\ &= m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4) \\ &\approx m + \frac{1}{2}m\mathbf{v}^2.\end{aligned}$$

Equivalently, the low-velocity series is

$$\begin{aligned}\mathcal{L}_{\text{free}} &= -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4), \\ \mathbf{P} &= m\mathbf{v} + O(\mathbf{v}^3), \\ \mathcal{H} &= m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4).\end{aligned}$$

This shows exactly how the relativistic action contains the Newtonian kinetic term together with the additive rest-energy constant. Classical mechanics therefore appears as the approximation, not as the starting point.

2.3.10 Flat Pullback Metric and Diagonalization

The map $X = f + \mathbf{g} \cdot \boldsymbol{\sigma}$ induces a flat metric in the original coordinate tuple

$$x = (t, r_1, r_2, r_3)^T, \quad dx = (dt, dr_1, dr_2, dr_3)^T.$$

Expanding df and $d\mathbf{g}$ in dx defines the symmetric pullback matrix $\mathbf{Q}(x)$:

$$\begin{aligned}ds^2 &= df^2 - d\mathbf{g}^2 \\ &= dx^T \mathbf{Q}(x) dx.\end{aligned}$$

If the map is locally invertible, Sylvester's law preserves the signature $(+, -, -, -)$. Position-dependent entries of $\mathbf{Q}$ can arise from curvilinear coordinates even though the target geometry remains flat; they do not by themselves represent curvature. If the map fails to be locally invertible, the pullback may instead be degenerate.

Pointwise, the real symmetric matrix has an orthogonal decomposition

$$\mathbf{Q} = \mathbf{N} \text{diag}(\lambda_0, \lambda_1, \lambda_2, \lambda_3) \mathbf{N}^T, \quad \mathbf{N}^T \mathbf{N} = \mathbf{I}.$$

Writing the columns of $\mathbf{N}$ as u_n , define the local coframe components

$$\eta_n := \text{sgn}(\lambda_n), \quad \omega_n := \sqrt{|\lambda_n|} (u_n \cdot dx).$$

Then

$$ds^2 = \sum_n \eta_n \omega_n^2.$$

The signs η_n cannot be dropped: diagonalization does not turn a Lorentzian interval into a Euclidean norm. When $\mathbf{Q}$ varies with position, the ω_n are local one-form components and need not be exact differentials of globally defined coordinates.

3 Covariant Transform Applications

Section 1.11 established the generic similarity law and the circular rotor law. Here the Lorentz action on real event, velocity, and momentum paravectors is separated from similarity and applied explicitly.

For $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, the passive event boost is $X' = L^* X L^*$, the $T = L$ specialization of Eq. (62). Its scalar and vector parts give the usual Lorentz formulas and preserve XX^* . Section 5 later applies the distinct similarity law (65) to the STA-bivector (or biparavector) representation carried by F . The spatial rotor convention used earlier is active; this boost convention is passive.

3.1 Lorentz Boost

The boost element comes from the same exponential calculus as the circular rotor, but its action on a space-time paravector is a congruence rather than the similarity action used for a spatial rotation. This distinction keeps a real event paravector real and allows its scalar and parallel-vector parts to mix. The auxiliary parameters β and γ are introduced first because they shorten the resulting formulas:

$$\begin{aligned}\theta &= \text{arctanh}(\beta), \\ \beta &= \tanh(\theta), \\ \gamma &= \cosh(\theta), \\ \gamma\beta &= \sinh(\theta), \\ \mathbf{u} &= (u_1, u_2, u_3), \mathbf{u}^2 = 1, \\ L(\theta, \mathbf{u}) &:= \exp(\tfrac{1}{2}\theta(\mathbf{u} \cdot \boldsymbol{\sigma})), \\ &= \cosh(\tfrac{1}{2}\theta) + \sinh(\tfrac{1}{2}\theta)(\mathbf{u} \cdot \boldsymbol{\sigma}). \\ L^{-1} &= \exp(-\tfrac{1}{2}\theta(\mathbf{u} \cdot \boldsymbol{\sigma})).\end{aligned}$$

Here $\theta, \gamma \in \mathbb{R}$, $\beta \in \mathbb{R}$ with $|\beta| < 1$, $\mathbf{u} \in \mathbb{R}^3$ is a unit real vector, and $L \in \mathbb{R} \oplus \mathbb{R}^3$ is the hyperbolic transform element. Its inverse satisfies $L^{-1} = L^* = L^{*\mathbf{R}}$. Elements along the same axis commute:

$$\text{com}(L(\theta, \mathbf{u}), L(\theta', \mathbf{u})) = 0.$$

Consequently, collinear boosts combine by addition of the parameter θ :

$$\begin{aligned}\theta' &= \theta_1 + \theta_2, \\ L(\theta', \mathbf{u}) &= L(\theta_1, \mathbf{u})L(\theta_2, \mathbf{u}), \\ \beta' &= (\beta_1 + \beta_2)/(1 + \beta_1\beta_2), \\ \gamma' &= \gamma_1\gamma_2(1 + \beta_1\beta_2).\end{aligned}$$

For the passive boost with the signs used below, the real space-time paravector $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, with $t \in \mathbb{R}$, $\mathbf{r} \in \mathbb{R}^3$, and $X \in \mathbb{R} \oplus \mathbb{R}^3$, transforms as

$$\begin{aligned}X &= t + \mathbf{r} \cdot \boldsymbol{\sigma}, \\ X' &= L^* X L^*, \\ &= t' + \mathbf{r}' \cdot \boldsymbol{\sigma}.\end{aligned}$$

$$\begin{aligned} t' &= t \cosh(\theta) - \sinh(\theta)(\mathbf{u} \cdot \mathbf{r}), \\ &= \gamma(t - \beta(\mathbf{u} \cdot \mathbf{r})). \end{aligned}$$

The vector part is clearest after using the axis functions in Eq. (31) to split $\mathbf{r}$ into components parallel and perpendicular to the chosen direction:

$$\begin{aligned} \text{par}(\mathbf{r}) &= (\mathbf{r} \cdot \mathbf{u})\mathbf{u} \quad \text{parallel to } \mathbf{u}, \\ \text{prp}(\mathbf{r}) &= \mathbf{r} - \text{par}(\mathbf{r}) \quad \text{perpendicular to } \mathbf{u}, \\ \mathbf{r} &= \text{par}(\mathbf{r}) + \text{prp}(\mathbf{r}), \\ \text{par}(\mathbf{r}') &= \gamma(\text{par}(\mathbf{r}) - \beta t\mathbf{u}), \\ \text{prp}(\mathbf{r}') &= \text{prp}(\mathbf{r}), \\ \mathbf{r}' &= \text{par}(\mathbf{r}') + \text{prp}(\mathbf{r}'), \\ &= \text{prp}(\mathbf{r}) + \gamma(\text{par}(\mathbf{r}) - \beta t\mathbf{u}), \\ &= \mathbf{r} + (\gamma(\mathbf{r} \cdot \mathbf{u} - \beta t) - \mathbf{r} \cdot \mathbf{u})\mathbf{u}. \end{aligned}$$

Equivalently, direct multiplication gives the full transformed paravector in one line:

$$L^* X L^* = \gamma(t - \beta \mathbf{u} \cdot \mathbf{r}) + (\text{prp}(\mathbf{r}) + \gamma(\text{par}(\mathbf{r}) - \beta t\mathbf{u})) \cdot \boldsymbol{\sigma}.$$

Thus the component parallel to $\mathbf{u}$ mixes with the scalar part while the perpendicular component is unchanged. In the small-parameter limit $\beta \rightarrow 0$,

$$\gamma \rightarrow 1, t' \rightarrow t, \mathbf{r}' \rightarrow \mathbf{r}.$$

The interval invariance then follows directly from $L^{-1} = L^*$:

$$\begin{aligned} \langle\langle X' \rangle\rangle^2 &= X' X'^*, \\ &= (L^* X L^*)(L^* X L^*)^*, \\ &= L^* X L^* L X^* L, \\ &= L^* X X^* L, \\ &= \langle\langle X \rangle\rangle^2 L^* L, \\ &= \langle\langle X \rangle\rangle^2. \end{aligned}$$

Length contraction follows only after the simultaneity condition is specified. Let a rod be at rest in the unprimed frame and aligned with $\mathbf{u}$, so $\Delta \text{prp}(\mathbf{r}) = 0$ and its proper length is $\ell_0 = \|\Delta \text{par}(\mathbf{r})\|$. The primed frame compares its endpoints at the same primed time, so $\Delta t' = 0$. The inverse boost then gives

$$\Delta \text{par}(\mathbf{r}) = \gamma(\Delta \text{par}(\mathbf{r}') + \beta \Delta t' \mathbf{u}) = \gamma \Delta \text{par}(\mathbf{r}'),$$

and therefore

$$\ell' = \frac{\ell_0}{\gamma}.$$

The perpendicular length is unchanged.

The same congruence acts on the four-momentum of a massive particle:

$$\begin{aligned} (mU)' &= L^*(mU)L^* \\ &= E' + \mathbf{P}' \cdot \boldsymbol{\sigma}. \end{aligned}$$

Consequently,

$$\begin{aligned} E' &= \gamma(E - \beta \mathbf{u} \cdot \mathbf{P}), \\ \text{par}(\mathbf{P}') &= \gamma(\text{par}(\mathbf{P}) - \beta E \mathbf{u}), \\ \text{prp}(\mathbf{P}') &= \text{prp}(\mathbf{P}). \end{aligned}$$

Because this is the same real-paravector transformation as for X , it preserves $E^2 - \mathbf{P}^2 = m^2$.

3.2 Compositions of Boosts and Rotors

The ordered products are listed here with their inverses. The order matters even when each factor separately has a simple inverse rule.

For one hyperbolic element and one circular element,

$$\begin{aligned} T &= LR, \\ T^{-1} &= T^{*R} = R^{*R} L^{*R} = R^R L^*, \\ T &= RL, \\ T^{-1} &= T^{*R} = L^{*R} R^{*R} = L^* R^R. \end{aligned}$$

For two factors of the same type,

$$\begin{aligned} T &= R_1 R_2, \\ T^{-1} &= T^{*R} = R_2^{*R} R_1^{*R}, \\ T &= L_1 L_2, \\ T^{-1} &= T^{*R} = L_2^{*R} L_1^{*R}. \end{aligned}$$

These formulas are simple order identities, but they matter because they fix the inverse of each ordered product without ambiguity.

4 Differential Operators

The electromagnetic and wave-equation sections use one fixed set of differential conventions. They are collected here once so the later sections can refer back to the same numbered equations instead of redefining the operators in multiple places.

For spatial derivatives with respect to $\mathbf{r} = (r_1, r_2, r_3)$, define

$$\partial_{\mathbf{r}} := \left\{ \frac{\partial}{\partial r_1}, \frac{\partial}{\partial r_2}, \frac{\partial}{\partial r_3} \right\}. \quad (83)$$

The gradient notation is then

$$\boldsymbol{\nabla} := \partial_{\mathbf{r}} = \left\{ \frac{\partial}{\partial r_1}, \frac{\partial}{\partial r_2}, \frac{\partial}{\partial r_3} \right\} = \{\partial_1, \partial_2, \partial_3\}. \quad (84)$$

For shorthand, the coordinate symbol $\mathbf{r}$ is dropped in the last form, so $\partial_i := \partial/\partial r_i$ for $i \in \{1, 2, 3\}$. The corresponding spatial Laplacian is

$$\nabla^2 := \partial_1^2 + \partial_2^2 + \partial_3^2. \quad (85)$$

The STA differential paravector and its conjugate are

$$\partial := \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla} = \partial_t + \sigma_1 \partial_1 + \sigma_2 \partial_2 + \sigma_3 \partial_3. \quad (86)$$

$$\partial^* := \partial_t - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla} = \partial_t - \sigma_1 \partial_1 - \sigma_2 \partial_2 - \sigma_3 \partial_3. \quad (87)$$

Because the coordinate derivatives commute on smooth fields, their ordered products give the scalar d'Alembertian

$$\square := \partial\partial^* \quad (88)$$

$$\begin{aligned} &= \partial^*\partial \\ &= \partial_t^2 - \nabla^2. \end{aligned} \quad (89)$$

4.1 Complex Fields

For a generic complex paravector field, write

$$\begin{aligned} \partial &= \partial_t + \nabla \cdot \sigma, \\ F &= (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \sigma. \end{aligned}$$

Assume

$$\begin{aligned} a : \mathbb{R} \times \mathbb{R}^3 &\rightarrow \mathbb{R}, & \mathbf{A} : \mathbb{R} \times \mathbb{R}^3 &\rightarrow \mathbb{R}^3, \\ b : \mathbb{R} \times \mathbb{R}^3 &\rightarrow \mathbb{R}, & \mathbf{B} : \mathbb{R} \times \mathbb{R}^3 &\rightarrow \mathbb{R}^3, \end{aligned}$$

so that

$$F : \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{C} \oplus \mathbb{C}^3.$$

The real/imaginary scalar/vector parts of ∂F and $i\partial F$ are then: Tables 19 and 20 record the scalar and vector parts of ∂F and $i\partial F$ for the generic complex paravector field $F = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \sigma$.

Table 19. Components of ∂F .

real scalar:	$\partial_t a + \nabla \cdot \mathbf{A}$
img scalar:	$\partial_t b + \nabla \cdot \mathbf{B}$
real vector:	$\nabla a + \partial_t \mathbf{A} - \nabla \times \mathbf{B}$
img vector:	$\nabla b + \nabla \times \mathbf{A} + \partial_t \mathbf{B}$

Table 20. Components of $i\partial F$.

real scalar:	$-\partial_t b - \nabla \cdot \mathbf{B}$
img scalar:	$\partial_t a + \nabla \cdot \mathbf{A}$
real vector:	$-\nabla b - \nabla \times \mathbf{A} - \partial_t \mathbf{B}$
img vector:	$\nabla a + \partial_t \mathbf{A} - \nabla \times \mathbf{B}$

5 Electromagnetic Fields

Electromagnetic fields, gauge structure, and the Lorentz force admit compact paravector forms. In physical content this is Maxwell theory in rationalized Heaviside–Lorentz units [2]; the packaging follows later STA presentations [12–14].

5.1 Fields, potentials, and Maxwell equations

The differential conventions are fixed once in Section 4; in particular, Eqs. (83)–(89) define the spatial derivative vector, the shorthand ∂_i , the paravector derivative, its conjugate, and the d'Alembertian. With those operators fixed, and with $V, \rho, \lambda, S, q \in \mathbb{R}$, $\mathbf{E}, \mathbf{B}, \mathbf{A}, \mathbf{J} \in \mathbb{R}^3$, the electromagnetic field vector and potential are written as

$$\mathbf{F} := \mathbf{E} + i\mathbf{B}, \quad F := \mathbf{F} \cdot \sigma, \quad \Phi = V + \mathbf{A} \cdot \sigma. \quad (90)$$

Here $\Phi \in \mathbb{R} \oplus \mathbb{R}^3$, $\mathbf{F} \in \mathbb{C}^3$, and $F \in \{0\} \oplus \mathbb{C}^3$ is a pure-vector complex paravector. Equation (90) packages electric and magnetic fields into F , and scalar and vector potentials into Φ .

The identity

$$\partial\Phi = S + F^*, \quad S := \partial_t V + \nabla \cdot \mathbf{A} \quad (91)$$

encodes

$$\mathbf{E} = -\partial_t \mathbf{A} - \nabla V, \quad \mathbf{B} = \nabla \times \mathbf{A}. \quad (92)$$

The compact Maxwell equation is

$$\partial F = \rho - \mathbf{J} \cdot \sigma. \quad (93)$$

Splitting real/imaginary scalar/vector parts of (93) reproduces

$$\nabla \cdot \mathbf{E} = \rho, \quad \nabla \cdot \mathbf{B} = 0, \quad (94)$$

$$\nabla \times \mathbf{B} = \mathbf{J} + \partial_t \mathbf{E}, \quad \nabla \times \mathbf{E} = -\partial_t \mathbf{B}. \quad (95)$$

Equations (94) and (95) are standard STA/GA electrodynamics up to sign conventions fixed by Eqs. (86), (87), (89), and (90); component by component they recover Maxwell's field equations in the present signature and field convention [2, 12, 13, 15]. Equation (93) avoids taking divergence and curl as separate primitive operators at this stage. They emerge automatically when the scalar and vector parts of the single STA equation are separated.

5.2 Boost transformation of electromagnetic fields

The same boost element L from Eq. (54) acts on the field paravector by

$$F' = L^* F L. \quad (96)$$

Equation (96) is deliberately different from the event congruence $X' = L^* X L^*$. Events carry the Minkowski-vector (real-paravector) representation, whereas F carries the STA-bivector/biparavector representation; the resulting similarity law gives the usual parallel/perpendicular mixing of $\mathbf{E}$ and $\mathbf{B}$.

5.3 Wave equations, gauge, and Lorentz force

Applying ∂^* to (93) gives

$$\square F = \partial^*(\rho - \mathbf{J} \cdot \sigma), \quad (97)$$

whose scalar part yields charge conservation,

$$\partial_t \rho + \nabla \cdot \mathbf{J} = 0. \quad (98)$$

Equation (98) is the continuity equation written in the present notation. For a real scalar gauge function $\lambda \in \mathbb{R}$, write the transformed quantities with a subscript to avoid confusing this gauge change with a Lorentz-frame prime:

$$\Phi_\lambda = \Phi + \partial^* \lambda, \quad F_\lambda = F. \quad (99)$$

This gauge transformation shows why the potential Φ is retained even when the field is physically primary: Φ carries the redundancy, while F carries the gauge-invariant derivative content.

For a particle of real mass $m > 0$, real charge q , and four-velocity U , the four-force formula is

$$m \frac{dU}{ds} = q \operatorname{real}(FU) = q\gamma(\mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}), \quad (100)$$

which packages both relativistic power and the Lorentz three-force. Since $d/ds = \gamma d/dt$, division by γ gives $dE/dt = q\mathbf{v} \cdot \mathbf{E}$ and $d\mathbf{P}/dt = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.

5.4 Lagrangian and energy-momentum remarks

The electromagnetic particle Lagrangian is

$$\mathcal{L} = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}), \quad (101)$$

and its Euler–Lagrange equations return $d\mathbf{P}/dt = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. Here the free action from Eq. (81) is supplemented by the potential term, and the field-strength form of the force law reappears after variation.

The frame-dependent field energy-density and flux paravector is

$$T := \frac{1}{2}FF^R = \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (102)$$

Equation (102) packages the energy density and Poynting vector into one frame-dependent paravector. The supporting subsection below records the precise caveat and the corresponding covariant stress-energy map.

5.5 Supporting Electromagnetic Derivations

The supporting derivations write out the field decomposition, potential reconstruction, source formulas, gauge transformations, force-law expansions, and variational identities underlying Eqs. (83)–(102).

5.5.1 Electricity and Magnetism

This subsection expands the compact field definitions in Eqs. (90) and (93) into explicit scalar, vector, and bivector parts. Rationalized Heaviside–Lorentz units are used, with $c = 1$. Here $\rho, S \in \mathbb{R}$, $\mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $\mathbf{F} = \mathbf{E} + i\mathbf{B} \in \mathbb{C}^3$, and $F = \mathbf{F} \cdot \boldsymbol{\sigma}$ is a pure-vector complex paravector, so $\operatorname{scalar}(F) = 0$. The spatial derivative conventions are

$$\begin{aligned} \boldsymbol{\nabla} &= \{\partial_1, \partial_2, \partial_3\}, \\ \partial_i &:= \partial/\partial r_i, \\ \nabla^2 &= \partial_1^2 + \partial_2^2 + \partial_3^2, \end{aligned}$$

and the differential paravector is

$$\begin{aligned} \partial &= \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla} \\ &= \partial_t + \sigma_1 \partial_1 + \sigma_2 \partial_2 + \sigma_3 \partial_3, \\ \partial^* &= \partial_t - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}. \end{aligned}$$

The generic real/imaginary scalar/vector decomposition of ∂F and $i\partial F$ has already been recorded in Subsection 4.1. With that operator bookkeeping fixed, the electric and magnetic fields can be combined without introducing any new multiplication rule.

Collect the electric and magnetic fields into the complex three-vector $\mathbf{F}$ and the field paravector F :

$$\begin{aligned} \mathbf{E} &= \{E_1, E_2, E_3\} && \text{real vector electric field,} \\ \mathbf{B} &= \{B_1, B_2, B_3\} && \text{real vector magnetic field,} \\ \mathbf{F} &= \mathbf{E} + i\mathbf{B} && \text{electromagnetic field,} \\ F &= \mathbf{F} \cdot \boldsymbol{\sigma} = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, && i = \sigma_1 \sigma_2 \sigma_3. \end{aligned}$$

$$\begin{aligned} F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= -\mathbf{F}^* \cdot \boldsymbol{\sigma}. \end{aligned}$$

The two simplest quadratic field combinations are

$$\begin{aligned} \frac{1}{2}F^2 &= \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) + i\mathbf{E} \cdot \mathbf{B}, \\ \frac{1}{2}FF^R &= \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

Thus $\frac{1}{2}F^2$ contains the proper-Lorentz invariants, while $\frac{1}{2}FF^R$ separates into the energy density and the Poynting vector. Under parity, $\mathbf{E} \cdot \mathbf{B}$ changes sign and is therefore a pseudoscalar rather than a scalar invariant of the full Lorentz group.

Define the real source paravector by

$$\begin{aligned} D &= \rho + \mathbf{J} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ D^* &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Here $\rho \in \mathbb{R}$ is the scalar charge density and $\mathbf{J} \in \mathbb{R}^3$ is the current-density vector. Maxwell's equations then read

$$\begin{aligned} \partial F &= D^* && \text{Maxwell's equations,} \\ &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \\ \partial^* F^* &= D. \end{aligned}$$

In this form, the scalar, sigma-number-one, and sigma-number-two parts of ∂F reproduce the four classical Maxwell equations in one step. Expanding the product shows where each component sits.

Expanding ∂F into scalar, pseudoscalar, vector, and bivector parts gives

$$\begin{aligned} \partial F &= (\partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla})((\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}), \\ &= (\boldsymbol{\nabla} \cdot \mathbf{E}), \\ &\quad + (\boldsymbol{\nabla} \cdot \mathbf{B})i, \\ &\quad + (\partial_t \mathbf{E} - \boldsymbol{\nabla} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &\quad + (\partial_t \mathbf{B} + \boldsymbol{\nabla} \times \mathbf{E}) \cdot (i\boldsymbol{\sigma}). \end{aligned}$$

The first two lines are the scalar and pseudoscalar parts. The last two lines are the sigma-number-one and sigma-number-two vector parts. Matching these components with $D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ immediately forces the magnetic-divergence term and Faraday term to vanish while fixing the electric source terms.

Accordingly, the component equations of $\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ are

$$\begin{aligned} \boldsymbol{\nabla} \cdot \mathbf{E} &= \rho, \\ \boldsymbol{\nabla} \cdot \mathbf{B} &= 0, \\ \partial_t \mathbf{E} - \boldsymbol{\nabla} \times \mathbf{B} &= -\mathbf{J}, \\ \partial_t \mathbf{B} + \boldsymbol{\nabla} \times \mathbf{E} &= 0. \end{aligned}$$

Separating the vector components then gives

$$\begin{aligned}\nabla \times \mathbf{B} &= \mathbf{J} + \partial_t \mathbf{E}, \\ \nabla \times \mathbf{E} &= -\partial_t \mathbf{B}.\end{aligned}$$

5.5.2 Lorentz Transform of Fields

This subsection specializes the boost law to the electromagnetic field paravector

$$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}.$$

For a constant inertial boost $L = \exp(\frac{1}{2}\theta \mathbf{u} \cdot \boldsymbol{\sigma})$ with $\mathbf{u}^2 = 1$, $\beta = \tanh(\theta)$, and $\gamma = \cosh(\theta)$, the field transforms by the general similarity (65), even though event and current paravectors transform by the congruence (62). In the inverse-first convention the left spin action is $L^{-1} = L^*$, so a bivector transforms as

$$F' = L^{-1}FL = L^*FL.$$

The final factor is L , rather than L^* , because a bivector carries the adjoint action $T^{-1}(\cdot)T$. A double- L^* sandwich would instead apply the event congruence to F and would not preserve its pure-vector form. For example, if $F = E_{\parallel} \mathbf{u} \cdot \boldsymbol{\sigma}$, then

$$L^*FL = F, \quad L^*FL^* = \gamma E_{\parallel} \mathbf{u} \cdot \boldsymbol{\sigma} - \gamma\beta E_{\parallel},$$

and the second expression has an inadmissible scalar part. Applying the general complex-vector result (66) to $\mathbf{V} = \mathbf{E} + i\mathbf{B}$ gives the field and its parallel/perpendicular parts:

$$\begin{aligned}F' &= L^*FL, \\ \text{par}(\mathbf{E}') &= \text{par}(\mathbf{E}), \\ \text{par}(\mathbf{B}') &= \text{par}(\mathbf{B}), \\ \text{prp}(\mathbf{E}') &= \gamma(\text{prp}(\mathbf{E}) + \beta \mathbf{u} \times \mathbf{B}), \\ \text{prp}(\mathbf{B}') &= \gamma(\text{prp}(\mathbf{B}) - \beta \mathbf{u} \times \mathbf{E}).\end{aligned}$$

Equivalently, the transformed fields can be written in the compact form

$$\begin{aligned}\mathbf{E}' &= \text{par}(\mathbf{E}) + \gamma(\text{prp}(\mathbf{E}) + \beta \mathbf{u} \times \mathbf{B}), \\ \mathbf{B}' &= \text{par}(\mathbf{B}) + \gamma(\text{prp}(\mathbf{B}) - \beta \mathbf{u} \times \mathbf{E}).\end{aligned}$$

The parallel components are unchanged, while the perpendicular components mix by the standard electric-magnetic cross terms. Because $LL^* = 1$ and F^2 is central, the field invariants are preserved directly:

$$F'^2 = L^*F(LL^*)FL = L^*F^2L = F^2.$$

Thus the real part of F^2 is a scalar and its imaginary part is a pseudoscalar under parity; both are invariant under proper Lorentz transformations.

For any constant normalized Lorentz element T , the potential and source are event-type paravectors, while the derivative carries the complementary action:

$$\Phi' = T^{*\text{R}}\Phi T^*, \quad D' = T^{*\text{R}}DT^*, \quad \partial' = T^{\text{R}}\partial T.$$

For the boost $T = L$, these reduce to

$$\Phi' = L^*\Phi L^*, \quad D' = L^*DL^*, \quad \partial' = L\partial L.$$

Consequently,

$$\begin{aligned}\partial' F' &= (T^{\text{R}}\partial T)(T^{*\text{R}}FT) \\ &= T^{\text{R}}(\partial F)T \\ &= T^{\text{R}}D^*T \\ &= D'^*.\end{aligned}$$

Thus Maxwell's equation is covariant. In components the current transformation is

$$\begin{aligned}\rho' &= \gamma(\rho - \beta \mathbf{u} \cdot \mathbf{J}), \\ \text{par}(\mathbf{J}') &= \gamma(\text{par}(\mathbf{J}) - \beta \rho \mathbf{u}), \\ \text{prp}(\mathbf{J}') &= \text{prp}(\mathbf{J}).\end{aligned}$$

5.5.3 Wave Operator and d'Alembertian

The operator conventions were fixed already in Eqs. (86)–(89). Here we use them only to derive the sourced wave consequences and to distinguish the scalar wave operator $\square$ from the full square ∂^2 :

$$\begin{aligned}\partial &= \partial_t + \boldsymbol{\sigma} \cdot \nabla, \\ \partial^* &= \partial_t - \boldsymbol{\sigma} \cdot \nabla, \\ \square &:= \partial_t^2 - \nabla^2 \\ &= \partial\partial^* = \partial^*\partial.\end{aligned}$$

With the sigma algebra one also has

$$\partial^2 = \partial_t^2 + \nabla^2 + 2\partial_t \nabla \cdot \boldsymbol{\sigma},$$

so $\partial^2 \neq \square$. With the conventions of Eqs. (86)–(89), applying ∂^* to Maxwell's equation gives

$$\begin{aligned}\square F &= \partial^* \partial F \\ &= \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma}) \\ &= (\partial_t \rho + \nabla \cdot \mathbf{J}) \\ &\quad + (-\partial_t \mathbf{J} - \nabla \rho + i \nabla \times \mathbf{J}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

Because F is pure vector, the scalar part of the left-hand side vanishes. The scalar and vector parts therefore give

$$\begin{aligned}\partial_t \rho + \nabla \cdot \mathbf{J} &= 0, \\ \square \mathbf{E} &= -\partial_t \mathbf{J} - \nabla \rho, \\ \square \mathbf{B} &= \nabla \times \mathbf{J}.\end{aligned}$$

In vacuum both source terms disappear and each field component satisfies the ordinary wave equation.

In vacuum, where $\rho = 0$ and $\mathbf{J} = 0$, this reduces to

$$\begin{aligned}\square \mathbf{E} = 0 &\Rightarrow \partial_t^2 \mathbf{E} = \nabla^2 \mathbf{E}, \\ \square \mathbf{B} = 0 &\Rightarrow \partial_t^2 \mathbf{B} = \nabla^2 \mathbf{B}.\end{aligned}$$

Tables 21 and 22 record the scalar and vector parts of Maxwell's equation and its sourced wave consequence.

Table 21. Components of $\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$.

real scalar:	$\boldsymbol{\nabla} \cdot \mathbf{E} = \rho$
img scalar:	$\boldsymbol{\nabla} \cdot \mathbf{B} = 0$
real vector:	$\partial_t \mathbf{E} - \boldsymbol{\nabla} \times \mathbf{B} = -\mathbf{J}$
img vector:	$\partial_t \mathbf{B} + \boldsymbol{\nabla} \times \mathbf{E} = 0$

Table 22. Components of $\square F = \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma})$.

real scalar:	$\partial_t \rho + \boldsymbol{\nabla} \cdot \mathbf{J} = 0$
img scalar:	$0 = 0$
real vector:	$\square \mathbf{E} = -\partial_t \mathbf{J} - \boldsymbol{\nabla} \rho$
img vector:	$\square \mathbf{B} = \boldsymbol{\nabla} \times \mathbf{J}$

5.5.4 Potential Fields

Eqs. (91) and (92) put gauge freedom, source reconstruction, and the electric/magnetic field split in one place. Here $V, S, \rho, \lambda \in \mathbb{R}$, $\mathbf{A}, \mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $\Phi \in \mathbb{R} \oplus \mathbb{R}^3$, $\mathbf{F} \in \mathbb{C}^3$, and F is the pure-vector complex paravector defined above:

$$\begin{aligned} F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ \Phi &= V + \mathbf{A} \cdot \boldsymbol{\sigma}, \\ \partial\Phi &= \partial_t V + \boldsymbol{\nabla} \cdot \mathbf{A}, \\ &\quad + (\partial_t \mathbf{A} + \boldsymbol{\nabla} V) \cdot \boldsymbol{\sigma}, \\ &\quad + i(\boldsymbol{\nabla} \times \mathbf{A}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

This expansion separates the potential gradient into one real scalar part and one pure-vector field part. The scalar piece becomes the Lorenz-gauge scalar S , and the remaining vector piece becomes the electromagnetic field:

$$\begin{aligned} S &:= \text{scalar}(\partial\Phi), \\ &= \partial_t V + \boldsymbol{\nabla} \cdot \mathbf{A}. \end{aligned}$$

The component identities extracted from $\partial\Phi$ are

$$\begin{aligned} \mathbf{E} &= -\partial_t \mathbf{A} - \boldsymbol{\nabla} V, \\ \mathbf{B} &= \boldsymbol{\nabla} \times \mathbf{A}. \end{aligned}$$

The identity is explicit:

$$\begin{aligned} \partial\Phi &= S + F^*, \\ F^* &= \partial\Phi - S, \\ \text{scalar}(F^*) &= \text{scalar}(\partial\Phi - S) = 0, \\ \text{vector}(F^*) &= \text{vector}(\partial\Phi). \end{aligned}$$

The field is therefore the non-scalar part of $\partial\Phi$. Under the gauge change below, only the scalar part of $\partial\Phi$ changes; the field paravector remains unchanged. Table 23 records the component content of the potential identity $\partial\Phi = S + F^*$.

Table 23. Components of $\partial\Phi = S + F^*$.

real scalar:	$\partial_t V + \boldsymbol{\nabla} \cdot \mathbf{A} = S$
img scalar:	$0 = 0$
real vector:	$\partial_t \mathbf{A} + \boldsymbol{\nabla} V = -\mathbf{E}$
img vector:	$\boldsymbol{\nabla} \times \mathbf{A} = \mathbf{B}$

The differential product $\partial\Phi$ is evaluated first; the involutions then act on the resulting field. No right-acting differential operator is implied. Equivalently, without introducing S , one may write

$$F^* = (\partial\Phi - (\partial\Phi)^{*R})/2.$$

5.5.5 Density from Potential

This subsection collects the derivation of the charge density and current density directly from the potential field in one place. Using the previously defined potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, scalar part $S = \partial_t V + \boldsymbol{\nabla} \cdot \mathbf{A}$, and identity $F^* = \partial\Phi - S$, one obtains

$$\begin{aligned} F &= \partial^* \Phi^* - S, \\ \partial F &= \partial \partial^* \Phi^* - \partial S, \\ &= \square \Phi^* - \partial S, \\ &= (\square V - \partial_t S) + (-\square \mathbf{A} - \boldsymbol{\nabla} S) \cdot \boldsymbol{\sigma}, \\ &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \end{aligned}$$

$$\begin{aligned} \rho &= \square V - \partial_t S, \\ \mathbf{J} &= \square \mathbf{A} + \boldsymbol{\nabla} S. \end{aligned}$$

These are the component source-reconstruction formulas hidden inside the compact Maxwell equation. The scalar equation determines charge density from the potential data, and the vector equation determines current density from the same data.

In Lorenz gauge, where $S = 0$, these relations reduce to

$$\begin{aligned} \partial_t V &= -\boldsymbol{\nabla} \cdot \mathbf{A}, \\ \rho &= \square V, \\ \mathbf{J} &= \square \mathbf{A}. \end{aligned}$$

5.5.6 Gauge Transform of the Potential

$$\begin{aligned} \text{scalar}(\partial\Phi) &= S, \\ S &= \partial_t V + \boldsymbol{\nabla} \cdot \mathbf{A}. \end{aligned}$$

Let λ be a real scalar gauge function. To distinguish a gauge change from the boost primes used above, write

$$\begin{aligned} \Phi &\longmapsto \Phi + \partial^* \lambda, \\ \partial\Phi &\longmapsto \partial\Phi + \square \lambda. \end{aligned}$$

Since $\square \lambda$ is again a scalar field, only the scalar part changes:

$$\begin{aligned} S &\longmapsto S + \square \lambda, \\ F &\longmapsto F. \end{aligned}$$

This makes the gauge split explicit. Transformations that preserve Lorenz gauge satisfy the residual condition $\square \lambda = 0$.

5.5.7 Four-Force

This subsection collects the relativistic work law, the Lorentz-force law, the component split of FU/γ , and the four-force equation in one place.

$$\begin{aligned} mU &= m \frac{dX}{ds}, \\ &= m\gamma + m\gamma \mathbf{v} \cdot \boldsymbol{\sigma}. \\ \frac{d}{ds} &= \gamma \frac{d}{dt}. \end{aligned}$$

The scalar part of $d(mU)/dt$ is the power equation and the vector part is the three-force equation. Writing both pieces in one paravector equation makes the usual split visible without abandoning the relativistic form.

The time component gives the relativistic work law,

$$\frac{d(m\gamma)}{dt} = q\mathbf{v} \cdot \mathbf{E}.$$

Here $\mathbf{E}$ denotes the electric field, not the energy operator E , and the magnetic field contributes no work term in the scalar component. The spatial component gives the relativistic Lorentz force law,

$$\begin{aligned} \mathbf{P} &:= m\gamma \mathbf{v}, \\ \frac{d\mathbf{P}}{dt} &= q\mathbf{E} + q\mathbf{v} \times \mathbf{B}. \end{aligned}$$

Together these assemble into

$$\frac{d(mU)}{dt} = q\mathbf{v} \cdot \mathbf{E} + q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.$$

To expose the paravector content explicitly, expand

$$\begin{aligned} FU/\gamma &= ((\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\ &= \mathbf{E} \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \boldsymbol{\sigma} + (\mathbf{E} \cdot \boldsymbol{\sigma})(\mathbf{v} \cdot \boldsymbol{\sigma}), \\ &\quad + i(\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{v} \cdot \boldsymbol{\sigma}), \\ &= \mathbf{E} \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \boldsymbol{\sigma} + \mathbf{E} \cdot \mathbf{v}, \\ &\quad + i(\mathbf{E} \times \mathbf{v}) \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \mathbf{v} - (\mathbf{B} \times \mathbf{v}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

The product FU naturally separates into real and imaginary parts. The real part reproduces the physical Lorentz-force content, while the imaginary part exchanges electric and magnetic roles and is therefore not used in the force law itself. Table 24 gives the component form of the product FU/γ used in the Lorentz-force law.

Table 24. Components of FU/γ .

real scalar:	$\mathbf{E} \cdot \mathbf{v}$
img scalar:	$\mathbf{B} \cdot \mathbf{v}$
real vector:	$\mathbf{E} + \mathbf{v} \times \mathbf{B}$
img vector:	$\mathbf{B} + \mathbf{E} \times \mathbf{v}$

$$\begin{aligned} \text{real}(FU) &= \frac{1}{2}(FU + UF^{\mathbf{R}}), \\ &= \gamma(\mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}). \end{aligned}$$

The physical four-force is therefore

$$\begin{aligned} m \frac{dU}{ds} &= q \text{real}(FU), \\ &= \frac{1}{2}q(FU + UF^{\mathbf{R}}). \end{aligned}$$

$$m \frac{d^2 X}{ds^2} = q \text{real}\left(F \frac{dX}{ds}\right) \quad \text{four-force.}$$

5.5.8 Lagrangian Derivation of the Lorentz Force

This subsection collects the minimally coupled action, the Euler–Lagrange derivation of the Lorentz force, the Hamiltonian, and the corresponding power identity in one place. Starting from the Lagrangian of Eq. (101), together with the potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, four-velocity $U = dX/ds$, and $\gamma = dt/ds$, the action takes the form

$$\begin{aligned} \text{action} &= -m \int ds - q \int \text{scalar}(\Phi^* U) ds \\ &= \int \left[-\frac{m}{\gamma} - q \text{scalar}((V - \mathbf{A} \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma})) \right] dt \\ &= \int \left(-\frac{m}{\gamma} - qV + q\mathbf{v} \cdot \mathbf{A} \right) dt. \\ \mathcal{L} &= -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}). \end{aligned}$$

This is the standard minimal-coupling Lagrangian written in the present paravector notation. The canonical momentum is shifted by the vector potential, exactly as in the ordinary component derivation.

The canonical momentum is

$$\begin{aligned} \partial_{\mathbf{v}} \mathcal{L} &= m\gamma \mathbf{v} + q\mathbf{A}, \\ &= \mathbf{P} + q\mathbf{A}. \end{aligned}$$

$$\begin{aligned} \frac{d}{dt} \partial_{\mathbf{v}} \mathcal{L} &= \frac{d\mathbf{P}}{dt} + q\partial_t \mathbf{A} + q(\mathbf{v} \cdot \nabla) \mathbf{A}, \\ \nabla \mathcal{L} &= -q\nabla V + q\nabla(\mathbf{v} \cdot \mathbf{A}). \end{aligned}$$

Using the vector-calculus identity

$$(\mathbf{v}(t) \cdot \nabla) \mathbf{A}(t, \mathbf{r}) = \nabla(\mathbf{v} \cdot \mathbf{A}) - \mathbf{v} \times (\nabla \times \mathbf{A}).$$

the Euler–Lagrange equation $(d/dt)\partial_{\mathbf{v}} \mathcal{L} = \nabla \mathcal{L}$ yields

$$\begin{aligned} \frac{d\mathbf{P}}{dt} &= -q\nabla V - q\partial_t \mathbf{A}, \\ &\quad + q(\nabla(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \nabla) \mathbf{A}), \\ &= q(-\nabla V - \partial_t \mathbf{A} + \mathbf{v} \times (\nabla \times \mathbf{A})), \\ &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \\ &= \frac{m}{\gamma} \text{vector}\left(\frac{dU}{ds}\right). \end{aligned}$$

The Hamiltonian is

$$\begin{aligned}
\mathcal{H} &= (\partial_{\mathbf{v}} \mathcal{L}) \cdot \mathbf{v} - \mathcal{L} \\
&= (m\gamma \mathbf{v} + q\mathbf{A}) \cdot \mathbf{v} + \frac{m}{\gamma} + qV - q\mathbf{v} \cdot \mathbf{A} \\
&= m\gamma \left(\mathbf{v}^2 + \frac{1}{\gamma^2} \right) + qV \\
&= m\gamma + qV.
\end{aligned}$$

Thus $\mathcal{H}$ is the mechanical relativistic energy plus the scalar-potential energy. Along a solution,

$$\begin{aligned}
\frac{d\mathcal{H}}{dt} &= -\partial_t \mathcal{L} = q\partial_t V - q\mathbf{v} \cdot \partial_t \mathbf{A}, \\
\frac{d\mathcal{H}}{dt} &= \frac{d(m\gamma)}{dt} + q\partial_t V + q\mathbf{v} \cdot \nabla V.
\end{aligned}$$

Equating these expressions gives

$$\begin{aligned}
\frac{d(m\gamma)}{dt} &= q\mathbf{v} \cdot (-\nabla V - \partial_t \mathbf{A}) \\
&= q\mathbf{v} \cdot \mathbf{E},
\end{aligned}$$

in agreement with the scalar component of the Lorentz-force equation.

5.5.9 Energy-Density and Poynting Paravector

Equation (102) packages the electromagnetic energy density and Poynting vector into one paravector. The algebra below records its components and the scalar energy balance implied by Maxwell's equation (93):

$$\begin{aligned}
F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\
F^{\mathbf{R}} &= (\mathbf{E} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\
T &= \frac{1}{2} F F^{\mathbf{R}}, \\
&= \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\
&= T_s + T_{\mathbf{v}} \cdot \boldsymbol{\sigma}. \\
\partial &= \partial_t + \boldsymbol{\sigma} \cdot \nabla.
\end{aligned}$$

The paravector $T = \frac{1}{2} F F^{\mathbf{R}}$ is the energy density and flux measured relative to the chosen time direction. It is not, by itself, the full electromagnetic stress-energy tensor. The covariant content is the linear map on real paravectors $a \in \mathbb{R} \oplus \mathbb{R}^3$,

$$a \mapsto \frac{1}{2} F a F^{\mathbf{R}},$$

whose value at $a = 1$ is the displayed paravector T [12, 14]. Indeed, for $a' = L^* a L^*$, $F' = L^* F L$, and $(F')^{\mathbf{R}} = L F^{\mathbf{R}} L^*$,

$$\frac{1}{2} F' a' (F')^{\mathbf{R}} = L^* \left(\frac{1}{2} F a F^{\mathbf{R}} \right) L^*,$$

so the map has the required event-paravector covariance.

Maxwell's equations immediately give the scalar energy balance:

$$\begin{aligned}
\partial_t T_s + \nabla \cdot T_{\mathbf{v}} &= \mathbf{E} \cdot \partial_t \mathbf{E} + \mathbf{B} \cdot \partial_t \mathbf{B} + \nabla \cdot (\mathbf{E} \times \mathbf{B}) \\
&= -\mathbf{J} \cdot \mathbf{E}.
\end{aligned}$$

This is Poynting's theorem in the present units: the field-energy loss equals the work done on charge. The two quadratic field products and their component content are collected below. They display the proper-Lorentz invariants and the frame-dependent energy-density/Poynting split without suggesting a separate field equation for F^2 . Table 25 lists the two proper-Lorentz invariants. Table 26 gives the frame-dependent energy-density/Poynting paravector $T = \frac{1}{2} F F^{\mathbf{R}}$.

Table 25. Components of $\frac{1}{2} F^2$.

real scalar:	$\frac{1}{2} (\mathbf{E}^2 - \mathbf{B}^2)$
img scalar:	$\mathbf{E} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
img vector:	$\mathbf{0}$

Table 26. Components of $\frac{1}{2} F F^{\mathbf{R}}$.

scalar:	$\frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2)$	energy density
vector:	$\mathbf{E} \times \mathbf{B}$	Poynting vector

6 Relativistic Wave Equations and Spin Dynamics

The same ordered product calculus gives the standard relativistic wave equations. This section separates three logically different carriers: a complex scalar for Klein–Gordon theory, one fixed minimal ideal for a Dirac field, and the surviving two-component field in the Pauli limit. Keeping those distinctions explicit prevents algebraic degrees of freedom from being mistaken for physical spinor components [7–10].

6.1 Klein–Gordon field and frequency splitting

For a complex scalar field $\psi \in \mathbb{C}$, the free Klein–Gordon equation is

$$(\partial \partial^* + m^2) \psi = (\square + m^2) \psi = 0. \quad (103)$$

On a suitable spatial domain, let

$$\begin{aligned}
\Omega &:= \sqrt{m^2 - \nabla^2}, \\
(\partial_t + i\Omega) \psi(+) &= 0, \quad (\partial_t - i\Omega) \psi(-) = 0, \\
\psi &= \psi(+) + \psi(-).
\end{aligned} \quad (104)$$

Here Ω is the positive self-adjoint Fourier multiplier with symbol $\sqrt{m^2 + \mathbf{k}^2}$. Thus

$$\psi(\pm, t) = \exp(\mp i\Omega(t - t_0)) \psi(\pm, t_0).$$

When Ω^{-1} exists on the chosen data, the two branches are

$$\psi(\pm) = \frac{1}{2}(\psi \pm i\Omega^{-1}\partial_t\psi).$$

A massless zero mode is treated separately. The scalar $|\psi|^2$ is not the relativistic Klein–Gordon probability density; the conserved current contains derivatives and is not positive on the full positive/negative-frequency solution space. The sign in $\psi(\pm)$ labels frequency in this subsection; it is unrelated to the spin-axis projection $\Pi(\pm)\psi$ used later.

6.2 Minimal coupling and ordered factorization

With the real potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, define

$$E := i\partial_t - qV, \quad \mathbf{P} := -i\nabla - q\mathbf{A}. \quad (105)$$

The minimally coupled scalar equation is

$$(-E^2 + \mathbf{P}^2 + m^2)\psi = 0, \quad (106)$$

or, after applying the Leibniz rule,

$$\left(\square + iqS - q^2\langle\langle\Phi\rangle\rangle^2 + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \nabla)\right)\psi = 0. \quad (107)$$

For smooth fields,

$$\begin{aligned} \text{com}(E, P_m) &= iqE_m, \\ \text{com}(P_m, P_n) &= iq(\partial_m A_n - \partial_n A_m). \end{aligned} \quad (108)$$

Consequently the written order matters:

$$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF, \quad (109)$$

$$(E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF^*. \quad (110)$$

These are ordered reverse-conjugate products of differential operators, not applications of the scalar-coefficient determinant formula.

6.3 One Dirac field

In the Pauli realization a paravector is an arbitrary 2×2 complex matrix. Left multiplication acts independently on its two columns. Therefore two unrestricted paravector blocks describe two degenerate Dirac fields, not one. A single Dirac field is obtained by selecting one matrix column. Intrinsically, fix

$$\Pi_0 := \frac{1}{2}(1 + \sigma_3),$$

and require

$$\boldsymbol{\Psi}_n = \boldsymbol{\Psi}_n \Pi_0, \quad n = 1, 2. \quad (111)$$

The carrier $\text{Cl}_{3,0}\Pi_0$ is a minimal left ideal. Every operator below acts from the left, so the restriction is preserved. The fixed right factor Π_0 selects a column; it is not a measurement of spin along the third axis. Equivalently, in the Pauli matrix realization each $\boldsymbol{\Psi}_n$ is an ordinary two-component Weyl column and $\boldsymbol{\Psi} = \{\boldsymbol{\Psi}_1; \boldsymbol{\Psi}_2\}$ has the usual four complex components.

Using the block map $\mathbf{W}$ from Eq. (74), define

$$\mathbf{D}_0 := \mathbf{W}(i\partial) - m\mathbf{I}, \quad \mathbf{D}_0\boldsymbol{\Psi} = 0, \quad (112)$$

$$\mathbf{D} := \mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I}, \quad \mathbf{D}\boldsymbol{\Psi} = 0. \quad (113)$$

The coupled system is equivalently

$$(E - \boldsymbol{\sigma} \cdot \mathbf{P})\boldsymbol{\Psi}_2 = m\boldsymbol{\Psi}_1, \quad (E + \boldsymbol{\sigma} \cdot \mathbf{P})\boldsymbol{\Psi}_1 = m\boldsymbol{\Psi}_2. \quad (114)$$

Multiplication by $\mathbf{D}' = \mathbf{W}(i\partial - q\Phi^*) + m\mathbf{I}$ and use of the two ordered factorizations gives

$$\begin{cases} (-E^2 + \mathbf{P}^2 + m^2 + iqF^*)\boldsymbol{\Psi}_1 = 0, \\ (-E^2 + \mathbf{P}^2 + m^2 + iqF)\boldsymbol{\Psi}_2 = 0. \end{cases} \quad (115)$$

Thus the first-order system squares to the expected Klein–Gordon operator plus the electromagnetic spin coupling.

Under the gauge law (99),

$$\boldsymbol{\Psi}_\lambda = \exp(-iq\lambda)\boldsymbol{\Psi}, \quad (116)$$

and the form of Eq. (113) is unchanged.

The reverse product supplies a positive time-slice norm, but its vector part is not a probability current. With the fixed-ideal convention above, the Dirac density and current may be written without component indices as

$$\begin{aligned} \rho_D &= 2 \text{ scalar}(\boldsymbol{\Psi}_1^R \boldsymbol{\Psi}_1 + \boldsymbol{\Psi}_2^R \boldsymbol{\Psi}_2), \\ \mathbf{j}_D &= 2 \text{ vector}(\boldsymbol{\Psi}_2 \boldsymbol{\Psi}_2^R - \boldsymbol{\Psi}_1 \boldsymbol{\Psi}_1^R), \end{aligned} \quad \partial_t \rho_D + \nabla \cdot \mathbf{j}_D = 0. \quad (117)$$

The normalization condition is the spatial integral $\int \rho_D d^3\mathbf{r} = 1$, not a pointwise equation. In the Pauli column realization these are the familiar Weyl-basis Dirac bilinears.

6.4 Pauli limit

A chiral block by itself is not the nonrelativistic large component. First remove the rest phase, $\boldsymbol{\Psi}_n = \exp(-imt)\tilde{\boldsymbol{\Psi}}_n$, and define the energy-basis combinations

$$\phi := \frac{\tilde{\boldsymbol{\Psi}}_1 + \tilde{\boldsymbol{\Psi}}_2}{\sqrt{2}}, \quad \chi := \frac{\tilde{\boldsymbol{\Psi}}_2 - \tilde{\boldsymbol{\Psi}}_1}{\sqrt{2}}. \quad (118)$$

Equation (114) then becomes exactly

$$E\phi = (\mathbf{P} \cdot \boldsymbol{\sigma})\chi, \quad (E + 2m)\chi = (\mathbf{P} \cdot \boldsymbol{\sigma})\phi. \quad (119)$$

At energies small compared with m ,

$$\chi = \frac{\mathbf{P} \cdot \boldsymbol{\sigma}}{2m}\phi + O(m^{-2}).$$

Using

$$(\mathbf{P} \cdot \boldsymbol{\sigma})^2 = \mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma},$$

one obtains

$$E\phi = \frac{\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m}\phi + O(m^{-2}). \quad (120)$$

Equivalently,

$$i\partial_t\phi = \left[\frac{(-i\nabla - q\mathbf{A})^2}{2m} + qV - \frac{q}{2m}\mathbf{B} \cdot \boldsymbol{\sigma} \right] \phi + O(m^{-2}). \quad (121)$$

Electric spin-orbit and Darwin terms enter at the next order; no electric term has been discarded while retaining another term of the same order.

If $\mathbf{B} \neq 0$ is uniform and constant, set $\mathbf{u} = \mathbf{B}/\|\mathbf{B}\|$ and

$$\phi = \phi(+) + \phi(-), \quad \phi(\pm) = \Pi(\pm)\phi. \quad (122)$$

Then

$$i\partial_t\phi(\pm) = \left(\frac{\mathbf{p}^2 \mp q\|\mathbf{B}\|}{2m} + qV \right) \phi(\pm). \quad (123)$$

For a varying axis the derivatives of $\Pi(\pm)$ and its commutator with the Hamiltonian couple the two projected fields, so this decoupling no longer holds.

6.5 Free energy and helicity projectors

For a free momentum mode, Eq. (114) is the Hamiltonian system

$$\begin{aligned} i\partial_t\Psi &= H_D\Psi, \\ H_D &:= \begin{pmatrix} -\boldsymbol{\sigma} \cdot \mathbf{p} & m \\ m & \boldsymbol{\sigma} \cdot \mathbf{p} \end{pmatrix}, \\ H_D^2 &= E_p^2\mathbf{I}, \quad E_p := \sqrt{m^2 + \mathbf{p}^2}. \end{aligned} \quad (124)$$

Its energy projectors are

$$\Lambda(\epsilon) := \frac{1}{2} \left(\mathbf{I} + \epsilon \frac{H_D}{E_p} \right), \quad \epsilon \in \{+1, -1\}.$$

For $\mathbf{p} \neq 0$, choose the helicity axis $\mathbf{u} = \mathbf{p}/\|\mathbf{p}\|$ and let

$$\Pi_h(s) := \begin{pmatrix} \Pi(s) & 0 \\ 0 & \Pi(s) \end{pmatrix}, \quad s \in \{+1, -1\}.$$

The two projector families commute, so

$$\Psi(\epsilon, s) := \Lambda(\epsilon)\Pi_h(s)\Psi, \quad \Psi = \sum_{\epsilon=\pm 1} \sum_{s=\pm 1} \Psi(\epsilon, s). \quad (125)$$

At rest any spin axis may be chosen. Away from rest, an arbitrary fixed axis need not commute with the energy projector; helicity is the notation-light simultaneous choice. Negative-frequency classical solutions are not identified with antiparticles until the field is quantized.

6.6 Supporting Wave-Equation Derivations

The supporting derivations give the ordered operator expansions, scalar frequency evolution, minimal-ideal interpretation, conserved bilinears, gauge covariance, squared Dirac system, and controlled Pauli approximation in detail.

6.6.1 Relativistic Quantum-Mechanical Setting

The wave-equation section uses one fixed operator language throughout. With the differential conventions from Section 4,

$$\begin{aligned} \partial &= \partial_t + \boldsymbol{\sigma} \cdot \nabla, \\ \partial^* &= \partial_t - \boldsymbol{\sigma} \cdot \nabla, \\ (i\partial)^* &= -i\partial^*. \end{aligned}$$

Recall also that the d'Alembertian is

$$\begin{aligned} \square &:= \partial_t^2 - \nabla^2 \\ &= \partial\partial^* = \partial^*\partial. \end{aligned}$$

All differential operators act on the fields to their right. In an ordered product the rightmost factor acts first, and the written order is retained whenever potentials and derivatives do not commute.

6.6.2 Algebra Blocks and the Physical Spinor Carrier

Throughout the spinful part of the wave-equation section, a wave field is not a space-time event but an algebra-valued amplitude field. A generic algebra block may be written

$$\psi : \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{C} \oplus \mathbb{C}^3, \quad \psi(t, \mathbf{r}) = \psi_s(t, \mathbf{r}) + \psi_v(t, \mathbf{r}) \cdot \boldsymbol{\sigma}.$$

The argument $(t, \mathbf{r})$ tells where the state is evaluated in space-time, while the value $\psi(t, \mathbf{r})$ lives in the internal complex-paravector algebra. This distinction matters. The real paravector

$$X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$$

describes an event or interval variable, whereas ψ is a quantum state amplitude defined *over* such events.

At each fixed point, an unrestricted complex paravector has four complex coefficients. In the Pauli matrix realization it is an arbitrary 2×2 complex matrix, and left multiplication acts independently on its two columns. It is therefore the regular algebra module, not one irreducible two-component spinor. To select one spinor column without adding component indices, fix once and for all

$$\Pi_0 := \frac{1}{2}(1 + \sigma_3)$$

and restrict a physical Weyl block by

$$\psi = \psi\Pi_0.$$

The set $(\mathbb{C} \oplus \mathbb{C}^3)\Pi_0$ is a minimal left ideal, selected by the fixed factor on the right. Every operator used below acts on the left, so the restriction is preserved. The factor Π_0 is only a column selector; it is not a measurement of spin along the third axis.

The scalar and vector coefficients of an algebra block are amplitude channels, not separately measurable particles or classical subfields. A bare component count also does not assign physical spin: the transformation law, field equation, and carrier restriction must be specified together.

The spin-0 Klein-Gordon subsections use $\psi \in \mathbb{C}$. The Dirac construction uses a two-block field

$$\Psi = \begin{Bmatrix} \Psi_1 \\ \Psi_2 \end{Bmatrix},$$

with $\Psi_n = \Psi_n \Pi_0$. Each block then carries two complex amplitudes and the pair carries the four complex amplitudes of one Dirac field. If the minimal-ideal restriction is omitted, the same equations remain algebraically valid but describe two independent Dirac fields, one in each matrix column.

6.6.3 Complex-Paravector Bilinears and Bra-Ket Extraction

To make the bra-ket notation precise in the paravector setting, write

$$\begin{aligned}\psi &= \psi_s + \psi_v \cdot \sigma, & O &= O_s + O_v \cdot \sigma, \\ Z &= Z_s + Z_v \cdot \sigma, & X &= X_s + X_v \cdot \sigma,\end{aligned}$$

with $\psi_s, Z_s \in \mathbb{C}$, $\psi_v, Z_v \in \mathbb{C}^3$, $X_s \in \mathbb{R}$, $X_v \in \mathbb{R}^3$, and with operator-valued components acting on fields to their right. The three involutions on the state are

$$\begin{aligned}\psi^* &= \psi_s^* - \psi_v^* \cdot \sigma, \\ \psi^R &= \psi_s^* + \psi_v^* \cdot \sigma, \\ \psi^{*R} &= \psi_s - \psi_v \cdot \sigma.\end{aligned}$$

The basic component products are collected below.

Table 27 records the direct action of a complex paravector operator on a complex paravector state.

Table 27. Components of $O\psi$.

scalar:	$O_s \psi_s + O_v \cdot \psi_v$
vector:	$O_s \psi_v + O_v \psi_s + i(O_v \times \psi_v)$

Tables 28–30 give the three basic bilinear sandwiches with a generic complex paravector $Z = Z_s + Z_v \cdot \sigma$.

Table 28. Components of $\psi^* Z \psi$.

scalar:	$Z_s(\psi_s ^2 - \psi_v^* \cdot \psi_v)$ $+ \psi_s^*(Z_v \cdot \psi_v) - \psi_s(Z_v \cdot \psi_v^*)$ $- i \psi_v^* \cdot (Z_v \times \psi_v)$
vector:	$Z_s(\psi_s^* \psi_v - \psi_s \psi_v^* - i \psi_v^* \times \psi_v)$ $+ (\psi_s ^2 + \psi_v^* \cdot \psi_v) Z_v$ $- (Z_v \cdot \psi_v^*) \psi_v - (Z_v \cdot \psi_v) \psi_v^*$ $+ i(\psi_s^* Z_v \times \psi_v + \psi_s Z_v \times \psi_v^*)$

Table 29. Components of $\psi^R Z \psi$.

scalar:	$Z_s(\psi_s ^2 + \psi_v^* \cdot \psi_v)$ $+ \psi_s^*(Z_v \cdot \psi_v) + \psi_s(Z_v \cdot \psi_v^*)$ $+ i \psi_v^* \cdot (Z_v \times \psi_v)$
vector:	$Z_s(\psi_s^* \psi_v + \psi_s \psi_v^* + i \psi_v^* \times \psi_v)$ $+ (\psi_s ^2 - \psi_v^* \cdot \psi_v) Z_v$ $+ (Z_v \cdot \psi_v^*) \psi_v + (Z_v \cdot \psi_v) \psi_v^*$ $+ i(\psi_s^* Z_v \times \psi_v - \psi_s Z_v \times \psi_v^*)$

Table 30. Components of $\psi^{*R} Z \psi$.

scalar:	$Z_s(\psi_s^2 - \psi_v \cdot \psi_v)$
vector:	$(\psi_s^2 + \psi_v \cdot \psi_v) Z_v$ $- 2(Z_v \cdot \psi_v) \psi_v + 2i \psi_s Z_v \times \psi_v$

Specializing to a real paravector $X = X_s + X_v \cdot \sigma$, with $X_s \in \mathbb{R}$ and $X_v \in \mathbb{R}^3$, gives Tables 31–33.

Table 31. Components of $\psi^* X \psi$.

scalar:	$X_s(\psi_s ^2 - \psi_v^* \cdot \psi_v)$ $+ \psi_s^*(X_v \cdot \psi_v) - \psi_s(X_v \cdot \psi_v^*)$ $- i \psi_v^* \cdot (X_v \times \psi_v)$
vector:	$X_s(\psi_s^* \psi_v - \psi_s \psi_v^* - i \psi_v^* \times \psi_v)$ $+ (\psi_s ^2 + \psi_v^* \cdot \psi_v) X_v$ $- (X_v \cdot \psi_v^*) \psi_v - (X_v \cdot \psi_v) \psi_v^*$ $+ i(\psi_s^* X_v \times \psi_v + \psi_s X_v \times \psi_v^*)$

Table 32. Components of $\psi^R X \psi$.

scalar:	$X_s(\psi_s ^2 + \psi_v^* \cdot \psi_v)$ $+ \psi_s^*(X_v \cdot \psi_v) + \psi_s(X_v \cdot \psi_v^*)$ $+ i \psi_v^* \cdot (X_v \times \psi_v)$
vector:	$X_s(\psi_s^* \psi_v + \psi_s \psi_v^* + i \psi_v^* \times \psi_v)$ $+ (\psi_s ^2 - \psi_v^* \cdot \psi_v) X_v$ $+ (X_v \cdot \psi_v^*) \psi_v + (X_v \cdot \psi_v) \psi_v^*$ $+ i(\psi_s^* X_v \times \psi_v - \psi_s X_v \times \psi_v^*)$

Table 33. Components of $\psi^{*R} X \psi$.

scalar:	$X_s(\psi_s^2 - \psi_v \cdot \psi_v)$
vector:	$(\psi_s^2 + \psi_v \cdot \psi_v) X_v$ $- 2(X_v \cdot \psi_v) \psi_v + 2i \psi_s X_v \times \psi_v$

For a scalar function S , the same three sandwiches reduce to Tables 34–36.

Table 34. Components of $\psi^* S \psi$.

scalar:	$S(\psi_s ^2 - \psi_v^* \cdot \psi_v)$
vector:	$S(\psi_s^* \psi_v - \psi_s \psi_v^* - i \psi_v^* \times \psi_v)$

Table 35. Components of $\psi^R S \psi$.

scalar:	$S(\psi_s ^2 + \psi_v^* \cdot \psi_v)$
vector:	$S(\psi_s^* \psi_v + \psi_s \psi_v^* + i \psi_v^* \times \psi_v)$

Table 36. Components of $\psi^{*R} S \psi$.

scalar:	$S(\psi_s^2 - \psi_v \cdot \psi_v)$
vector:	0

For $S \in \mathbb{R}$, the reverse sandwich is

$$\psi^R S \psi = S \psi^R \psi,$$

whose scalar part is

$$\rho_\psi := \text{scalar}(\psi^R \psi) = |\psi_s|^2 + \psi_v^* \cdot \psi_v \geq 0.$$

In the matrix realization scalar extraction is one half of the trace, so on a selected column the conventional column norm is $2\rho_\psi$. The factor is shown explicitly below when the physical Dirac density is defined. This positive scalar supplies the time-slice

Hilbert norm. The vector part of $\psi^R\psi$, however, is not automatically a spatial probability current. In fact, under the active spinor convention $\psi' = R^R\psi$, it is invariant:

$$\psi'^R\psi' = \psi^R R R^R \psi = \psi^R\psi.$$

Spatial spin or polarization information instead occurs in bilinears with the opposite order, such as $\psi\psi^R$, or in operator sandwiches fixed by the field equation. More generally, $\psi^R O \psi$ contains four complex, hence eight real, coefficient channels before an extraction is chosen. No generic physical interpretation is assigned to its nonscalar channels.

At fixed time, the algebra-block normalization is

$$\langle \psi | \psi \rangle := \int_{\mathbb{R}^3} \text{scalar}(\psi^R \psi) d^3r = \int_{\mathbb{R}^3} \rho_\psi d^3r = 1.$$

For a complex-paravector operator O , define

$$\langle \psi | O | \psi \rangle := \int_{\mathbb{R}^3} \text{scalar}(\psi^R O \psi) d^3r.$$

If the state is not already normalized, the corresponding expectation map is

$$\mathbb{E}_\psi(O) := \frac{\int_{\mathbb{R}^3} \text{scalar}(\psi^R O \psi) d^3r}{\int_{\mathbb{R}^3} \text{scalar}(\psi^R \psi) d^3r}.$$

If $O\psi = \phi_s + \phi_v \cdot \sigma$, then the extracted scalar channel is simply

$$\text{scalar}(\psi^R O \psi) = \psi_s^* \phi_s + \psi_v^* \cdot \phi_v.$$

When O is reverse-self-adjoint under the spatial integral and the boundary terms are benign, this scalar expectation is real. For the Dirac field the physical norm in the conventional column normalization used below is twice the sum of this expression over its two restricted blocks; the field equation below supplies the associated conserved spatial current.

By contrast,

$$\psi^{*R}\psi = \psi_s^2 - \psi_v \cdot \psi_v = \langle\langle\psi\rangle\rangle^2$$

is the determinant bilinear. It is algebraically useful, but it is not positive-definite and so it should not be used as the probabilistic normalization. In this framework the single bra-ket bracket $\langle \cdot | \cdot \rangle$ is therefore best reserved for the scalar expectation map built from reversion, while $\langle\langle \cdot \rangle\rangle^2$ continues to denote the determinant form.

6.6.4 Carrier Spaces, Pairings, and Spin Assignment

This subsection sharpens the interpretation of the carrier spaces already used above and connects them to the projector, commutator, and adjoint formulas in Eqs. (50), (35), and (126). The goal is to keep the algebraic carrier, the physical spin assignment, and the observable bilinears conceptually separate. Standard physics does not assign spin from component count alone. The physically relevant data are

(transformation law, field equation, constraints).

In standard Lorentz language,

$$\begin{aligned} \text{spin } 0 &: (0, 0), \\ \text{spin } \frac{1}{2} &: (\frac{1}{2}, 0) \oplus (0, \frac{1}{2}), \\ \text{spin } 1 &: (1, 0) \text{ or } (0, 1) \quad \text{for one chiral } F, \\ &: (1, 0) \oplus (0, 1) \quad \text{for } F \text{ together with its conjugate,} \\ &: (\frac{1}{2}, \frac{1}{2}) \quad \text{for } A^\mu. \end{aligned}$$

So the algebraic carrier spaces

$$\mathbb{C}, \quad \mathbb{C} \oplus \mathbb{C}^3, \quad \{0\} \oplus \mathbb{C}^3$$

do not determine the physical spin by themselves. They become spin sectors only after the operator law is fixed.

Bilinear pairings. The reverse product supplies the positive scalar norm

$$\rho_\psi := \text{scalar}(\psi^R \psi) = |\psi_s|^2 + \psi_v^* \cdot \psi_v.$$

Its vector part is an algebraic column-space channel, not a universal space-time current. Currents are identified only after a field equation and transformation law have been fixed. For two fields $\phi, \psi \in \mathbb{C} \oplus \mathbb{C}^3$, the three scalar pairings are

$$\begin{aligned} \text{scalar}(\phi^R \psi) &= \phi_s^* \psi_s + \phi_v^* \cdot \psi_v && \text{Hermitian-type pairing,} \\ \text{scalar}(\phi^{*R} \psi) &= \phi_s \psi_s - \phi_v \cdot \psi_v && \text{metric pairing,} \\ \text{scalar}(\phi^* \psi) &= \phi_s^* \psi_s - \phi_v^* \cdot \psi_v && \text{indefinite companion.} \end{aligned}$$

For $\phi = \psi$, the metric pairing reduces to

$$\text{scalar}(\psi^{*R} \psi) = \psi^{*R} \psi = \langle\langle\psi\rangle\rangle^2.$$

Only the reverse pairing is being used here as the quantum bracket pairing. The determinant pairing is the intrinsic quadratic form used in factorization identities, and the companion pairing is algebraically well-defined but does not carry an independent standard quantum interpretation here.

Accordingly, for the spatially integrated bracket

$$\langle \phi | \psi \rangle := \int_{\mathbb{R}^3} \text{scalar}(\phi^R \psi) d^3r, \quad \langle \phi | O \psi \rangle = \langle O^\dagger \phi | \psi \rangle, \quad (126)$$

the operator adjoint $O^\dagger$ is the reverse-adjoint determined by this pairing, up to the usual boundary terms when derivatives are present. For left multiplication by a fixed paravector Z , the adjoint is left multiplication by Z^R , since $(Z^R \phi)^R = \phi^R Z$. Nonscalar parts of $\psi^R O \psi$ remain available as algebraic bilinears, but a current, flux, or polarization interpretation must be established separately for the chosen operator and field equation.

Generalized operator products. The following identities are algebraic. They describe operator composition on a paravector-valued state, but by themselves they do not assign a physical spin.

For operator-valued components, the commutator no longer collapses to the pure cross-product law of Eq. (35). Write

$$O = O_s + O_v \cdot \sigma, \quad O' = O'_s + O'_v \cdot \sigma,$$

with component operators commuting with the fixed basis elements and acting on the state to their right. Then, with dot and cross products taken componentwise and in the written operator order,

$$\begin{aligned} \text{com}(O, O') \psi &= \left(\text{com}(O_s, O'_s) + O_v \cdot O'_v - O'_v \cdot O_v \right) \psi \\ &+ \left(\text{com}(O_s, O'_v) + \text{com}(O_v, O'_s) \right. \\ &\quad \left. + i(O_v \times O'_v - O'_v \times O_v) \right) \cdot \sigma \psi, \end{aligned} \quad (127)$$

where the scalar-vector commutators are understood componentwise. Unlike the purely algebraic paravector commutator, Eq. (127) generally has both scalar and vector parts. Only when all component operators mutually commute does it reduce to the simpler law

$$\text{com}(O, O') \psi = 2i(O_v \times O'_v) \cdot \sigma \psi.$$

The matching anticommutator is

$$\begin{aligned} \text{acom}(O, O') \psi &= (OO' + O'O) \psi \\ &= \left(\text{acom}(O_s, O'_s) + O_v \cdot O'_v + O'_v \cdot O_v \right) \psi \\ &+ \left(\text{acom}(O_s, O'_v) \right. \\ &\quad \left. + \text{acom}(O_v, O'_s) \right. \\ &\quad \left. + i(O_v \times O'_v + O'_v \times O_v) \right) \cdot \sigma \psi, \end{aligned} \quad (128)$$

so that

$$OO' = \frac{1}{2}(\text{acom}(O, O') + \text{com}(O, O')).$$

This is the natural companion to the operator commutator formula and is also the algebraic starting point for covariance and uncertainty identities once a state, expectation functional, and operator domain have been specified.

Carrier closure. The same algebra shows which carrier spaces are actually closed under which operators:

$$\begin{aligned} \partial : \mathbb{C} &\rightarrow \mathbb{C} \oplus \mathbb{C}^3, \\ \partial : \{0\} \oplus \mathbb{C}^3 &\rightarrow \mathbb{C} \oplus \mathbb{C}^3, \\ \partial : \mathbb{C} \oplus \mathbb{C}^3 &\rightarrow \mathbb{C} \oplus \mathbb{C}^3, \\ \partial : (\mathbb{C} \oplus \mathbb{C}^3)\Pi_0 &\rightarrow (\mathbb{C} \oplus \mathbb{C}^3)\Pi_0. \end{aligned} \quad (129)$$

Indeed,

$$\begin{aligned} \partial \psi_s &= (\partial_t \psi_s) + (\nabla \psi_s) \cdot \sigma, \\ \partial(\psi_v \cdot \sigma) &= (\nabla \cdot \psi_v) + (\partial_t \psi_v + i \nabla \times \psi_v) \cdot \sigma. \end{aligned}$$

The full complex-paravector space is the smallest algebra containing the scalar and vector channels, but it is not the smallest left module. The minimal left ideal $(\mathbb{C} \oplus \mathbb{C}^3)\Pi_0$, selected by a fixed right factor, is a two-complex-dimensional left module and

is also closed because every differential and potential operator acts on the left. The scalar subspace closes only after passing to scalar second-order combinations such as $\square + m^2$. A pure-vector specialization is algebraically meaningful, but it is not preserved by generic first-order paravector action and therefore requires Maxwell- or Proca-type constraints if it is to describe a spin-1 field. A real-paravector specialization is not a general quantum carrier because it is not closed under multiplication by the phase $\exp(i\alpha)$.

With those algebraic preliminaries fixed, the organizing principle is this: one first chooses a carrier field for ψ , then restricts to the operators that preserve that carrier or define its field equation, and then lists the bilinear pairings that extract the physically relevant observables from that carrier. The bilinears need not take values in the same carrier field; in general they land in scalar, vector, or paravector observable channels.

With that understood, the physically consistent sector-by-sector summary is the following.

1. Spin 0: complex scalar Klein–Gordon sector

carrier:	$\psi : \mathbb{R}^{1,3} \rightarrow \mathbb{C}$,
representation:	$(0, 0)$,
field equation:	$(\square + m^2)\psi = 0$,
operator class:	$O_s : \mathbb{C} \rightarrow \mathbb{C}$,
examples:	$\square + m^2, \partial_t \pm i\Omega,$ $-E^2 + \mathbf{P}^2 + m^2, \dots$,
spin projector:	$\Pi(\pm)\psi \notin \mathbb{C}$.

The scalar pairings reduce to

$$\begin{aligned} \text{scalar}(\phi^R \psi) &= \phi^* \psi, \\ \text{scalar}(\phi^{*R} \psi) &= \phi \psi, \\ \text{scalar}(\phi^* \psi) &= \phi^* \psi. \end{aligned}$$

So the reverse and sigma-conjugate pairings coincide on the scalar sector, while the metric pairing is different. The local quantity $|\psi|^2$ is positive but is not the time component of the conserved relativistic current. For $m > 0$, the derivative-built current can be kept in scalar/vector form using the covariant operators already defined:

$$\begin{aligned} \rho_{\text{KG}} &:= \frac{1}{2m} (\psi^* E \psi + (E \psi)^* \psi), \\ \mathbf{j}_{\text{KG}} &:= \frac{1}{2m} (\psi^* \mathbf{P} \psi + (\mathbf{P} \psi)^* \psi), \end{aligned}$$

$$\partial_t \rho_{\text{KG}} + \nabla \cdot \mathbf{j}_{\text{KG}} = 0.$$

For the free field set $q = 0$. The density is conserved but is not positive on the full positive- and negative-frequency solution space. A one-particle interpretation therefore requires a frequency restriction, while quantum field theory interprets the two branches as particle and antiparticle modes.

2. Spin $\frac{1}{2}$: Dirac/Pauli sector

algebraic block: $\psi : \mathbb{R}^{1,3} \rightarrow \mathbb{C} \oplus \mathbb{C}^3$,

physical field: $\Psi = \begin{Bmatrix} \Psi_1 \\ \Psi_2 \end{Bmatrix}$, $\Psi_n \in (\mathbb{C} \oplus \mathbb{C}^3)\Pi_0$,

representation: $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$,

field equations: $(\mathbf{W}(i\partial) - m\mathbf{I})\Psi = 0$,
 $(\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi = 0$,

operator class: $O = O_s + O_v \cdot \sigma$,
 acting on each paravector block,

examples: ∂, ∂^* ,
 $E \pm \sigma \cdot \mathbf{P}, \Pi(\pm)$,

spin projector: $\Pi(\pm)$ acts from the left on each block.

The basic one-block pairings are

$$\begin{aligned} \text{scalar}(\phi^R \psi) &= \phi_s^* \psi_s + \phi_v^* \cdot \psi_v, \\ \text{scalar}(\phi^* \psi) &= \phi_s \psi_s - \phi_v \cdot \psi_v, \\ \text{scalar}(\phi^* \psi) &= \phi_s^* \psi_s - \phi_v^* \cdot \psi_v. \end{aligned}$$

For the two-block Dirac carrier, the natural extension is

$$\langle \{\Xi_1; \Xi_2\} \mid \{\Psi_1; \Psi_2\} \rangle := 2 \int_{\mathbb{R}^3} \sum_{n=1}^2 \text{scalar}(\Xi_n^R \Psi_n) d^3r.$$

The reverse pairing is the positive Hermitian-type pairing used for time-slice expectations. The metric pairing is the algebraic form used in identities such as ZZ^* , and the companion pairing is indefinite. The minimal-ideal restriction makes each block one Weyl spinor. Without it, the two matrix columns give two independent copies of every displayed Dirac and Pauli equation.

The coupled first-order equation gives the conserved Dirac density and current directly:

$$\begin{aligned} \rho_D &:= 2 \text{scalar}(\Psi_1^R \Psi_1 + \Psi_2^R \Psi_2), \\ \mathbf{j}_D &:= 2 \text{vector}(\Psi_2 \Psi_2^R - \Psi_1 \Psi_1^R), \\ \partial_t \rho_D + \nabla \cdot \mathbf{j}_D &= 0. \end{aligned}$$

The opposite product order in $\mathbf{j}_D$ is essential: it is the order that transforms as a spatial vector under left spinor transformations.

3. Spin 1: vector or field-strength sector

carrier: $F : \mathbb{R}^{1,3} \rightarrow \{0\} \oplus \mathbb{C}^3$,

main example: $F = (\mathbf{E} + i\mathbf{B}) \cdot \sigma$,

representation: $(1, 0)$ or $(0, 1)$ for F ,
 $(1, 0) \oplus (0, 1)$ for F and its conjugate,
 $(\frac{1}{2}, \frac{1}{2})$ for a vector potential A^μ ,

field equation: $\partial F = D^*$,

operator class: $\partial, \partial^*, \square$, and Maxwell/Proca constraints,

closure facts: $\partial, \partial^* : \{0\} \oplus \mathbb{C}^3 \rightarrow \mathbb{C} \oplus \mathbb{C}^3$,
 $\square : \{0\} \oplus \mathbb{C}^3 \rightarrow \{0\} \oplus \mathbb{C}^3$,

spin projector: $\Pi(\pm)F \notin \{0\} \oplus \mathbb{C}^3$.

So the two-projector split $\Pi(\pm)$ is not the standard spin-1 decomposition. The standard polarization counts are

$$\begin{aligned} \text{massive spin 1 : } m_s &= -1, 0, 1, \\ \text{massless spin 1 : } \lambda &= \pm 1. \end{aligned}$$

The bilinear quantities already present in the manuscript are

$$\begin{aligned} \frac{1}{2}F^2 &= \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) + i\mathbf{E} \cdot \mathbf{B}, \\ \frac{1}{2}F F^R &= \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \sigma. \end{aligned}$$

These have direct standard physical interpretations: $\frac{1}{2}F^2$ packages the Lorentz invariants, while $\frac{1}{2}F F^R$ packages the energy density and the Poynting vector. A simple bra-ket built from $F^R F$ is not the standard Maxwell-field pairing used here; the physically relevant quantities are the field bilinears together with the field equations and their gauge or subsidiary constraints.

The consistent summary is therefore this:

spin $\neq$ component count alone,
 spin = (carrier, operator law, physical bilinears, constraints).

Within that summary, $(\cdot)^R$ is the pairing used for quantum expectations, $(\cdot)^{*R}$ supplies the determinant pairing used for algebraic factorization, and the pure sigma-conjugate pairing remains an auxiliary companion.

6.6.5 Spin-0 Wave Mechanics

The scalar relativistic wave equation can be read directly from the mass-shell identity developed earlier in the kinematic section. Here $\psi \in \mathbb{C}$ is a complex scalar field, $m \in \mathbb{R}$, and the free-particle operators are defined using the differential conventions from Section 4. Start from

$$\langle mU \rangle^2 = m^2 \quad \Rightarrow \quad -E^2 + \mathbf{P}^2 + m^2 = 0.$$

For a free particle,

$$E = i\partial_t, \quad \mathbf{P} = -i\nabla,$$

and equivalently

$$E + \mathbf{P} \cdot \sigma = i\partial^*.$$

Acting on a scalar field ψ , the operator chain is

$$\begin{aligned} (-E^2 + \mathbf{P}^2 + m^2)\psi &= (-(i\partial_t)^2 - \nabla^2 + m^2)\psi \\ &= (\partial_t^2 - \nabla^2 + m^2)\psi \\ &= (\partial\partial^* + m^2)\psi \\ &= (\square + m^2)\psi. \end{aligned}$$

Thus the free spin-0 relativistic wave equation is

$$(\square + m^2)\psi = 0 \quad \text{Klein-Gordon equation.}$$

This is the scalar wave equation recorded in Eq. (103). The dispersion relation $\omega^2 = \mathbf{k}^2 + m^2$ and the later positive/negative-frequency split are consequences of this same operator identity.

6.6.6 Positive and Negative Frequency Sectors

The frequency decomposition is one of the central interpretive steps in the free scalar theory. On a spatial domain where the positive square root is defined, set

$$\begin{aligned}\Omega &= \sqrt{m^2 - \nabla^2} \\ &= \langle\langle m + \boldsymbol{\sigma} \cdot \nabla \rangle\rangle.\end{aligned}$$

Accordingly,

$$\begin{aligned}\square + m^2 &= \partial_t^2 + \Omega^2 \\ &= \partial_t^2 + (m^2 - \nabla^2) \\ &= (\partial_t - i\Omega)(\partial_t + i\Omega) \\ &= (\partial_t + i\Omega)(\partial_t - i\Omega).\end{aligned}$$

The positive- and negative-frequency sectors are then defined by

$$\begin{aligned}(\partial_t \pm i\Omega)\psi(\pm) &= 0, \\ \psi(\pm, t) &= \exp(\mp i\Omega(t - t_0))\psi(\pm, t_0), \\ \psi &= \psi(+) + \psi(-).\end{aligned}$$

Here the parenthesized sign labels frequency and is not the spin-axis label in $\Pi(\pm)\psi$. When Ω^{-1} exists, the two parts of a Klein–Gordon solution at any time can equivalently be extracted as

$$\psi(\pm) = \frac{1}{2} (\psi \pm i\Omega^{-1}\partial_t\psi).$$

For a massless field the spatial zero mode is treated separately. The split separates the two frequency branches; it does not turn the negative-frequency part into a negative probability. In a relativistic quantum field, the two branches acquire the particle–antiparticle interpretation [16].

6.6.7 Fourier Meaning of the Frequency Operator

The nonlocal square root Ω is most transparent one spatial Fourier mode at a time. With the standard phase convention, write schematically

$$\begin{aligned}\psi &= \sum_n c_n f_n, \\ f_n &= \exp(-i\omega_n t + i\mathbf{k}_n \cdot \mathbf{r}).\end{aligned}$$

Then

$$\begin{aligned}\square\psi &= \sum_n (\mathbf{k}_n^2 - \omega_n^2) c_n f_n, \\ (\square + m^2)\psi &= \sum_n (\mathbf{k}_n^2 + m^2 - \omega_n^2) c_n f_n, \\ \Omega\psi &= \sum_n \sqrt{m^2 + \mathbf{k}_n^2} c_n f_n.\end{aligned}$$

For on-shell Klein–Gordon modes,

$$\mathbf{k}_n^2 + m^2 = \omega_n^2, \quad \omega_n = \pm \sqrt{\mathbf{k}_n^2 + m^2},$$

and Ω multiplies each spatial mode by its positive energy magnitude. This defines the only square-root operator needed in the frequency split and avoids assigning an unnecessary branch to $\sqrt{\square}$.

6.6.8 Klein–Gordon with Potential Fields

To keep the minimally coupled scalar theory in one self-contained place, start from the charged-particle Lagrangian

$$\mathcal{L} = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}).$$

Here E denotes the energy operator, not the electric field $\mathbf{E}$. The Hamiltonian and canonical momentum are then

$$i\partial_t = \mathcal{H} = E + qV, \quad -i\nabla = \partial_{\mathbf{v}}\mathcal{L} = \mathbf{P} + q\mathbf{A},$$

so the covariant energy and momentum operators are

$$E = i\partial_t - qV, \quad \mathbf{P} = -i\nabla - q\mathbf{A}.$$

Acting on a smooth test field ψ , the Leibniz rules needed below are

$$\begin{aligned}\partial_t(V\psi) &= (\partial_t V)\psi + V\partial_t\psi, \\ \partial_m(A_n\psi) &= (\partial_m A_n)\psi + A_n\partial_m\psi.\end{aligned}$$

From these one obtains

$$\begin{aligned}E^2\psi &= E(i\partial_t\psi - qV\psi) \\ &= (-\partial_t^2 - iq\partial_t V - 2iqV\partial_t + q^2 V^2)\psi, \\ \mathbf{P}^2\psi &= \mathbf{P} \cdot (-i\nabla\psi - q\mathbf{A}\psi) \\ &= (-\nabla^2 + iq\nabla \cdot \mathbf{A} + 2iq\mathbf{A} \cdot \nabla + q^2 \mathbf{A}^2)\psi.\end{aligned}$$

Recall also

$$\begin{aligned}\square &= \partial_t^2 - \nabla^2, \\ \Phi &= V + \mathbf{A} \cdot \boldsymbol{\sigma}, \\ S &= \partial_t V + \nabla \cdot \mathbf{A}, \\ \langle\langle \partial \rangle\rangle^2 &= \square, \\ \langle\langle \Phi \rangle\rangle^2 &= V^2 - \mathbf{A}^2.\end{aligned}$$

Substituting the operator expansions gives the minimally coupled Klein–Gordon equation in explicit form:

$$\begin{aligned}0 &= (-E^2 + \mathbf{P}^2 + m^2)\psi \\ &= \left(\square + iq(\partial_t V + \nabla \cdot \mathbf{A}) + 2iq(V\partial_t + \mathbf{A} \cdot \nabla) \right. \\ &\quad \left. - q^2(V^2 - \mathbf{A}^2) + m^2 \right)\psi \\ &= \left(\square + iqS - q^2\langle\langle \Phi \rangle\rangle^2 + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \nabla) \right)\psi.\end{aligned}$$

The electromagnetic fields and commutators enter through

$$\begin{aligned}\mathbf{E} &= -\partial_t \mathbf{A} - \nabla V, \\ \mathbf{B} &= \nabla \times \mathbf{A}, \\ F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma},\end{aligned}$$

and

$$\begin{aligned}
\text{com}(E, P_m) \psi &= (i\partial_t - qV)(-i\partial_m \psi - qA_m \psi) \\
&\quad - (-i\partial_m - qA_m)(i\partial_t \psi - qV\psi) \\
&= -iq((\partial_t A_m)\psi + A_m \partial_t \psi) \\
&\quad - iq((\partial_m V)\psi + V\partial_m \psi) \\
&\quad + iqV\partial_m \psi + iqA_m \partial_t \psi \\
&= -iq(\partial_t A_m + \partial_m V)\psi \\
&= iqE_m \psi, \\
\text{com}(P_m, P_n) \psi &= (-i\partial_m - qA_m)(-i\partial_n \psi - qA_n \psi) \\
&\quad - (-i\partial_n - qA_n)(-i\partial_m \psi - qA_m \psi) \\
&= iq(\partial_m A_n - \partial_n A_m)\psi.
\end{aligned}$$

Hence

$$\text{com}(E, \mathbf{P}) = iq\mathbf{E}, \quad \mathbf{P} \times \mathbf{P} = iq\mathbf{B}.$$

Substituting those commutators into the ordered products yields

$$\begin{aligned}
(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) &= E^2 - \mathbf{P}^2 \\
&\quad + \boldsymbol{\sigma} \cdot (-\text{com}(E, \mathbf{P}) - i\mathbf{P} \times \mathbf{P}) \\
&= E^2 - \mathbf{P}^2 - iqF, \\
(E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P}) &= E^2 - \mathbf{P}^2 \\
&\quad + \boldsymbol{\sigma} \cdot (\text{com}(E, \mathbf{P}) - i\mathbf{P} \times \mathbf{P}) \\
&= E^2 - \mathbf{P}^2 - iqF^*.
\end{aligned}$$

Since

$$E + \boldsymbol{\sigma} \cdot \mathbf{P} = i\partial^* - q\Phi, \quad E - \boldsymbol{\sigma} \cdot \mathbf{P} = i\partial - q\Phi^* = (E + \boldsymbol{\sigma} \cdot \mathbf{P})^{*\text{R}},$$

the two ordered reverse-conjugate products are

$$(E \pm \boldsymbol{\sigma} \cdot \mathbf{P})(E \pm \boldsymbol{\sigma} \cdot \mathbf{P})^{*\text{R}} = E^2 - \mathbf{P}^2 + \boldsymbol{\sigma} \cdot (\mp \text{com}(E, \mathbf{P}) - i\mathbf{P} \times \mathbf{P}).$$

Unlike the determinant form defined for commuting complex coefficients, these operator products retain a vector part. Operator ordering therefore matters, and the two first-order products remain distinct. Consequently,

$$\begin{aligned}
-(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P})^{*\text{R}} &= (\partial^* + iq\Phi)(\partial + iq\Phi^*), \\
-(E + \boldsymbol{\sigma} \cdot \mathbf{P})^{*\text{R}}(E + \boldsymbol{\sigma} \cdot \mathbf{P}) &= (\partial + iq\Phi^*)(\partial^* + iq\Phi).
\end{aligned}$$

The same factorizations can be rewritten directly in derivative form:

$$\begin{aligned}
(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P})\psi &= -(\partial^* + iq\Phi)(\partial + iq\Phi^*)\psi \\
&= -\square\psi - iq\partial^*(\Phi^*\psi) \\
&\quad - iq\Phi\partial\psi \\
&\quad + q^2\langle\langle\Phi\rangle\rangle^2\psi \\
&= -\square\psi - iqS\psi + q^2\langle\langle\Phi\rangle\rangle^2\psi \\
&\quad - 2iq(V\partial_t + \mathbf{A} \cdot \boldsymbol{\nabla})\psi \\
&\quad - iqF\psi, \\
(E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P})\psi &= -(\partial + iq\Phi^*)(\partial^* + iq\Phi)\psi
\end{aligned}$$

$$\begin{aligned}
&= -\square\psi - iq\partial(\Phi\psi) \\
&\quad - iq\Phi^*\partial^*\psi \\
&\quad + q^2\langle\langle\Phi\rangle\rangle^2\psi \\
&= -\square\psi - iqS\psi + q^2\langle\langle\Phi\rangle\rangle^2\psi \\
&\quad - 2iq(V\partial_t + \mathbf{A} \cdot \boldsymbol{\nabla})\psi \\
&\quad - iqF^*\psi.
\end{aligned}$$

Equivalently,

$$\begin{aligned}
\partial^*(\Phi^*\psi) + \Phi\partial\psi &= S\psi + 2(V\partial_t + \mathbf{A} \cdot \boldsymbol{\nabla})\psi + F\psi, \\
\partial(\Phi\psi) + \Phi^*\partial^*\psi &= S\psi + 2(V\partial_t + \mathbf{A} \cdot \boldsymbol{\nabla})\psi + F^*\psi,
\end{aligned}$$

and their difference isolates the electric contribution:

$$\begin{aligned}
(\partial^*(\Phi^*\psi) + \Phi\partial\psi) - (\partial(\Phi\psi) + \Phi^*\partial^*\psi) \\
&= ((\partial\Phi)^* - \partial\Phi)\psi \\
&= (F - F^*)\psi \\
&= 2(\mathbf{E} \cdot \boldsymbol{\sigma})\psi.
\end{aligned}$$

In summary,

$$\begin{aligned}
E &:= i\partial_t - qV, & \mathbf{P} &:= -i\boldsymbol{\nabla} - q\mathbf{A}, \\
\partial &:= \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, & \partial^* &:= \partial_t - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, \\
\Phi &:= V + \mathbf{A} \cdot \boldsymbol{\sigma}, & F &:= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\
S &:= \partial_t V + \boldsymbol{\nabla} \cdot \mathbf{A}, & \mathbf{E} &:= -\partial_t \mathbf{A} - \boldsymbol{\nabla} V, \\
\mathbf{B} &:= \boldsymbol{\nabla} \times \mathbf{A},
\end{aligned}$$

and the scalar Klein–Gordon operator can be recovered from either ordered first-order product:

$$\begin{aligned}
0 &= (-E^2 + \mathbf{P}^2 + m^2)\psi \\
&= \left(\square + iqS - q^2\langle\langle\Phi\rangle\rangle^2 + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \boldsymbol{\nabla})\right)\psi \\
&= (\partial^* + iq\Phi)(\partial + iq\Phi^*)\psi - iqF\psi + m^2\psi \\
&= (\partial + iq\Phi^*)(\partial^* + iq\Phi)\psi - iqF^*\psi + m^2\psi.
\end{aligned}$$

The ordered first-order products themselves obey

$$\begin{aligned}
(E \pm \boldsymbol{\sigma} \cdot \mathbf{P})(E \mp \boldsymbol{\sigma} \cdot \mathbf{P}) &= E^2 - (\boldsymbol{\sigma} \cdot \mathbf{P})^2 \mp \text{com}(E, \boldsymbol{\sigma} \cdot \mathbf{P}) \\
&= E^2 - \mathbf{P}^2 - iq(\pm\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}.
\end{aligned}$$

Taking the scalar part therefore recovers

$$0 = \text{scalar}((\partial + iq\Phi^*)(\partial^* + iq\Phi))\psi + m^2\psi.$$

Only the scalar part reproduces the Klein–Gordon operator itself; the full ordered products retain the non-scalar field terms $-iqF$ and $-iqF^*$ that distinguish the two factorizations.

6.6.9 Spin- $\frac{1}{2}$ Wave Mechanics

Before the electromagnetic field is turned on, the same auxiliary 2×2 map from Eq. (74) gives a self-contained first-order version of the relativistic mass shell. Consider the two-block field

$$\Psi = \begin{Bmatrix} \Psi_1 \\ \Psi_2 \end{Bmatrix}.$$

For one Dirac field both blocks obey $\Psi_n = \Psi_n \Pi_0$. If they are left unrestricted, every formula below acts independently on the two matrix columns and therefore describes two identical Dirac fields. For any complex paravector or paravector-valued operator Z , define

$$\mathbf{W}(Z) := \begin{bmatrix} 0 & Z \\ Z^{*\mathbf{R}} & 0 \end{bmatrix},$$

$$\mathbf{W}(Z)^2 = \text{diag}(ZZ^{*\mathbf{R}}, Z^{*\mathbf{R}}Z).$$

The map is complex-linear, so in particular

$$\mathbf{W}(i\partial) = i \begin{bmatrix} 0 & \partial \\ \partial^* & 0 \end{bmatrix} = i\mathbf{W}(\partial),$$

$$\mathbf{W}(i\partial)^2 = -\text{diag}(\partial\partial^*, \partial^*\partial)$$

$$= -\square \mathbf{I}.$$

Define the free Dirac operator matrix and its mass conjugate by

$$\mathbf{D} := \mathbf{W}(i\partial) - m\mathbf{I}, \quad \mathbf{D}' := \mathbf{W}(i\partial) + m\mathbf{I}.$$

Then

$$\mathbf{D}'\mathbf{D} = \mathbf{D}\mathbf{D}' = (\mathbf{W}(i\partial) + m\mathbf{I})(\mathbf{W}(i\partial) - m\mathbf{I})$$

$$= \mathbf{W}(i\partial)^2 - m^2\mathbf{I}$$

$$= -(\square + m^2)\mathbf{I}.$$

Thus the free Dirac equation

$$\mathbf{D}\Psi = 0$$

forces each block to satisfy the Klein–Gordon operator. No determinant of algebra-valued blocks is needed: Eq. (75) supplies the relevant minimal polynomial directly.

The equivalence with the usual Weyl form is explicit in the Pauli matrix realization. For a plane wave

$$\exp(-iEt + i\mathbf{p} \cdot \mathbf{r})$$

and one selected matrix column ξ_n of each block, the two equations are

$$(E - \mathbf{p} \cdot \boldsymbol{\sigma})\xi_2 = m\xi_1, \quad (E + \mathbf{p} \cdot \boldsymbol{\sigma})\xi_1 = m\xi_2.$$

Their product gives $E^2 = \mathbf{p}^2 + m^2$. These are precisely the two Weyl equations for one four-component Dirac spinor: the two selected columns carry the conjugate Weyl transformation laws, and their pair realizes $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$. Retaining both matrix columns gives two independent copies of this system.

Writing out $\mathbf{D}\Psi = 0$ gives the two coupled equations

$$i\partial\Psi_2 - m\Psi_1 = 0,$$

$$i\partial^*\Psi_1 - m\Psi_2 = 0.$$

For $m > 0$, solving the first equation for the upper component and substituting into the second yields

$$\Psi_1 = \frac{i\partial\Psi_2}{m},$$

$$0 = i\partial^*\Psi_1 - m\Psi_2$$

$$= -\frac{\partial^*\partial\Psi_2}{m} - m\Psi_2$$

$$= -\frac{(\square + m^2)\Psi_2}{m}.$$

Hence the lower component is a Klein–Gordon solution, and the upper component is obtained from it by one first-order derivative.

Conversely, let ψ' be a scalar Klein–Gordon solution and let $a, b \in (\mathbb{C} \oplus \mathbb{C}^3)\Pi_0$ be constant polarization elements. Apply the mass-conjugate operator to the ideal-valued seed:

$$\Psi := \frac{1}{m}\mathbf{D}' \begin{Bmatrix} \psi'a \\ \psi'b \end{Bmatrix}.$$

Because the constants occur on the right, derivatives act only on ψ' , and the two blocks are

$$\Psi_1 = \psi'a + \frac{i}{m}(\partial\psi')b, \quad \Psi_2 = \frac{i}{m}(\partial^*\psi')a + \psi'b.$$

Using $\mathbf{D}\mathbf{D}' = -(\square + m^2)\mathbf{I}$ gives

$$\mathbf{D}\Psi = -\frac{1}{m}(\square + m^2)\psi' \begin{Bmatrix} a \\ b \end{Bmatrix} = 0.$$

The constants a and b carry the spin polarization; a bare scalar solution alone does not. For example, $a = \Pi_0, b = 0$ gives

$$\Psi = \begin{Bmatrix} \psi'\Pi_0 \\ \frac{i}{m}(\partial^*\psi')\Pi_0 \end{Bmatrix}.$$

This is the first-order lift from the Klein–Gordon mass shell into the chosen Dirac spinor module.

6.6.10 Dirac with Potential Field

Consider again the two-block field

$$\Psi = \begin{Bmatrix} \Psi_1 \\ \Psi_2 \end{Bmatrix}.$$

The one-field interpretation continues to impose $\Psi_n = \Psi_n \Pi_0$; all potential and derivative factors act from the left and preserve the ideal. For any complex paravector or paravector-valued operator Z , define

$$\mathbf{W}(Z) := \begin{bmatrix} 0 & Z \\ Z^{*\mathbf{R}} & 0 \end{bmatrix},$$

$$\mathbf{W}(Z)^2 = \text{diag}(ZZ^{*\mathbf{R}}, Z^{*\mathbf{R}}Z),$$

$$\mathbf{W}(Z)^* = \mathbf{W}(Z^*),$$

$$\mathbf{W}(Z)^{\mathbf{R}} = \mathbf{W}(Z^{\mathbf{R}}).$$

The Dirac operator with potential is

$$\begin{aligned}\mathbf{D} &:= \mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I} \\ &= i\mathbf{W}(\partial) - q\mathbf{W}(\Phi^*) - m\mathbf{I}.\end{aligned}$$

Because

$$\begin{aligned}\mathbf{W}(\partial)^2 &= \square\mathbf{I}, \\ \mathbf{W}(\partial)\mathbf{W}(\Phi^*) &= \begin{bmatrix} 0 & \partial \\ \partial^* & 0 \end{bmatrix} \begin{bmatrix} 0 & \Phi^* \\ \Phi & 0 \end{bmatrix} = \text{diag}(\partial\Phi, \partial^*\Phi^*), \\ \mathbf{W}(\Phi^*)\mathbf{W}(\partial) &= \begin{bmatrix} 0 & \Phi^* \\ \Phi & 0 \end{bmatrix} \begin{bmatrix} 0 & \partial \\ \partial^* & 0 \end{bmatrix} = \text{diag}(\Phi^*\partial^*, \Phi\partial), \\ \mathbf{W}(\Phi^*)^2 &= \begin{bmatrix} 0 & \Phi^* \\ \Phi & 0 \end{bmatrix}^2 = \text{diag}(\Phi^*\Phi, \Phi\Phi^*) \\ &= \langle\langle\Phi\rangle\rangle^2\mathbf{I},\end{aligned}$$

one obtains

$$\begin{aligned}\mathbf{W}(i\partial - q\Phi^*)^2 &= -\mathbf{W}(\partial)^2 \\ &\quad - iq(\mathbf{W}(\partial)\mathbf{W}(\Phi^*) + \mathbf{W}(\Phi^*)\mathbf{W}(\partial)) \\ &\quad + q^2\mathbf{W}(\Phi^*)^2 \\ &= -\square\mathbf{I} - iq \text{diag}(\partial\Phi + \Phi^*\partial^*, \partial^*\Phi^* + \Phi\partial) \\ &\quad + q^2\langle\langle\Phi\rangle\rangle^2\mathbf{I} \\ &= -\text{diag}\left((\partial + iq\Phi^*)(\partial^* + iq\Phi),\right. \\ &\quad \left.(\partial^* + iq\Phi)(\partial + iq\Phi^*)\right).\end{aligned}$$

The Dirac equation

$$\mathbf{D}\Psi = 0$$

therefore becomes the two coupled equations

$$\begin{aligned}-m\Psi_1 + (i\partial - q\Phi^*)\Psi_2 &= 0, \\ (i\partial^* - q\Phi)\Psi_1 - m\Psi_2 &= 0.\end{aligned}$$

Equivalently, their time evolution is

$$\begin{aligned}i\partial_t\Psi_1 &= (qV - \boldsymbol{\sigma} \cdot \mathbf{P})\Psi_1 + m\Psi_2, \\ i\partial_t\Psi_2 &= (qV + \boldsymbol{\sigma} \cdot \mathbf{P})\Psi_2 + m\Psi_1.\end{aligned}$$

Reverse these equations, multiply on the opposite side, and take scalar parts. The real potential terms cancel within each block, the two mass terms cancel in the sum, and the derivative terms combine into a divergence. Consequently,

$$\partial_t \text{scalar}(\Psi_1^R\Psi_1 + \Psi_2^R\Psi_2) + \nabla \cdot \text{vector}(\Psi_2\Psi_2^R - \Psi_1\Psi_1^R) = 0.$$

This verifies the density and current stated above and explains why the vector part of a single $\psi^R\psi$ is not the Dirac current. Solving the first equation for the upper component and substituting into the second gives

$$\begin{aligned}\Psi_1 &= \frac{(i\partial - q\Phi^*)\Psi_2}{m}, \\ 0 &= (i\partial^* - q\Phi)\Psi_1 - m\Psi_2 \\ &= -\frac{((\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2)\Psi_2}{m}.\end{aligned}$$

Thus each block satisfies a Klein–Gordon-type operator with the appropriate ordered field coupling.

The mass-conjugate operator is

$$\mathbf{D}' := \mathbf{W}(i\partial - q\Phi^*) + m\mathbf{I}.$$

Multiplying by $\mathbf{D}'$ yields

$$\begin{aligned}0 &= \mathbf{D}'\mathbf{D}\Psi = (\mathbf{W}(i\partial - q\Phi^*)^2 - m^2\mathbf{I})\Psi \\ &= -\text{diag}\left((\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2,\right. \\ &\quad \left.(\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2\right)\Psi.\end{aligned}$$

Equivalently,

$$\begin{aligned}0 &= ((\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2)\Psi_1 \\ &= (-E^2 + \mathbf{P}^2 + m^2 + iqF^*)\Psi_1, \\ 0 &= ((\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2)\Psi_2 \\ &= (-E^2 + \mathbf{P}^2 + m^2 + iqF)\Psi_2.\end{aligned}$$

Once the square is taken, the coupled Dirac system separates into two Klein–Gordon-type equations whose field terms differ only by the conjugation pattern fixed by the operator ordering.

6.6.11 Gauge Transform of the Dirac System

Let $\lambda \in \mathbb{R}$ be a real scalar gauge field and define

$$\Phi_\lambda = \Phi + \partial^*\lambda.$$

Because λ is real,

$$\partial\lambda = (\Phi_\lambda - \Phi)^*.$$

Transform the two-block field by

$$\Psi = \exp(iq\lambda)\Psi_\lambda.$$

Then the derivative of the phase factor is

$$\partial(\exp(iq\lambda)\Psi_\lambda) = \exp(iq\lambda)(iq\partial\lambda + \partial)\Psi_\lambda.$$

Therefore,

$$\begin{aligned}0 &= (\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi \\ &= (\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\exp(iq\lambda)\Psi_\lambda \\ &= \exp(iq\lambda)(\mathbf{W}(i\partial - q\partial\lambda - q\Phi^*) - m\mathbf{I})\Psi_\lambda \\ &= \exp(iq\lambda)(\mathbf{W}(i\partial - q(\Phi_\lambda - \Phi)^* - q\Phi^*) - m\mathbf{I})\Psi_\lambda \\ &= \exp(iq\lambda)(\mathbf{W}(i\partial - q\Phi_\lambda^*) - m\mathbf{I})\Psi_\lambda.\end{aligned}$$

Hence

$$(\mathbf{W}(i\partial - q\Phi_\lambda^*) - m\mathbf{I})\Psi_\lambda = 0.$$

This is the gauge-covariance statement: the phase change of the Dirac field is exactly compensated by the shift of the potential, so the coupled Dirac equation keeps the same form in the transformed gauge.

6.6.12 Dirac Equation Components

For the component form of the coupled Dirac equation, write

$$\begin{aligned}\partial &:= \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, \\ \partial^* &:= \partial_t - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, \\ \Phi &:= V + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ \Phi^* &:= V - \mathbf{A} \cdot \boldsymbol{\sigma}, \\ \psi &:= \psi_s + \psi_v \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3.\end{aligned}$$

For the field term, use

$$\begin{aligned}F &:= \mathbf{F} \cdot \boldsymbol{\sigma}, \\ \text{scalar}(F) &= 0, \\ \mathbf{F} &:= \mathbf{E} + i\mathbf{B} \in \mathbb{C}^3, \\ F^* &= -\mathbf{F}^* \cdot \boldsymbol{\sigma}.\end{aligned}$$

The supporting tables then give the scalar and vector parts of $\partial\psi$, $\partial^*\psi$, $\Phi\psi$, $\Phi^*\psi$, $iqF\psi$, and $iqF^*\psi$ explicitly. They are identities in the full algebra; imposing $\psi = \psi\Pi_0$ selects the physical submodule without changing those identities. Tables 37 and 38 record the direct component action of ∂ and ∂^* on a complex paravector test field.

Table 37. Components of $\partial\psi$.

scalar:	$\partial_t\psi_s + \boldsymbol{\nabla} \cdot \psi_v$
vector:	$\partial_t\psi_v + \boldsymbol{\nabla}\psi_s + i(\boldsymbol{\nabla} \times \psi_v)$

Table 38. Components of $\partial^*\psi$.

scalar:	$\partial_t\psi_s - \boldsymbol{\nabla} \cdot \psi_v$
vector:	$\partial_t\psi_v - \boldsymbol{\nabla}\psi_s - i(\boldsymbol{\nabla} \times \psi_v)$

Tables 39 and 40 then show how the potential Φ and its conjugate act on the same field in components.

Table 39. Components of $\Phi\psi$.

scalar:	$V\psi_s + \mathbf{A} \cdot \psi_v$
vector:	$V\psi_v + \psi_s\mathbf{A} + i(\mathbf{A} \times \psi_v)$

Table 40. Components of $\Phi^*\psi$.

scalar:	$V\psi_s - \mathbf{A} \cdot \psi_v$
vector:	$V\psi_v - \psi_s\mathbf{A} - i(\mathbf{A} \times \psi_v)$

Tables 41 and 42 complete the Dirac-block operator tables by recording the field-coupling terms $iqF\psi$ and $iqF^*\psi$.

Table 41. Components of $iqF\psi$.

scalar:	$iq(\mathbf{F} \cdot \psi_v)$
vector:	$iq\psi_s\mathbf{F} - q(\mathbf{F} \times \psi_v)$

Table 42. Components of $iqF^*\psi$.

scalar:	$-iq(\mathbf{F}^* \cdot \psi_v)$
vector:	$-iq\psi_s\mathbf{F}^* + q(\mathbf{F}^* \times \psi_v)$

6.6.13 Low-Energy Spin-0 Limit

For the minimally coupled scalar theory, use the covariant operators

$$E = i\partial_t - qV, \quad \mathbf{P} = -i\boldsymbol{\nabla} - q\mathbf{A}.$$

Remove the fast rest-mass oscillation by writing

$$\psi' = \exp(-imt)\psi.$$

Then

$$\begin{aligned}E^2\psi' &= \exp(-imt)(E + m)^2\psi, \\ \mathbf{P}^2\psi' &= \exp(-imt)\mathbf{P}^2\psi.\end{aligned}$$

Substituting into the minimally coupled Klein–Gordon equation gives

$$\begin{aligned}0 &= (-E^2 + \mathbf{P}^2 + m^2)\psi' \\ &= \exp(-imt)(-(E + m)^2 + \mathbf{P}^2 + m^2)\psi,\end{aligned}$$

hence

$$0 = (-E^2 - 2mE + \mathbf{P}^2)\psi.$$

For the free theory, or more generally when the operators in the iteration commute, this gives

$$\begin{aligned}E &= \frac{\mathbf{P}^2 - E^2}{2m}, \\ &= \frac{\mathbf{P}^2}{2m} - \frac{\mathbf{P}^4}{8m^3} + \dots.\end{aligned}$$

For noncommuting or time-dependent potentials the higher-order terms also contain commutators; the leading low-energy relation remains

$$E\psi \approx \frac{\mathbf{P}^2}{2m}\psi.$$

Solving for $i\partial_t\psi$ yields the Schrödinger equation

$$i\partial_t\psi \approx \left(\frac{(-i\boldsymbol{\nabla} - q\mathbf{A})^2}{2m} + qV \right) \psi.$$

If $\mathbf{A} = 0$, this reduces to

$$i\partial_t\psi \approx \left(-\frac{\boldsymbol{\nabla}^2}{2m} + qV \right) \psi.$$

The scalar low-energy limit is therefore already encoded in the relativistic Klein–Gordon theory.

6.6.14 Low-Energy Spin- $\frac{1}{2}$ and Pauli Limit

Start from the minimal-left-ideal two-block field

$$\Psi' = \begin{Bmatrix} \Psi'_1 \\ \Psi'_2 \end{Bmatrix}$$

and the potential-coupled Dirac equation

$$(\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I})\Psi' = 0.$$

Its two block equations are

$$\begin{aligned}(i\partial - q\Phi^*)\Psi'_2 - m\Psi'_1 &= 0, \\ (i\partial^* - q\Phi)\Psi'_1 - m\Psi'_2 &= 0.\end{aligned}$$

Introduce the covariant operators

$$E := i\partial_t - qV, \quad \mathbf{P} := -i\nabla - q\mathbf{A}.$$

The system is then

$$\begin{aligned}(E - \mathbf{P} \cdot \boldsymbol{\sigma})\Psi'_2 - m\Psi'_1 &= 0, \\ (E + \mathbf{P} \cdot \boldsymbol{\sigma})\Psi'_1 - m\Psi'_2 &= 0.\end{aligned}$$

Peel off the fast rest-mass oscillation with

$$\Psi'_n = \exp(-imt)\Psi_n.$$

Since

$$E\Psi'_n = \exp(-imt)(E + m)\Psi_n,$$

the reduced system becomes

$$\begin{aligned}(E + m - \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_2 - m\Psi_1 &= 0, \\ (E + m + \mathbf{P} \cdot \boldsymbol{\sigma})\Psi_1 - m\Psi_2 &= 0.\end{aligned}$$

In the Weyl basis neither block is small at rest. The energy-basis combinations are instead

$$\phi := \frac{\Psi_1 + \Psi_2}{\sqrt{2}}, \quad \chi := \frac{\Psi_2 - \Psi_1}{\sqrt{2}}.$$

Adding and subtracting the reduced equations gives the exact system

$$E\phi = (\mathbf{P} \cdot \boldsymbol{\sigma})\chi, \quad (E + 2m)\chi = (\mathbf{P} \cdot \boldsymbol{\sigma})\phi.$$

Thus ϕ is the large positive-energy field and χ is suppressed by one power of $1/m$. Formally,

$$\begin{aligned}\chi &= (E + 2m)^{-1}(\mathbf{P} \cdot \boldsymbol{\sigma})\phi, \\ E\phi &= (\mathbf{P} \cdot \boldsymbol{\sigma})(E + 2m)^{-1}(\mathbf{P} \cdot \boldsymbol{\sigma})\phi \\ &= \frac{(\mathbf{P} \cdot \boldsymbol{\sigma})^2}{2m}\phi + O(m^{-2}),\end{aligned}$$

where the expansion assumes energies small compared with $2m$ and smooth external fields on the Compton scale.

The magnetic term follows directly from the covariant-momentum commutator:

$$\begin{aligned}\text{com}(P_m, P_n) &= iq(\partial_m A_n - \partial_n A_m), \\ \mathbf{P} \times \mathbf{P} &= iq\mathbf{B}, \\ (\mathbf{P} \cdot \boldsymbol{\sigma})^2 &= \mathbf{P}^2 + i(\mathbf{P} \times \mathbf{P}) \cdot \boldsymbol{\sigma} \\ &= \mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}.\end{aligned}$$

Consequently,

$$E\phi = \frac{\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m}\phi + O(m^{-2}),$$

and the Pauli equation is

$$i\partial_t\phi = \left(\frac{(-i\nabla - q\mathbf{A})^2}{2m} + qV \right) \phi - \frac{q}{2m}(\mathbf{B} \cdot \boldsymbol{\sigma})\phi + O(m^{-2}).$$

Electric-field, spin-orbit, and Darwin terms arise at the next order through the noncommutation of E and $\mathbf{P}$; they must not be obtained by discarding the electric term in the second-order equation for a single Weyl block. The leading spin coupling is therefore a controlled consequence of the Dirac system [9, 17].

6.6.15 Spin Up and Spin Down

Where $\mathbf{B} \neq \mathbf{0}$, let its direction define the spin axis,

$$\mathbf{u} := \frac{\mathbf{B}}{\|\mathbf{B}\|},$$

and use the projector pair from Eq. (50),

$$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad (\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm\Pi(\pm).$$

Decompose the Pauli state by

$$\begin{aligned}\phi(\pm) &:= \Pi(\pm)\phi, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})\phi(\pm) &= \pm\phi(\pm), \\ \phi &= \phi(+) + \phi(-),\end{aligned}$$

so $\phi(+)$ and $\phi(-)$ are the spin-up and spin-down parts along the field axis. These left projectors preserve the fixed minimal-ideal condition $\phi = \phi\Pi_0$.

At leading order, write the Pauli equation as

$$\begin{aligned}\mathbf{P} &:= -i\nabla - q\mathbf{A}, \\ i\partial_t\phi &= \left(\frac{\mathbf{P}^2}{2m} + qV \right) \phi - \frac{q}{2m}\|\mathbf{B}\|(\mathbf{u} \cdot \boldsymbol{\sigma})\phi.\end{aligned}$$

Equivalently,

$$i\partial_t\phi = H\phi,$$

with

$$H := \frac{\mathbf{P}^2}{2m} + qV - \frac{q}{2m}\|\mathbf{B}\|\mathbf{u} \cdot \boldsymbol{\sigma}.$$

Applying $\Pi(\pm)$ to the time evolution gives

$$\begin{aligned}i\partial_t\phi(\pm) &= H\phi(\pm) + (i\partial_t\Pi(\pm) + \text{com}(\Pi(\pm), H))\phi \\ &= H\phi(\pm) + \left(i\partial_t\Pi(\pm) + \frac{\text{com}(\Pi(\pm), \mathbf{P}^2)}{2m} \right) \phi.\end{aligned}$$

Here qV and the local field-axis term commute pointwise with $\Pi(\pm)$; spatial variation enters through the displayed kinetic commutator. This term, together with $\partial_t\Pi(\pm)$, couples the two local spin sectors. If $\mathbf{B}$ is uniform and static, then $\Pi(\pm)$ is constant and the coupling term vanishes, so the two spin sectors decouple. In that case

$$(-\mathbf{B} \cdot \boldsymbol{\sigma})\phi(\pm) = \mp\|\mathbf{B}\|\phi(\pm),$$

and the Zeeman effect appears as

$$i\partial_t\phi(\pm) = \left(\frac{\mathbf{P}^2 \mp q\|\mathbf{B}\|}{2m} + qV \right) \phi(\pm).$$

The field direction therefore selects a preferred axis, and the two projector parts evolve with opposite Zeeman shifts.

6.6.16 Free Energy and Helicity Projector Checks

For the free momentum-mode Hamiltonian in Eq. (124), direct block multiplication gives

$$H_D^2 = \begin{pmatrix} (\boldsymbol{\sigma} \cdot \mathbf{p})^2 + m^2 & -m\boldsymbol{\sigma} \cdot \mathbf{p} + m\boldsymbol{\sigma} \cdot \mathbf{p} \\ -m\boldsymbol{\sigma} \cdot \mathbf{p} + m\boldsymbol{\sigma} \cdot \mathbf{p} & (\boldsymbol{\sigma} \cdot \mathbf{p})^2 + m^2 \end{pmatrix} \\ = (m^2 + \mathbf{p}^2)\mathbf{I} = E_{\mathbf{p}}^2\mathbf{I}.$$

It follows immediately that

$$\Lambda(\epsilon)^2 = \frac{1}{4} \left(\mathbf{I} + 2\epsilon \frac{H_D}{E_{\mathbf{p}}} + \frac{H_D^2}{E_{\mathbf{p}}^2} \right) = \Lambda(\epsilon), \\ H_D \Lambda(\epsilon) = \epsilon E_{\mathbf{p}} \Lambda(\epsilon).$$

The complementary relations are

$$\Lambda(+)\Lambda(-) = 0, \quad \Lambda(+) + \Lambda(-) = \mathbf{I}.$$

For $\mathbf{p} \neq \mathbf{0}$, the helicity axis obeys

$$(\boldsymbol{\sigma} \cdot \mathbf{p})\Pi(s) = s\|\mathbf{p}\|\Pi(s), \quad \text{com}(\boldsymbol{\sigma} \cdot \mathbf{p}, \Pi(s)) = 0.$$

The mass entries commute with every $\Pi(s)$, so

$$\text{com}(H_D, \Pi_h(s)) = 0, \quad \text{com}(\Lambda(\epsilon), \Pi_h(s)) = 0.$$

Thus the commuting product in Eq. (125) is a projector onto a simultaneous energy–helicity sector.

Finally, all entries of H_D , $\Lambda(\epsilon)$, and $\Pi_h(s)$ act from the left. Hence, whenever a block satisfies the fixed-ideal condition $\Psi_n = \Psi_n \Pi_0$, every projected block satisfies it as well, since

$$(O\Psi_n)\Pi_0 = O(\Psi_n\Pi_0) = O\Psi_n$$

for each such left-acting block entry O . The energy and helicity decomposition therefore introduces no second spinor column and preserves the one-Dirac-field carrier selected in Eq. (111).

7 Conclusions

The algebra used here is the real Pauli algebra $\text{Cl}_{3,0}$, written as one complex scalar plus one complex three-vector. Its economy comes from a structural fact rather than a new physical postulate: the central pseudoscalar provides the complex unit, the Pauli product combines dot and cross products, and the reverse-conjugate product is the 2×2 determinant. On real event paravectors that determinant is the Minkowski interval.

Within this framework the manuscript derives, with one notation, spatial rotations, Lorentz boosts, four-velocity and four-momentum, Maxwell’s equations, the Lorentz force, scalar relativistic waves, the Dirac factorization, and the leading Pauli Hamiltonian. The distinctions between representations are essential. Events transform by Lorentz congruence, the electromagnetic STA-bivector/biparavector carried by F transforms by similarity, and one Dirac field occupies a fixed minimal left ideal (one Pauli-matrix column). Once these distinctions are enforced, the compact notation agrees with the standard component theories.

The reformulation also makes several complementary structures easy to see. The real and imaginary parts of ∂F yield the four Maxwell equations; the two ordered momentum products yield the conjugate spin couplings F and F^* ; and the same axis projectors organize elementary functions, Pauli spin sectors, and free helicity sectors. These are connections among established equations, not evidence for an additional interaction.

7.1 Scope and outlook

The results in this paper assume flat Minkowski space, smooth fields, and domains on which the displayed differential operators and square roots are defined. The electromagnetic discussion is classical, and the wave-equation discussion is first-quantized. A negative-frequency solution becomes an antiparticle statement only after second quantization. Likewise, $FF^R/2$ is the energy-density/Poynting paravector seen in one frame; the full electromagnetic stress-energy object is the linear map $a \mapsto \frac{1}{2}FaF^R$ on real paravectors a .

A nonlinear coordinate map or algebraic change of basis in flat space changes the displayed components but does not by itself create curvature. A genuine extension to general relativity would require additional geometric structure—a position-dependent frame, a compatible connection, curvature, and covariant field equations. Complex space-time, dark-sector, and cosmological claims are outside what has been derived here.

Natural next steps are narrower and testable: a systematic treatment of operator domains and boundary conditions; a local-frame extension with independent curvature; constitutive maps for media; and a second-quantized construction in which the minimal-ideal carrier, charge conjugation, and multiparticle state space are specified explicitly.

A Verification and Formula Status

Table 43 separates exact algebraic identities from physical statements that require a stated representation or approximation. The accompanying symbolic report is generated by `tools/verify_sta_results.py`.

Table 43. Status of the principal constructions.

Statement	Standard correspondence	Status	Qualification
Eqs. (5) and (33)	Real Pauli/Clifford algebra [12, 14]	✓	Associative scalar–vector form of $\text{Cl}_{3,0}$.
Eqs. (58)–(62)	Similarity transport and the $\text{SL}(2, \mathbb{C})$ event action	✓	Algebra elements transform by similarity; reversion-self-adjoint events transform by congruence. The reverse conjugate is the inverse only after determinant-one normalization.
Eqs. (41) and (42)	Hilbert–Schmidt norm and 2×2 determinant	✓	The first is positive; the second is generally complex and its square root is branch-dependent.
Eqs. (57) and (54)	Spatial rotations and Lorentz boosts [4, 5]	✓	For spatial rotors the event congruence reduces to similarity; nonunitary event boosts require congruence.
Eq. (77)	Minkowski interval [6]	✓	Valid on real event paravectors with signature $(+, -, -, -)$.
Eqs. (93) and (96)	Maxwell electrodynamics [2, 15]	✓	Signs follow $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ and the passive-boost convention.
Eq. (100)	Relativistic Lorentz force [15]	✓	Division by γ gives the coordinate-time power and force equations.
Eq. (102)	Electromagnetic energy density and flux	✓	T is the chosen-observer paravector; the full stress-energy object is the linear map on real paravectors.
Eqs. (103) and (106)	Klein–Gordon equation [7, 8]	✓	The frequency split assumes the stated spatial operator domain.
Eqs. (113) and (111)	Weyl-basis Dirac equation [10, 16]	✓	One fixed minimal left ideal gives one Dirac field; unrestricted algebra blocks give two copies.
Eq. (115)	Squared minimally coupled Dirac system	✓	The upper and lower blocks carry $+iqF^*$ and $+iqF$, respectively.
Eqs. (119)–(121)	Leading Pauli reduction [9, 17]	✓	The expansion is in inverse mass after the large/small energy-basis split.
Eq. (125)	Free energy and helicity projectors	✓	Joint sectors commute for helicity (or any spin axis at rest), not for an arbitrary fixed axis.

Python Verification of Main Algebraic and Operator Identities

The identities below were checked symbolically in Python using SymPy and the standard Pauli-matrix realization of the sigma basis. Algebraic identities were reduced directly in matrix form, while differential identities were evaluated on generic test functions $\psi(t, x, y, z)$ with symbolic potentials and field components. The verification script is `tools/verify_sta_results.py`.

Check	Status	Detail
Eqs. (1)–(2)	✓	All nine basis products satisfy $\sigma_n \sigma_m = \delta_{nm} + \epsilon_{nmk} i \sigma_k$ in the Pauli realization.
Eq. (32)	✓	The product $(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma})$ reduces to $\mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$ for symbolic components A_n, B_n .

Eqs. (33)–(42)	✓	The complex paravector product, the determinant form $ZZ^{*\mathbf{R}} = S^2 - \mathbf{V}^2$, and the Euclidean scalar part $\text{scalar}(ZZ^{\mathbf{R}}) = S ^2 + \ \mathbf{V}\ ^2$ all reduce correctly.
Eqs. (58)–(60)	✓	For generic complex $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $T = f + \mathbf{g} \cdot \boldsymbol{\sigma}$, multiplication through by $\Delta_T = f^2 - \mathbf{g}^2$ reproduces the stated scalar-preserving vector formula for $T^{-1}ZT$. This directly checks the full complex-paravector similarity calculation.
Eqs. (54) and (96)	✓	For a boost along the first axis, the event congruence $X' = L^*XL^*$ gives the standard time/longitudinal mixing, leaves the transverse coordinates real and unchanged, and preserves $\det(X)$. It also satisfies the event intertwiner $LX' = XL^*$. The rejected similarity L^*XL does not give that event boost. The field similarity $F' = L^*FL$ gives the standard electric–magnetic mixing, whereas the rejected double-star field sandwich introduces a scalar part.
Eqs. (50), (78), (79)	✓	The idempotents $\Pi(\pm)$ satisfy the projector identities, $UU^{*\mathbf{R}} = 1$, and $E^2 - \mathbf{P}^2 = m^2$ follows from $mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}$.
Eqs. (91), (92), (93)	✓	The potential identity gives $S = \partial_t V + \nabla \cdot \mathbf{A}$, $\mathbf{E} = -\partial_t \mathbf{A} - \nabla V$, and $\mathbf{B} = \nabla \times \mathbf{A}$, while ∂F splits into Gauss, Faraday, and Ampère–Maxwell component equations with the manuscript’s signs.
Eqs. (99) and (100)	✓	The gauge law $\Phi_\lambda = \Phi + \partial^* \lambda$ gives $V_\lambda = V + \partial_t \lambda$ and $\mathbf{A}_\lambda = \mathbf{A} - \nabla \lambda$, and the product FU splits into the expected power term $\gamma \mathbf{v} \cdot \mathbf{E}$ and force term $\gamma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.
Eqs. (107), (108), and (109)	✓	Acting on a generic test function reproduces the expanded Klein–Gordon operator, the commutators $\text{com}(E, P_m) = iqE_m$ and $\text{com}(P_m, P_n) = iq(\partial_m A_n - \partial_n A_m)$, and the factorization $(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF$.
Eqs. (115), (120), and (123)	✓	Eliminating one Dirac component yields the two squared equations; after the common overall sign is chosen as in Eq. (115), the upper component carries $+iqF^*$ and the lower component carries $+iqF$. The exact Pauli identity $(\boldsymbol{\sigma} \cdot \mathbf{P})^2 = \mathbf{P}^2 - q \mathbf{B} \cdot \boldsymbol{\sigma}$ holds on a test function; and for uniform $\mathbf{B} = B_0 \mathbf{u}$ the projectors satisfy $(\mathbf{B} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm B_0 \Pi(\pm)$.
Eq. (69)	✓	The quaternion correspondence satisfies $QQ^{\mathbf{R}} = a^2 + \mathbf{b}^2$ and $Q^2 = a^2 - \mathbf{b}^2 - 2ia \mathbf{b} \cdot \boldsymbol{\sigma}$.

These symbolic checks complement the literature comparison in Section A. They separate purely algebraic and operator-theoretic identities from statements whose interpretation still depends on physical conventions.

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